



BASE II Clearing Visa Clearing Exchange (VCX)

User Guide

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Visa Supplemental Requirements

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Chapter 1

VCX Overview

This is an overview of Visa Clearing Exchange (VCX) processing. It describes the software, responsibilities of administrators and operators, and the support for operating systems and regional settings. It also explains how to sign on, configure VCX settings, browse and print VCX reports, and briefly explains how to use online Help.

This information is intended for system administrators and operators responsible for configuring and running routine jobs.

Important: VCX must be installed within the client's internal network and protected by their firewall. It is designed to operate as a secure, internal application and must not be exposed to the public internet or deployed in an internet-facing configuration.

VCX Description

The BASE II Clearing Visa Clearing Exchange (VCX), Release 1.0 is a software application supplied by Visa for processing transaction files sent to, and received from, VisaNet. VCX is the interface between the client processing center's internal payment processing system and an Extended Access Server (EAS), Direct Exchange (DEX), or Visa File Gateway (VFG). It prepares the client processing center's outgoing interchange for transmission to VisaNet. VCX also receives and reports incoming interchange files from Visa, preparing it for the client processing center's Host system.

VCX includes these features:

- Linux operating system
- Windows operating system
- Client-server architecture
- Secure framework
- Browser-based user interface
- Flexible reporting options
- Batch procedure customization via run control options

VCX, Release 1.0 is not backwards-compatible with previous versions of the Visa Clearing Exchange software or Edit Package Release 4.0.

VCX, Release 1.0 offers a variety of run control options to allow client's processing centers to customize VCX for their needs. VCX also offers a user-friendly and intuitive user interface, easy navigation, and online help.

VCX Key Features

Visa Clearing Exchange (VCX) provides flexibility in application configuration and daily processing.

Key Features:

- System administration
- Client-server architecture
- Format-independent architecture
- Use of rules engine and value tables
- Run control options
- Secure file transfer support
- Run automation

Format-Independent Architecture

VCX software uses a common internal format for its basic core processing, validation, and reporting. VCX fully supports processing of BASE II transactions.

Rules Engine and Value Tables

VCX transaction validation is performed using a common internal format and business rules defined in rules and value tables. The business rules and value tables have an effective date that supports the testing of future business enhancements by setting the table date to the effective date of the release. When new value tables and rules are developed, the changes will be distributed via TC 54 – Visa Clearing Exchange Table Update Record Transactions and will be automatically applied to the endpoint's environment. There will be no endpoint action required. If changes to rules and value tables are not received, they can be requested from Visa to the processing center through incoming interchange via TC 54.

VCX Tables include: Account Range Definition (ARDEF) table, Visa Identifier (VID) table, value tables, and rules tables.

Note: Endpoints that choose not to use VCX but want to receive VID, ARDEF, and other value table updates can choose to implement the TC 54 – Visa Clearing Exchange Table Update Record Transactions.

ARDEF Table:

The ARDEF Table contains all valid ARDEF entries, namely: the account prefix range, its associated issuing identifier, card-number length indicator, check-digit indicator, product ID, and account funding source. The ARDEF file is used to verify account numbers received in interchange.

VID Table:

An acquirer ID and issuer ID are six-digit system number numerics used by Visa to identify endpoints. The VID table contains all valid acquirer and issuer IDs and their BASE II processing status codes. The VID table file is used to associate acquirer and issuer identifiers in interchange with their account numbers received in interchange.

Value Tables:

The repository value tables are made up of various tables used in the validation of BASE II transactions. Examples include:

- Country Code
- Currency Code
- Merchant Category Code (MCC)
- Fee Reason Code
- Return Reason Code

Rule Tables:

The Rules tables contain business validation logic used by VCX to validate transaction data. The rules table is internal to VCX processing and layouts for this file are not provided.

Automatic Table Updates

Visa can send updates required to keep the endpoint's repository tables current on a daily basis. If an endpoint's repository tables are out of date, repository table update transactions sent by Visa can bring the tables up to date. The VCX software will determine and create the table updates required to bring processing center tables up to date, based on the current version of the tables at Visa and the endpoint's current table versions.

Note: The processing center must run the VCX outgoing process to provide Visa with the current status of its tables in order to receive table updates tailored to its current table levels. Otherwise, the default set of table updates will be sent.

Run Control Options

The use of run control options allows processing centers to customize VCX software to their specific processing needs. Users can create various run control profiles to satisfy their daily processing needs, such as testing, production, and special runs.

Run control options allow the selection of various functions, including, but not limited to:

- Establishing duplicate batch level detection settings.
- Splitting incoming ITF files into multiple CTF files.
- Setting tolerance checking limits at the item levels.

See the *Run Options* chapter for a complete list of supported run control options.

Configurable Parameters within the Config File

Users can use the Config file to configure various parameters, like customizing archive folders, setting user and password expiration times, and archiving and retention periods for files and logs. For additional details on configurable parameters, see the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

System Administration

The administrative function provides configuration tools for these functions:

- User Setup – Establishes administrator and operator IDs, passwords, operator privileges, approving operator IDs, resets passwords.
- BASE II CIB Setup – Set up CIBs for test and production.
- File Transfer Setup – Set up file transfer interface connection to VCX via Direct Exchange (DEX) Open File Delivery (OFD), Extended Access Server (EAS), or Visa File Gateway (VFG).
- Batch Procedure – Set up and maintain batch procedures.

VCX Administrator Responsibilities

Processing centers with multiple VCX users must designate one or more persons as the system administrator for VCX, Release 1.0. The system administrator will have these responsibilities:

- Ensure that there are at least two administrators for VCX, Release 1.0
- Managing a multi-operator environment
 - Adding and removing operator IDs
 - Designating operator modes and access level
 - Viewing all operators and administrators for their installation
- Storing a VCX application key

Note: The VCX application key must be securely stored in case the VCX administrator cannot log in to VCX. If the application key is not available to restore access, the VCX software must be completely reinstalled.
- Setting up a secure interface for EAS, DEX OFD, or VFG
- Setting up BASE II CIB data folders
- Ensuring that run control options are set correctly
- Ensuring that table and file updates are performed in a timely manner by monitoring the dashboard that contains the table and file update information on a regular basis
- Mentoring new operators
- Assisting operators to resolve problems

VCX Operator Responsibilities

A VCX operator may perform these tasks:

- Initiating incoming edit run and interchange transaction files (ITFs) from VisaNet
- Setting up temporary run control options that override permanent options for a single processing run
- Facilitating reprocessing of any transactions that were returned from VisaNet and needed to be validated by a subsequent outgoing edit run
- Receiving the outgoing center transaction files (CTFs) from the processing center's pre-edit software so VCX can validate the information during an outgoing edit run
- Facilitating reprocessing of any transactions rejected by the outgoing edit run and resubmitting them into outgoing edit run
- Preparing outgoing interchange for transmission to VisaNet
- Reviewing or printing reports

Formatting

One of the functions of VCX is to format CTF data into ITF data during an outgoing cycle and to format ITF data into CTF data during an incoming cycle. The processing center controls some formatting functions so that incoming files and transactions can be easily processed by the center's internal systems.

Validation

Validation of interchange files is a major function of VCX. In addition to standard validation, processing centers can select additional forms of validation, such as the setting of tolerance levels at the item levels and duplicate file and batch detection. Additional validation functions are controlled through run control options.

Reporting

VCX provides various reports. Formats for dates, time, and numeric values can be modified according to national standards and are available using run control options.

For details on the reports, see the *BASE II Clearing Visa Clearing Exchange (VCX) Reports Reference Guide* or the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

Reports generated by VCX will be identified by the prefix VX, followed by a three-digit number.

Report Types

Report Series	Report Type
100-199	Outgoing reports
200-299	Incoming reports
300-399	On-request reports
700-799	Optional transaction-based reports
999	Index of reports

For reports that provide totals at different levels, the report ID will include an alphabetic extension to represent the totaling levels:

- A – Transaction detail level
- B – Batch/acquirer/issuer ID level
- C – File/center level
- D – Central Processing Date (CPD) level
- E – Run level
- F – Cycle level

Secure File Transfer Support

BASE II interchange files can be securely sent and received to and from Visa, using these delivery channels:

- DEX OFD – Visa will provide the option for endpoints to use File Transfer Protocol Secure (FTPS) to transfer files to DEX OFD for VCX, Release 1.0. The FTPS Client for OFD is included with the VCX software installation package. Acquirers and Issuers may use this software for secure file transfer on outgoing runs. It is still required for endpoints to use third-party secure FTP server software for incoming runs through OFD. Additionally, FTPS products that are Requests for Comments (RFC) 4217-compliant and support a command line (scripting) interface and Transport Layer Security (TLS) 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from OFD.

See *Visa Direct Exchange Open File Delivery Service Implementation Guide* for information about channel encryption.

- EAS – Visa will provide the option for endpoints to use an FTPS connection to an EAS for VCX, Release 1.0. The FTPS Client for Extended Access is included with the VCX software installation package. Acquirers and issuers may use this software for secure file transfer. Additionally, FTPS products that are RFC 4217-compliant and support a command line (scripting) interface and TLS 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from EAS.

See *Visa Extended Access Server Endpoint Guide* and *Visa Extended Access Server Centralized (EAS-C) Client Implementation Guide* for information about EAS setup.

- VFG – Visa will provide the option for endpoints to use an FTPS to VFG for VCX, Release 1.0. The FTPS Client for VFG is included in the VCX software installation package. Acquirers and issuers may use this software for secure file transfer on outgoing runs. It is still required for endpoints to use third-party secure FTP server software for incoming runs through VFG. Additionally, FTPS products that are RFC 4217-compliant and support a command line (scripting) interface and TLS 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from VFG.

See *Visa File Gateway Client Implementation Guide* for information about VFG setup.

- Visa Cloud Connect – Support for Cloud Connect will be announced later.

Note: DEX OFD delivery channel will be replaced by a new Visa File Gateway (VFG) in future releases.

BASE II TC 54 File Delivery

Acquirers and issuers may receive the TC 54–Visa Clearing Exchange Table Update Record Transactions in either UNDIFF file type or by using BASE II customized file delivery. Direct Exchange, Visa File Gateway (VFG), or an Extended Access (EA) server can be used for either file delivery method.

The **VCX54** customized delivery file type enables the TC 54–Visa Clearing Exchange Table Update Record Transactions delivery as a custom file type. This allows both Visa Clearing Exchange (VCX) users and non-VCX users to receive the TC 54–Visa Clearing Exchange Table Update Record Transactions in a separate file.

BASE II Customized Delivery Type Field Value

BASE II customized delivery files are identified by a unique value in the BASE II Customized Delivery Type field, positions 109–113, of TCR 0–Incoming ITF records, in TC 90–Header Records.

Direct Exchange, VFG, and EA Server File Type Value

File type values identify the kinds of files that are listed on the available incoming file download lists for Direct Exchange, VFG, and EA Server file transfers.

The file type of **VCX54** identifies files containing TC 54–Visa Clearing Exchange Table Update Record Transactions.

Tools

- Initialize history files – used to initialize the history files. See [History File](#).
- EAS File Services (EFS) tool – used to view ready or purge files under a specific EAS profile and BASE II CIB. Additionally, users can cancel ready files or restore purge files. If the user's cancel ready or restore purge files request fails, the errors can be viewed in the file transfer protocol (FTP) log.
- View Application Log tool – lists all actions completed by the user in VCX as well as their timestamps.

Batch Procedures

Users can set up and maintain batch procedures, submit them for immediate execution or schedule for a given date or time using the scheduler, and view completed jobs and batch job history. Users can create multiple batch procedures. The View Batch Logs function allows users to view the log after the batch procedure is run. Users will be able to view, edit, duplicate, or delete the current batch procedures that can be run or create a new batch procedure. Batch procedures define the set of parameters for a job that users will run. A batch job is the execution of the batch procedure.

Users will be able to automate the execution of the batch procedure by running a script on a Linux or Windows server. The script can call a scheduler to run a batch at a specific date and time.

Batch procedures may be used for these processes:

- Incoming File Transfer (EAS, DEX OFD, or VFG)
- Incoming Run
- Outgoing Run
- Outgoing File Transfer (EAS, DEX OFD, or VFG)
- Table Extracts
- Table Reports

A single batch procedure can include a combination of these processes.

Note: VCX will not be backwards compatible with any previous version of VCX or Edit Package.

History File

The History File is used to store the history of the outgoing and incoming processing runs, and to control reruns and assign batch numbers for multiple daily runs.

VCX Benefits

Visa Clearing Exchange (VCX), Release 1.0 provides extensive flexibility using run control options. These options allow processing centers to execute VCX software in a way best suited to their individual processing needs. A default set of Run Control Options is supplied with the VCX software that contains Incoming and Outgoing settings. The processing center can update the default options at any time, either permanently or temporarily.

Users must choose run control options and report application configuration options the first time the main application is run after installation. Permanent settings, options, and table overrides may be changed at any time.

VCX offers these benefits:

- Helps ensure file and data integrity of BASE II clearing files submitted by endpoints into VisaNet.
- Helps ensure secure transmissions between VisaNet and endpoints.
- Supports batch automation for "Incoming" and "Outgoing" processing.
- Supports report generation and table extraction.
- Supports Linux OS.
- Supports Windows OS.
- Provides format translation from EBCDIC to ASCII and from ASCII to EBCDIC.

VCX Support

For the production environment, VCX support is available for clients should they encounter any issues. Visa clients should contact their Visa account representative or email the Visa Operations Command Center – Clearing and Settlement Operation Support. If a client has Priority 1 issue that requires urgent attention, they may call 1-800-VISANET (1-800-847-2638). If the issue cannot be handled immediately, VCX Support personnel will be called.

For migration related support Visa will respond to tickets based on priority. For example, when the Action portion of an error message description in Visa Clearing Exchange (VCX) says to contact Visa for support, you should contact your Visa representative. Before contacting Visa Technical Support, you should have this user and system information available for the support technician.

- Your name, phone number, and email address.
- Basic information about your processing center, such as its name, BASE II CIB, location, supervisor, phone number, and email address.
- A full description of the problem, including the exact steps and operations you performed before the problem occurred, as well as any steps taken to resolve the problem. The more complete and concise the description is, the more effective Visa Technical Support will be in assisting you.
- The exact text, number, logs, and source of all error messages related to the problem.
- Information about the Visa Clearing Exchange application, such as the release version, specification file version, table file versions, and server operating system. For easy access to this information, see the About Page, which is found by navigating from **Help > About**.
- Any other information that could be pertinent or useful in resolving the problem. For example, if the problem involves file transfers, your EAS version and IP address could be essential to resolving the problem.

About Page in VCX

This describes the **About** page in Visa Clearing Exchange (VCX).

The **About** page lists the version, user interface version, and server version information for VCX. Additionally, the **About** page gives more information about access and the proprietary nature of the VCX software.

VCX Documentation

This documentation is about, or in relation to, Visa Clearing Exchange (VCX).

VCX Documentation

- *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Client Implementation Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Messages Reference Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Reports Reference Guide*

BASE II Documentation

- *BASE II Clearing Data Codes*
- *BASE II Clearing Interchange Formats, TC 01 to TC 49*
- *BASE II Clearing Interchange Formats, TC 50 to TC 92*
- *BASE II Clearing Overview*
- *BASE II Clearing Services*

VCMS Testing Documentation

- *VisaNet Certification Management Service (VCMS) BASE II Testing Guide*

Other Documentation

- *Visa Core Rules and Visa Product and Service Rules*
- *Visa Europe Operating Regulations – Processing*
- *Visa Merchant Data Standards Manual*

Chapter 2

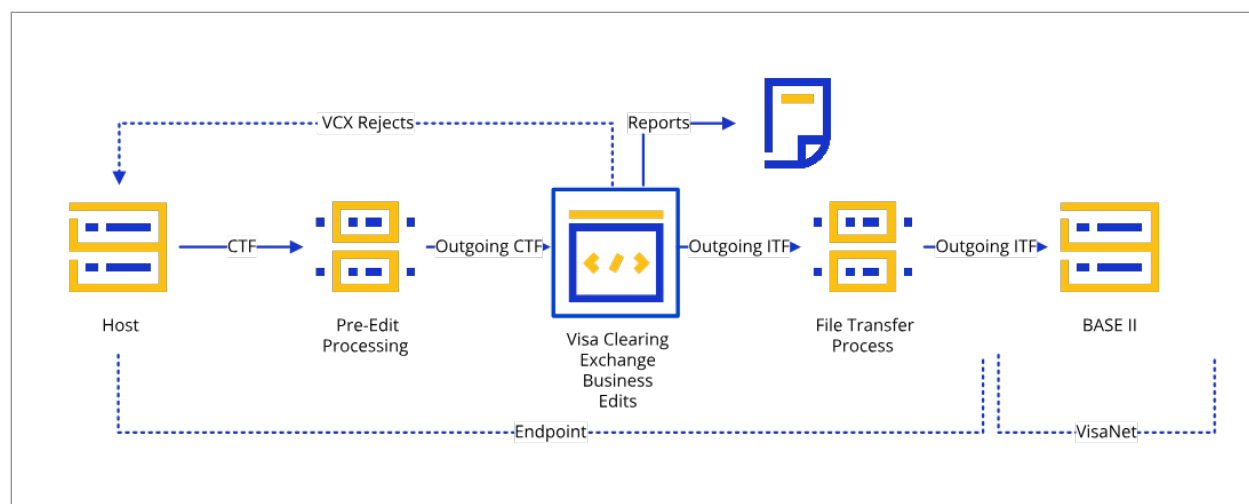
VCX Process Flow

Visa Clearing Exchange (VCX) has two processing phases: outgoing interchange processing and incoming interchange processing.

VCX Outgoing Processing

The term "outgoing" refers to interchange sent from the client's processing center to VisaNet.

VCX Outgoing Processing



Pre-Edit Processing

The first part of the outgoing process is performed by software belonging to the client's processing center. This is usually referred to as pre-edit processing since it is performed before processing by Visa Clearing Exchange (VCX). Pre-edit processing involves the capture, editing, and formatting of interchange data. Transactions submitted into interchange are processed through the client processing center's pre-edit software that places this information into a Center Transaction File (CTF). Once pre-editing is completed, the data is ready to be run through VCX.

VCX Processing

Outgoing runs process each transaction found in the input CTF file. The run control profile is used to identify a named set of options defined in the repository table for Outgoing processing. At the end of processing, the control program produces the necessary reports. VCX then produces an Interchange Transaction File (ITF) that is transferred to an Extended Access Server (EAS), Direct Exchange (DEX) Open File Delivery (OFD), or Visa File Gateway platform, where it is held until a predetermined collection time.

BASE II System Processing

At a predetermined time, the BASE II system collects the interchange files from EAS, DEX OFD, or VFG. As the interchange transactions are received by BASE II, clearing occurs. The transactions are validated, converted to the appropriate currencies, and totaled and sorted by destination. Fees are then calculated. At the end of each BASE II cycle, settlement positions are computed for all clients with either incoming or outgoing transactions.

Outgoing Processing Errors

If an error is encountered during an outgoing VCX run, the VX-199 Log Report and/or the VX-100A/B/C Validation Exception Detail Reports will be generated. The outgoing run results will also be displayed on the dashboard. The endpoint will be able to review the VX-199 and VX-100 reports to take appropriate actions or contact Visa support.

Outgoing Return Codes

Return Code	Description
0	Successful, no errors. ITF and reports created.
2	Transaction-level warnings. ITF and reports created. Transactions with warnings are included on the VX-100A Validation Exception Detail Report.
4	Transactions rejected. ITF and reports created. Rejected transactions are included on VX-100A Validation Exception Detail Report and omitted from output ITF.
16	Severe errors. ITF not created. VX-100A Validation Exception Detail Report and VX-199 Log File Report created.
20	The input file is not present in the designated file location on the server.

Duplicate File Detection

Duplicate file detection offers processing centers the ability to check for potential file duplicates prior to processing the CTF.

The duplicate file detection feature is based on checking a TC 90 header record in the outgoing CTF. The header record is used for duplicate file detection within the same processing date. The provision of an outgoing CTF TC 90 header record is optional.

Processing centers that use this feature must generate the TC 90 header record in pre-edit software that creates outgoing CTFs destined for outgoing interchange.

Duplicate file detection is based on two data elements in the TC 90 header record:

- Center Processing Date – This is the processing date of the client's pre-edit system.
- File ID – This is a three-digit number that must be unique for each CTF processed by VCX for any given Center Processing Date. File IDs do not need to be in an ascending sequence.

The outgoing VCX run compares the combination of Center Processing Date and File ID against the VCX History File. If a match is found, the VCX run is immediately terminated.

Processing centers can produce a TC 90 header record and bypass duplicate detection by applying a value of zero in the File ID field.

Important: If multiple pre-edit systems create outgoing CTFs destined for outgoing interchange, Visa recommends that a predefined range be allocated to each pre-edit system to ensure uniqueness of File IDs and to avoid potential conflict.

Processing Collection-Only Transactions

During an outgoing run, Visa Clearing Exchange (VCX) identifies collection-only transactions in the input file and writes these transactions into a separate outgoing interchange file.

The outgoing process will split the accepted ITF into two separate files. The first output file will contain financial and non-financial transactions excluding collection only transactions and the second output file will contain collection only transactions. Each file will have a separate unique file ID.

The collection-only split ITF file will have a different unique file ID (file name).

VCX builds the unique file ID as nnnnnn0yyyymmddMrrff00 for ITF.

Where:

- nnnnnn is CIB value
- 0 is a separator (always zero)
- yyyymmdd – run date
- M – is the run mode (P for prod and T for Test)
- rr is the Run number
- ff is the File number, where 01 represents financial transactions, and 02 represents collection-only transactions
- 00 is the trailing constant, where 00 is with financial transactions and CO is collection-only transactions
- There are some processors that may have multiple data centers and may opt to use the File ID Modifier run control option, which will replace the run mode rr – run number and ff – file number 00

The collection-only file unique-file ID will be the same as the other file with the exception that file number will be 02 and the trailing constant 00 will instead be the characters CO. The file number can be 02 because currently for an outgoing run there is never more than one file per run.

Each of the two output ITF files will start with batch number 1. The VCX History file will be updated with information for the two ITF files created. This will enable reconciliation of Batch Acknowledgments (TC 44).

Note: In certain regions (for example, CEMEA/AP), Visa has mandated that all domestic original 'On-Us' Visa point of sale (POS) transactions be submitted to VisaNet daily. Clients must submit all "on-us" settled transactions as collection only transactions.

Collection-Only (CONLY) File Type

File type 'CONLY' has been defined to identify collection-only transactions. The first output file will contain financial and non-financial transactions excluding collection-only transactions with file type of 'UNDIF' in the file header (TC 90, TCR 0, positions 74-78 and positions 109-113). The second output file will contain collection-only transactions with file type of 'CONLY' in the file header (TC 90, TCR 0, positions 74-78 and positions 109-113).

VCX will use the file type 'CONLY' to populate the custom file type for inclusion in the ITF header file that is required when sending compressed files to Extended Access Server (EAS).

Collection-Only Reports

Collection-only transaction counts and amounts are segregated from other transactions on all outgoing VCX summary reports such as the VX-110B, VX-110C, VX-110E, VX-110F, and VX-120F.

Outgoing Interchange Summary by Currency Code reports (VX-111C, VX-111E) which totals the financial transaction types by currency codes will not include collection-only transactions.

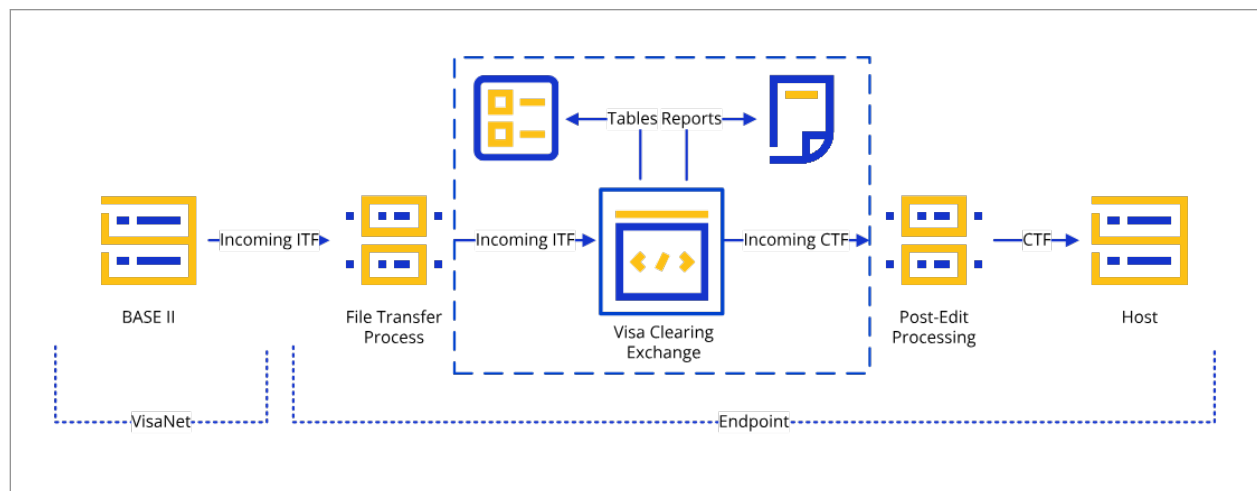
Outgoing Validation Exception reports (VX-100A, VX-100B, VX-100C) which list transactions found in error will include collection-only.

Incoming Batch Acknowledgment reports (VX-220, VX-221B, VX-221D, VX-221E) support the reporting of two outgoing files and the corresponding batch acknowledgments.

VCX Incoming Processing

The term incoming refers to interchange coming into the client's processing center from VisaNet.

VCX Incoming Processing



VCX Processing

An incoming VCX run processes each transaction found in the input ITF file. You can define a run control profile to be used to identify a named set of options defined in the repository table for incoming processing.

At the end of processing, the control program produces the necessary reports and makes the CTF file available for use by the client's processing center if there are no severe errors in the interchange file.

Incoming VCX files contain:

- Financial and nonfinancial interchange transactions
- VSS settlement reports
- VCX table updates

Note: To ensure proper application of table update transactions, all incoming files must be processed. Files must be processed in sequence by the Central Processing Date (CPD) on which they were created to properly update the repository tables and code-related files.

Post-Edit Processing

Post-edit software, provided by the client's processing center or by an outside vendor, reformats, and processes incoming interchange received from VisaNet. This software prepares interchange so that it can be processed by cardholder- and merchant-posting programs and used for other processing center purposes, such as fraud analysis.

Input File Sequence

The Input Sequence feature sequences incoming files automatically based on their contents. This functionality allows continuation files to be sequenced without manual intervention.

Incoming Processing Errors

If an error is encountered during an incoming VXC run, the VX-299 Log Report and/or VX-200A Validation Exception Detail Report will be generated. The incoming run results will also be displayed on the dashboard. The endpoint can review the VX-299 and VX-200A reports to take appropriate actions or contact Visa for support.

Incoming Return Codes

Return Code	Description
0	No errors were encountered during the incoming process and the table update process finished successfully.
1	Successful with Delta TC54 updates.
2	Successful with informational message.

Incoming Return Codes

Return Code	Description
5	No errors were encountered during the incoming process and TC 54 updates records were not present in the input file.
6	Processing of TC 54 updates failed.
8	Failed, TC 54 data problem.
16	Failed, errors were encountered.
20	The input file is not present in the designated file location on the server.

Chapter 3

Before You Begin with VCX

This chapter includes the prerequisites for Visa Clearing Exchange (VCX), as well as administrator and operator responsibilities. See the *VisaNet Global Technical Letter and Implementation Guide* or your Visa Representative for the most recently supported versions for system requirements.

System and Operational Requirements for VCX

Before installing Visa Clearing Exchange (VCX), clients must ensure that they can meet the necessary system and operational requirements.

Important: The most up-to-date system requirements for VCX are available in the most recent version of the *VisaNet Business Enhancements Global Technical Letter and Implementation Guide*.

Installation Hardware Requirements:

These are the minimum hardware requirements for installing VCX:

- **CPU** – 4CPU/4 CORE, 64bit
- **Disk storage** – 500 GB

- **Memory** – 32 GB
- **Communication Protocol** – HTTPS

Note: VCX only supports HTTPS communication between client and server machines. This requires a TLS certificate. During part of the installation process, you will have an option to install self-signed certificates generated by the OpenSSL application or a certificate issued by any recognized certificate issuing provider.

- **File Transfer Protocol** – FTPS

Operating System Requirements:

Visa supports the VCX application running on Linux OS or Windows OS:

- **Red Hat Enterprise Linux** – RHEL 8.10 (minimum) or 9.4 (maximum)¹
- **Windows Server 2022** – Standard Core

Runtime Library:

VCX requires these libraries, which the Automated Prerequisite Software Installer installs:

- **NodeJS** – Version 20¹
- **Open JDK/Java** – Version 21¹
- **Open SSL** – 1.1.1¹

Browser Requirements:

VCX can be used with these browsers:

- **Google Chrome** – Version 120¹ (minimum) or Version 139 (maximum)
- **Microsoft Edge** – Version 120¹ (minimum) or Version 139 (maximum)

Databases:

VCX requires this database with a generic setup:

PostgreSQL – Version 16.8¹

Notes:

PostgreSQL setup will be included in the Prerequisite Install script.

VCX is supported on the listed version and may be supported on a higher version provided the higher version maintains backward compatibility with the listed version.

¹ VCX is supported on the listed version and may be supported on a higher version provided the higher version maintains backward compatibility with the listed version.

S-Nail:

By default RHEL 9.0 or above uses S-Nail

Note: S-Nail replaces Mailx from RHEL 9.0 or above

VCX requires the use of Mailx for two-factor authentication (2FA) for Linux:

Mailx–Version 12.5 or above (this is the default version with Linux 8 or above)

SMTP Mail:

VCX requires the use of SMTP mail for multifactor authentication (MFA) for Windows:

File Transfer Program (FTP):

VCX requires the use of FTP for Windows:

FTPS – or File Transfer Protocol Secure is a secure file transfer protocol that uses encryption to secure file transfers over a network using Transport Layer Security (TLS) protocol.

PORT:

Ensure that ports 3000 and 8000 are open and available for VCX services to run.

More Information

Contact a regional Visa representative or your client support team to learn more.

VCX Secure Connection

This describes the three secure connections you can establish for Visa Clearing Exchange (VCX).

VCX Secure EAS Interface

The option to use a secure File Transfer Protocol Secure (FTPS) connection to an Extended Access Server (EAS) is available in VCX. Most FTPS products that are RFC 4217-compliant and support a command line (scripting) interface should be able to be configured to work with VCX. For setup information, see the latest *EAS Endpoint Guide*.

VCX Secure DEX OFD Interface

The option to use FTPS to transfer files to Direct Exchange (DEX) Open File Delivery (OFD) is available in VCX. FTPS products that are RFC4217/RFC 2228 compliant and support a command-line (scripting) interface and Transport Layer Security (TLS) 1.2 or later can be configured to work with VCX. For more information about channel encryption using FTPS, see the latest *Visa Direct Exchange Open File Delivery Service Implementation Guide*.

VCX Secure VFG Interface

The option to use FTPS to transfer files to Visa File Gateway (VFG) is available in VCX. FTPS products that are RFC4217/RFC 2228 compliant and support a command-line (scripting) interface and TLS 1.2 or later can be configured to work with VCX.

VCX Administrator Responsibilities

Processing centers with multiple operators must designate one or more persons as the System Administrator for Visa Clearing Exchange (VCX).

The administrator has these responsibilities:

- Managing a multi-user (multi-operator) environment
 - Adding and removing operator IDs
 - Designating operator modes and access levels
 - Viewing all operators and administrators for their installation
- Store a VCX Application Key
- Setting up a secure interface for Extended Access Server (EAS), Direct Exchange (DEX) Open File Delivery (OFD), or Visa File Gateway (VFG)
- Setting up BASE II CIB data folders
- Ensuring that run control options are set correctly
- Ensuring that table and file updates are performed in a timely manner
- Mentoring new operators
- Assisting operators in resolving problems.

The primary administrator for the installation of VCX is the first administrator and can to approve, promote, and manage other users of VCX. After the primary administrator is established, other users may register using the online interface. From there, the primary administrator or any other administrator may approve, promote, or manage other users. Administrators can also limit and mix and match the functionality available to other administrators.

VCX Operator Responsibilities

A Visa Clearing Exchange (VCX) operator may perform these tasks. Operators must register via the online interface, described in *Getting Started with VCX*.

These are the activities that a VCX operator may perform:

- Initiating incoming edit run of ITFs from VisaNet
- Setting up temporary run control options that override permanent options for a single processing run
- Facilitating reprocessing of any transactions that were returned from VisaNet and needed to be validated by a subsequent outgoing edit run
- Receiving the outgoing CTFs from the processing center's pre-edit software so VCX can validate the information during an outgoing edit run
- Facilitating reprocessing of any transactions rejected by the outgoing edit run and resubmitting them into outgoing edit run
- Preparing outgoing interchange for transmission to VisaNet
- Reviewing or printing reports

Validate Checksum for the VCX Package

The checksum for the build package (.tgz) file is provided in the **ReadMe.txt** file. There is one for Windows and another for Linux. This procedure describes how to generate the checksum for the Windows or Linux installation of Visa Clearing Exchange (VCX) and validate that the checksum value is present in the **ReadMe.txt** file.

1. Generate the checksum of the downloaded file.
2. **For Linux:** Use the following command to generate the checksum of the package.

```
sha256sum /path/to/received/file > new_checksum.txt
```

Replace `/path/to/received/file` with the actual path to the downloaded file.

3. **For Windows:** Use the following command to generate the checksum of the package.

```
CertUtil -hashfile path\to\received\file SHA256 > new_checksum.txt
```

Replace `path\to\received\file` with the actual path to the downloaded file.

4. Compare the Checksums by comparing the checksum values in the **VCX_Full_Install_ReadMe.rtf** file and **new_checksum.txt**. If the checksum values match, the file was not altered during the transfer. If the checksum values do not match, the file may have been tampered with or corrupted during the transfer. Contact your Visa Representative for assistance.

VCX Deployment on AWS – Network Settings

To maintain a secure VCX deployment in AWS, adhere to the following network configuration recommendations.

Inbound Traffic

Allow inbound access only on the port configured for VCX (default: TCP 3000). Restrict all other inbound ports unless explicitly required for your environment.

Outbound Traffic

Limit outbound access to:

- The EAS/VFG control port and passive data ports required for file transfer operations.
- SMTP ports used for outbound email notifications.

Note: If using SMTP port 25, AWS may throttle outbound traffic by default. To avoid delays or delivery issues, submit a request to AWS Support to remove the restriction, or configure VCX to use an alternative SMTP port (such as 587 or 465) supported by your mail relay.

These settings help ensure secure communication while maintaining compatibility with VCX's operational requirements.

Note: For specific VCX installation on the Linux OS, refer to [VCX Download and Install Procedures for Linux OS](#). For Windows, refer to [VCX Download and Install Procedures for Windows OS](#).

Updating CONFIG Settings for Mainframe-to-VCX Conversions

When a Mainframe client is converting to VCX, the Account Manager must update the CONFIG settings in VCMS for testing. This step prevents issues with HASH values in the VCMS/TEST environment.

Steps to Update CONFIG Settings

1. Log in to CORE / CONFIG
 - a. Open the CORE / CONFIG application.
 - b. From the Environments selection menu, choose **VCMS**.
2. Expand the Clearing / VSS Section
3. Click the Application Processor under Clearing/VSS Section.
4. Enter the CIB (Center Identification Block) number of the bank in both (Start Key & End Key fields)
5. Click **Search**.
6. From the search results, click on the first entry in the list.
7. Update Details
 - a. On the Application Processor details screen:
Set Verify/Generate Hash Totals to **No**.
Update Processor Platform/OS to the correct value for the conversion.

Chapter 4

VCX Prerequisite Software Installer

This chapter describes the download and install procedures for Visa Clearing Exchange Prerequisite Software Installer on Linux & Windows OS.

You will also need to decide if the PostgreSQL DB will be installed on the same server as the VCX application.

- If you want to install the PostgreSQL DB on the same server, select the "yes" option for PostgreSQL DB Install when executing the VCX-Pre-requisite Software Installer.
- If you want to install the PostgreSQL DB on a separate server, select the "no" option for PostgreSQL DB Install when executing the VCX-Pre-requisite Software Installer.

Note: When the "no" option for PostgreSQL DB Install is selected the PostgreSQL database installation and related activities are not managed by the VCX Prerequisite Software Installer. You will need to manage these processes separately.

Create a PostgreSQL DB account, database, and schema on the server of your choice for use by the VCX application and ensure that a network connection is established between the PostgreSQL Server and the VCX Application Server.

Once you have created the PostgreSQL DB account, database, and schema, set up the network connection between the PostgreSQL Server and the VCX Application Server. Run the following command on your VCX Application Server to verify that the connection is successful.

- In your VCX Application Server, open the terminal and type the following command:

```
curl -v https://DBhostname:port (where  
"DBhostname:port"=Hostname and port of PostgreSQL server)
```

Results

Successful connection response

```
* Trying DBhostname...  
* Connected to DBhostname (DBhostname) port 5432 (#0)  
* ALPN, offering h2  
* ALPN, offering http/1.1  
* CAfile: /etc/pki/tls/certs/ca-bundle.crt  
* TLSv1.0 (OUT), TLS header, Certificate Status (22):  
* TLSv1.3 (OUT), TLS handshake, Client hello (1):  
* OpenSSL SSL_connect: SSL_ERROR_ZERO_RETURN in connection to  
DBhostname:5432  
* Closing connection 0  
curl: (35) OpenSSL SSL_connect: SSL_ERROR_ZERO_RETURN in  
connection to DBhostname:5432
```

Unsuccessful connection response

```
* Trying DBhostname...  
* TCP_NODELAY set  
* connect to DBhostname port 5432 failed: Connection timed out  
* Failed to connect to DBhostname port 5432: Connection timed  
out  
* Closing connection 0  
curl: (7) Failed to connect to DBhostname port 5432:  
Connection timed out
```

If you would like to enable SSL on a PostgreSQL server please follow these instructions:

1. Navigate to the PostgreSQL data folder where PostgreSQL is installed, for example on Linux: - /var/lib/pgsql/data and on Windows: C:\Program Files\PostgreSQL\<version>\data\.
2. Open the pg_hba.conf file in a text editor, find and update the following information:
 - DATABASE, (database name)
 - USER, (PostgreSQL username)
 - ADDRESS (IP address of the server where the VCX application is installed)
 - METHOD, (authentication method)

3. Open the `postgresql.conf` file in a text editor, find, uncomment, and update the following information:
 - `listen_addresses`, (IP address of the server where the VCX application is installed)
 - `ssl`, (on)
 - `ssl_ca_file`, (`ca.crt`)
 - `ssl_cert_file`, (`server.crt`)
 - `ssl_key_file`, (`server.key`)
 - `ssl_ciphers_file`, (`HIGH : MEDIUM : +3DES : ! aNULL # allowed SSL ciphers`)
 - `ssl_prefer_server_ciphers`, (on)
 - `ssl_esdh_curve`, (`prime256v1`)
 - `ssl_min_protocol_version`, (`TLSv1.2`)
 - `ssl_max_protocol_version`, (`TLSv1.3`)
4. To apply the configuration changes, restart the server by entering these commands:
 - on Linux -
`sudo systemctl reload <postgresql service name, ex: postgresql>`
`sudo systemctl restart <postgresql service name, ex: postgresql>`
 - on Windows - Using Powershell
`Restart-Service -Name <postgresql service name, ex: postgresql-x64-15>`

LINUX Prerequisites Installer

The Linux Prerequisite Installer is a launcher using the `install_prereqs.sh` script, to install essential prerequisites for your application on Linux systems.

Directory Structure

Linux-Prerequisite-Software.tgz —rhel/

```
| └─ packages/
|   | └─ node-v20.17.0-linux-x64.tar.gz
|   | └─ microsoft-jdk-21.0.4-linux-x64.tar.gz
|   └─ install_prereqs.sh
| └─ licenses/
```

Getting Started

Prerequisites

- **Administrative Privileges:** You must have root access to run the script.

Installation Steps

1. Create a folder and place the Linux-Prerequisite-Software.tgz file

```
mkdir <folder_name>
cd <folder_name>
mv Linux-Prerequisite-Software.tgz <folder_name>
```

2. Extract the Installer Package

Make sure to extract the entire `Linux-Prerequisite-Software` directory to a local directory `<folder_name>`, for example, `/opt/Linux-Prerequisite-Software`.

```
tar -xvf Linux-Prerequisite-Software.tgz
```

3. Update the `install_prereqs.sh` file with execution permission using the following command.

```
cd ./package/rhel
chmod +x ./install_prereqs.sh
```


4. Execute the Script with Root Privileges

```
./install_prereqs.sh
=====
| VCX Prerequisites Installation Begins |
=====

Node.js is not recognized on your system and may not be installed.
Would you like to install Node.js version 20.17.0 now? [Y/N]: Y
Installing Node.js. Please wait...
Node.js has been successfully installed.
Java is not recognized on your system and needs to be installed.
Would you like to install Java version 21.0.4 now? [Y/N]: Y
Installing Java. Please wait...
Java has been successfully installed.
OpenSSL is already installed on your system (version 1.0.2).
A newer version of OpenSSL (3.3.2) is available. Do you want to
upgrade? (Y/N): Y
PostgreSQL is not recognized on your system.
Would you like to install PostgreSQL version 16 on this server?
[Y/N]: Y
PostgreSQL server installed successfully.
These credentials will be used for authenticating the VCX
application to the PostgreSQL server.
Enter the PostgreSQL username for the VCX application: vcxuser
Enter the PostgreSQL password for user 'vcxuser':
Confirm password:
VCX application PostgreSQL user setup completed successfully.

=====
| VCX Prerequisites Installation Complete |
=====
```

Note: When the "yes" option for PostgreSQL DB install is selected, the script will install the PostgreSQL Server on the same server including the **vcxdb** database along with the **vcx** schema.

When the "no" option for PostgreSQL DB install is selected the PostgreSQL DB installation and related activities are not managed by the VCX Prerequisite Software Installer. You will need to manage these processes separately.

If no errors are detected during the installation, a similar message confirming that the installation is complete will be displayed:

```
Installation ended at Sat Nov 9 18:28:50 GMT 2024.
```

Note: Installation details can be reviewed in <folder_name>/rhel/
install_prereqs.log located in the script directory.

5. To refresh the PATH environment variables, you can use the following command.

```
source /etc/profile
```

Windows Prerequisite Installer

The Windows Prerequisite Installer is a PowerShell launcher script designed to install essential prerequisites for your application on Windows systems. The prerequisites include:

- Node.js
- OpenJDK
- OpenSSL
- PostgreSQL 16

All necessary installer files are included in the `installer's` directory.

Directory Structure

Windows-Prerequisite-Software.tgz/

```
|— windows/  
| |— installers/  
| | |— node-v20.17.0-x64.msi  
| | |— microsoft-jdk-21.0.4-windows-x64.msi  
| | |— postgresql-16.8-1-windows-x64.exe  
| | |— FireDaemon-OpenSSL-x64-3.3.2.exe  
| |— install_prereqs.ps1  
| |— licenses/
```

Getting Started

Prerequisites

- **Administrator Privileges:** Are required to install software and modify system environment variables.

Installation Steps

1. Create a folder and place the Windows-Prerequisite-Software.tgz file

```
mkdir <folder_name>  
cd <folder_name>  
mv Windows-Prerequisite-Software.tgz <folder_name>
```

2. Extract the Installer Package

Make sure to extract the entire Windows-Prerequisite-Software directory to a local directory <folder_name>, for example, /opt/Windows-Prerequisite-Software .

```
tar -xvf Windows-Prerequisite-Software.tgz
```

3. Run the Launcher Script

a. Open PowerShell as an Administrator

Using GUI:

Right-click the PowerShell icon and select **Run as Administrator**.

Using non-GUI:

Open PowerShell as an administrator from the Command Prompt.

```
powershell -Command "Start-Process PowerShell -Verb RunAs"
```

b. Execute the Launcher Script:

```
cd package/windows/  
./installer prereqs.psl  
=====
```

| VCX Prerequisites Installation Begins |

```
=====
```

Node.js is not installed. Do you want to install it? (Y/N): **Y**
Installing Node.js, it may take a while, please wait...
Node.js installed successfully.
A newer version of OpenJDK (21.0.4) is available. Do you want to upgrade? (Y/N): **Y**
Installing OpenJDK, it may take a while, please wait...
OpenJDK installed successfully.
Added 'C:\Program Files\Microsoft\jdk-21.0.4' to system PATH.
OpenSSL is already installed on your system (version 1.0.2).
A newer version of OpenSSL (3.3.2) is available. Do you want to upgrade? (Y/N): **Y**
Installing OpenSSL, it may take a while, please wait...
OpenSSL installed successfully.
PostgreSQL 16 is not installed. Do you want to install it on this server? (Y/N): **Y**
Please provide a password for the database superuser (postgres).
password : ***
Retype password : ***
Please save your PostgreSQL superuser password securely for future logins.
Installing PostgreSQL, it may take a while, please wait...
PostgreSQL installed successfully.
'C:\Program Files\PostgreSQL\16\bin' is already present in system PATH.
PostgreSQL installed and service 'postgresql-x64-16' is running on port 5432.

```
=====
```

| VCX Prerequisites Installation Complete |

```
=====
```

Note: Turkish clients should set their locale to **English_United States** during PostgreSQL setup.

When the "yes" option for PostgreSQL DB install is selected, the script will install the PostgreSQL Server on the same server including the **vcxdb** database along with the **vcx** schema.

When the "no" option for PostgreSQL DB install is selected the PostgreSQL DB installation and related activities are not managed by the VCX Prerequisite Software Installer. You will need to manage these processes separately.

If no errors are detected during the installation, a message confirming that the installation is complete will be displayed:

```
Installation process completed.
```

Note: Installation details can be reviewed in <folder_name>/windows/Pre_req_install.log located in the script directory.

Chapter 5

VCX Download and Install Procedures for Linux OS

This chapter describes the download and install procedures for Visa Clearing Exchange (VCX) on Linux OS.

Download VCX

Users should download Visa Clearing Exchange (VCX) prior to registering and logging in for the first time.

VCX can be downloaded via Visa Access. Contact your Visa representative if you require assistance finding the VCX download page.

Note: If upgrading to the latest version of VCX, place the new version in the same folder as the previous installation.

Install VCX on Linux

Visa Clearing Exchange (VCX) comes as a software installation package. Once the software is installed, it is available for all users on the machine.

VCX requires certain prerequisites to run on a server. See *Before You Begin with VCX* for system requirements for installation.

Installation Steps

1. Create a folder and place the VCX-Linux-nn-nn-nn.tgz file

```
mkdir <folder_name>
cd <folder_name>
mv VCX-Linux-nn-nn-nn.tgz <folder_name>
```

2. Extract the Software Package

Make sure to extract the entire VCX-Linux-nn-nn-nn directory to a local directory <folder_name>, for example, /opt/app/VCX.

```
cd <folder_name>
tar -xvf VCX-Linux-nn-nn-nn.tgz
```

3. Service Accounts and Port Configuration

- a. **Service Accounts Setup:** During installation, two service accounts are created to ensure the application runs efficiently:
 - **VCXService Account:** Launches the VCX Application on **PORT 8000**.
 - **VCXProxyService Account:** Manages the VCX Proxy Application on **PORT 3000**.
- b. **Port Availability:** Ensure that **PORT 3000** and **PORT 8000** are not in use by other applications.
- c. **Updating Port Settings (Optional)**
If you need to change the default ports, follow these steps before installation:
 - **Locate the Configuration File:** Find the vcxClient.Config.json file in the config folder of the VCX package.
 - **Modify Port Numbers:**
 - For **VCXService**, update the port value, "vcxservice": { "port": <ENTER PORT NUMBER> }
 - For **VCXProxyService**, change the port value, "vcxproxyservice": { "port": <ENTER PORT NUMBER> }
 - **Save the File:** After making your changes, save the file to apply the new port settings.

4. Execute the Script with Root Privileges

- a. Execute the script.

```
cd <folder_name>  
node Installerbundle.js
```

- b. Checking for prerequisite software.

```
Welcome to the Visa Clearing Exchange (VCX) Installation  
process.  
Checking for prerequisite software...  
✓ NodeJS: Installed (Version: 21.6)  
✓ JAVA: Installed (Version: 17.0)  
✓ OpenSSL: Installed (Version: 1.1.1)  
✓ PostgreSQL: Installed (Version: 16) All prerequisite  
software is installed.
```

Note: If PostgreSQL is unavailable on the server, the application displays the following warning message.

```
PostgreSQL: Not Installed or Incorrect Version  
WARNING: If setting up the VCX application and database on the  
same server, ensure PostgreSQL is installed.
```

- c. **Select the installation type:** Upgrade or New Installation.

```
What do you want to do? > New install
```

- d. Select the default path to use for VCX Application.

```
✓ Do you want to use the default base path as VCX base  
path(/opt/app/vcx)? > (Y/n)...  
<CONFIRM>  
You selected: /opt/app/vcx  
Total files to copy: 2537  
Files copied successfully!  
Creating and copying directories and files...  
Files copied successfully.  
Files copied successfully.  
Script executed successfully.  
-----
```

- e. During the installation process, the user will be prompted to generate HTTPS self-signed cert or use their own HTTPS certs.

Note: For self-signed certificate, the Common Name (CN) should match with server FQDN (Fully Qualified Domain Name), VCX will create a cert with FQDN automatically.

```
Fully Qualified Domain Name (FQDN): <HOSTNAME DETAILS>
✓ Do you want to use HTTPS certificate? If not, generate a
self-signed certificate. (y/N)
<CONFIRM>
generate self-signed certificate for me
Generating a self-signed certificate using OpenSSL...
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'key.pem'
-----
Self-signed certificate generated successfully. Please
replace these certificates with your own.
-----
```

- f. Configure PostgreSQL DB

```
Please enter the PostgreSQL database configuration.
? Enter the PostgreSQL host: <ENTER POSTGRES SQL HOST; For
Default Setting Click ENTER>
? Enter the PostgreSQL port: <ENTER POSTGRES SQL PORT; For
Default Setting Click ENTER>
? Enter the PostgreSQL VCX database: <ENTER VCX DB; vcxdb>
? Enter the PostgreSQL VCX schema: <ENTER VCX SCHEMA; vcx>
? Enter the PostgreSQL username: <ENTER POSTGRES SQL USER>
? Enter the PostgreSQL password: <ENTER POSTGRES SQL USER
PASSWORD>
Running VCX Protection using Protect Config
<folder_name> /config/VCXConfig_AutoGen.json-
<folder_name>config/VCX_ProtectedConfig_AutoGen.json
Completed VCX Protection
Database setup is complete, and your configuration has been
securely encrypted and stored in a file named <folder_name>/
database/pgDbConfig.cnf
Group vcxservice created successfully
User vcxservice created successfully
User vcxproxyservice created successfully
VCXProxyService uninstalled. Now installing again...
VCXProxyService installed and started successfully on Linux
```

Note: If the PostgreSQL username or password you enter is incorrect, the application will keep asking you to provide the correct credentials and will not move forward until valid details are entered.

To stop the process, press CTRL + C on your keyboard. This will close the installation. You can then restart the installation and try again with the correct credentials.

- g. If PostgreSQL Server SSL is enabled.

```
Please enter the PostgreSQL database configuration.  
? Select PostgreSQL SSL/TLS mode: (Use arrow keys)
```

- h. If PostgreSQL Server SSL is enabled and the full verify option is selected, enter the absolute path of the `ca.crt`.

```
Please enter the PostgreSQL database configuration.  
? Select PostgreSQL SSL/TLS mode: verify-full (CA cert  
required, Encrypt + validate PostgreSQL server)  
? enter full absolute path yo PostgreSQL server CA  
certificate file: <Example: /tmp/cert/ca.crt>
```

- i. Setup **Primary Admin Email ID**

The user will be prompted to enter the email address of the primary administrator of the application.

```
Running Primary Admin Utility...  
? Please enter your email id <ENTER PRIMARY ADMIN EMAIL ID>
```

Note: The primary administrator must be registered before logging in to VCX.

- j. The user will be prompted to enable/disable multi-factor authentication (MFA).

```
Two-Factor Verification Opt-Out Waiver
? Visa strongly recommends using Two-Step Verification as an
additional layer of security to help prevent unauthorized
access to Visa Clearing Exchange.

By choosing not to enable Two-Step Verification for your Visa
Clearing Exchange user accounts, I certify that I have the
authority to do so on behalf of the entity that licensed this
instance of Visa
Clearing Exchange.

The licensee is responsible for and Visa disclaims any and all
liability with regards to the chosen authentication method.

Would you like to ENABLE Two-Step Verification(requires mail
server)? <CONFIRM>
You entered: <CONFIRMATION>
MFA configuration has been updated successfully
? Would you like to ENABLE e-mail communication(requires mail
server)? <CONFIRM>
You entered: <CONFIRMATION>
E-mail configuration has been updated successfully
```

If MFA is disabled, then the application key is printed on the console.

```
Your Visa Clearing Exchange Application Key is <XXXXXX-XXXXX-
XXXXXX-XXXXXX-XXXXXX-XXXXXX> is in this format. Please keep this
key in a safe place -- you will need it to run the admin
utility on the web server.
IMPORTANT: If you lose this key, there is no way to retrieve
the key from the application. If you reinstall the application
without the application key, you will lose all historical data!
? Please enter APPLICATION KEY? <ENTER APPLICATION KEY; for
example: 2613C-5DGAI-UXYX9-D3488-EFAGI-C02C6>
```

If MFA is enabled, the user will be prompted to enter a valid email address to receive a one-time password. Once the one-time password is validated, the system will send out an application key to that email.

```
? Please enter Verification Code sent to your email? <ENTER
CODE; for example: 032329>
```

Important:

- Email notifications are sent only to Admin Users when MFA is enabled.
- The application key is unique to your installation. Keep the application key in a safe place, as it will be provided to you only once.

Once you enter the application key, the system will validate the application key and create a primary administrator account.

k. Setup Primary Admin Account

```
Running Primary account management utility
? Please select an action from the below Create new primary
account
? Enter First Name: <ENTER PRIMARY ADMIN FIRST NAME>
? Enter Last Name: <ENTER PRIMARY ADMIN LAST NAME>
? Enter primary account login id (Login ID is expected to be
an email ID): <ENTER PRIMARY EMAIL ID>
? Enter password <ENTER PASSWORD>
? Confirm password <RE-ENTER PASSWORD>
Setting up new primary account with the login id <PRIMARY
ADMIN EMAIL ID>
New primary account has been setup successfully
Please login to VCX at: <URL>
```

l. Restart the application

```
systemctl restart VCXService
```

Access the VCX application by navigating to the URL mentioned.

Note:

- Installation of certificates only must occur once. Replace the TLS certificate if they are about to expire. Visa does not recommend using self-signed certificates.
- If you are using external certificates, put the certificates in the / config folder, where the cert is saved as cert.pem and the key is saved as key.pem. Make sure the key is not encrypted, as VCX does not support encrypted keys.

Chapter 6

VCX Download and Install Procedures for Windows OS

This chapter describes the download and install procedures for Visa Clearing Exchange (VCX) on the current version of Windows OS.

Download VCX

Users should download Visa Clearing Exchange (VCX) prior to registering and logging in for the first time.

VCX can be downloaded via Visa Access. Contact your Visa representative if you require assistance finding the VCX download page.

Note: If upgrading to the latest version of VCX, place the new version in the same folder as the previous installation.

Install VCX on Windows

Visa Clearing Exchange (VCX) comes as a software installation package. Once the software is installed, it is available for all users on the machine.

VCX requires certain prerequisites to run on a server. See *Before You Begin with VCX* for system requirements for installation.

Installation Steps

1. Create a folder and place the VCX-Windows-nn-nn-nn.tgz file

```
mkdir <folder_name>
cd <folder_name>
mv VCX-Windows-nn-nn-nn.tgz <folder_name>
```

2. Extract the Software Package

Make sure to extract the entire VCX-Windows-nn-nn-nn directory to a local directory <folder_name>, for example, C:\Program Files\VCX.

```
cd <folder_name>
tar -xvf VCX-Windows-nn-nn-nn.tgz
```

3. Service Accounts and Port Configuration

- a. **Service Accounts Setup:** During installation, two service accounts are created to ensure the application runs efficiently:
 - **VCXService Account:** Launches the VCX Application on **PORT 8000**.
 - **VCXProxyService Account:** Manages the VCX Proxy Application on **PORT 3000**.
- b. **Port Availability:** Ensure that **PORT 3000** and **PORT 8000** are not in use by other applications.
- c. **Updating Port Settings (Optional)**
If you need to change the default ports, follow these steps before installation:
 - **Locate the Configuration File:** Find the `vcxService.Config.json` file in the (package/package/) of the VCX package.
 - **Modify Port Numbers:**
 - For **VCXService**, update the port value, "vcxservice": { "port": <ENTER PORT NUMBER> }
 - For **VCXProxyService**, change the port value, "vcxproxyservice": { "port": <ENTER PORT NUMBER> }
 - **Save the File:** After making your changes, save the file to apply the new port settings.

4. Execute the Script with Admin Privileges

a. Execute the script

```
cd <folder_name>  
node Installerbundle.js
```

b. Checking for prerequisite software.

```
Welcome to the Visa Clearing Exchange (VCX) Installation  
process.  
Checking for prerequisite software...  
✓ NodeJS: Installed (Version: 21.6)  
✓ JAVA: Installed (Version: 17.0)  
✓ OpenSSL: Installed (Version: 1.1.1)  
✓ PostgreSQL: Installed (Version: 16) All prerequisite  
software is installed.
```

Note: If PostgreSQL is unavailable on the server, the application displays the following warning message.

```
PostgreSQL: Not Installed or Incorrect Version  
WARNING: If setting up the VCX application and database on the  
same server, ensure PostgreSQL is installed.
```

c. Select the installation type: Upgrade or New Installation.

```
What do you want to do? > New install
```

d. Select the default path to use for VCX Application.

```
✓ Do you want to use the default base path as VCX base  
path(C:/vcx)? > (Y/n)...  
<CONFIRM>  
You selected: C:/vcx  
Total files to copy: 6830  
Files copied successfully
```

- e. During the installation process, the user will be prompted to generate HTTPS self-signed cert or use their own HTTPS certs.

Note: For self-signed certificate, the Common Name (CN) should match with server FQDN (Fully Qualified Domain Name), VCX will create a cert with FQDN automatically.

```
Fully Qualified Domain Name (FQDN): <HOSTNAME DETAILS>
✓ Do you want to use HTTPS certificate? If not, generate a
self-signed certificate. (y/N)
<CONFIRM>
generate self-signed certificate for me
Generating a self-signed certificate using OpenSSL...
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'key.pem'
-----
Self-signed vertificate generated successfully. Please
replace these vertificates with your own.
```

- f. Configure PostgreSQL DB

```
Please enter the PostgreSQL database configuration.
? Enter the PostgreSQL host: <ENTER POSTGRESQL HOST; For
Default Setting Click ENTER>
? Enter the PostgreSQL port: <ENTER POSTGRESQL PORT; For
Default Setting Click ENTER>
? Enter the PostgreSQL VCX database: <ENTER VCX DB; vcxdb>
? Enter the PostgreSQL VCX schema: <ENTER SCHEMA; vcx
? Enter the PostgreSQL username: <ENTER POSTGRESQL USER>
? Enter the PostgreSQL password: [hidden] <ENTER POSTGRESQL
USER PASSWORD>
Running VCX Protection using Protect Config
<folder_name> /config/VCXConfig_AutoGen.json-
<folder_name>config/VCX_ProtectedConfig_AutoGen.json
Completed VCX Protection
Database setup is complete, and your configuration has been
securely encrypted and stored in a file named <folder_name>/
database/pgDbConfig.cnf
VCXProxyService started successfully.
VCXService started successfully.
```

Note: If the PostgreSQL username or password you enter is incorrect, the application will keep asking you to provide the correct credentials and will not move forward until valid details are entered.

To stop the process, press CTRL + C on your keyboard. This will close the installation. You can then restart the installation and try again with the correct credentials.

- g. If PostgreSQL Server SSL is enabled.

```
Please enter the PostgreSQL database configuration.
? Select PostgreSQL SSL/TLS mode: (Use arrow keys)
```

- h. If PostgreSQL Server SSL is enabled and the full verify option is selected, enter the absolute path of the `ca.crt`.

```
Please enter the PostgreSQL database configuration.
? Select PostgreSQL SSL/TLS mode: verify-full (CA cert
required, Encrypt + validate PostgreSQL server)
? enter full absolute path to PostgreSQL server CA
certificate file: <Example: /tmp/cert/ca.crt>
```

- i. Setup **Primary Admin Email ID**

The user will be prompted to enter the email address of the primary administrator of the application.

```
Running Primary Admin Utility...
? Please enter your email id <ENTER PRIMARY ADMIN EMAIL ID>
```

Note: The primary administrator must be registered before logging in to VCX.

- j. The user will be prompted to enable/disable multi-factor authentication (MFA).

```
Two-Factor Verification Opt-Out Waiver
? Visa strongly recommends using Two-Step Verification as an
additional layer of security to help prevent unauthorized
access to Visa Clearing Exchange.

By choosing not to enable Two-Step Verification for your Visa
Clearing Exchange user accounts, I certify that I have the
authority to do so on behalf of the entity that licensed this
instance of Visa
Clearing Exchange.

The licensee is responsible for and Visa disclaims any and all
liability with regards to the chosen authentication method.

Would you like to ENABLE Two-Step Verification(requires mail
server)? Currently, you can enable this option after VCX is
installed successfully from the UI (browser).
You entered: <CONFIRMATION>
MFA configuration has been updated successfully
? Would you like to ENABLE e-mail communication(requires mail
server)? <CONFIRM>
You entered: <CONFIRMATION>
E-mail configuration has been updated successfully
```

Known Issue: If you are installing VCX for the first time on Windows and have Multi-Factor Authentication (MFA) enabled, the installation might fail. The **workaround** for this issue is: When asked, select "No" for both "Would you like to ENABLE Two-Step Verification (requires mail server)?" and "Would you like to ENABLE e-mail communication (requires mail server)?".

Note: You can enable these options after the installation from the application settings page in the browser-based UI.

If MFA is disabled, then application key is printed on the console.

```
Your Visa Clearing Exchange Application Key is <XXXXXX-XXXXX-XXXXXX-XXXXXX-XXXXXX> is in this format. Please keep this key in a safe place -- you will need it to run the admin utility on the web server.
```

IMPORTANT: If you lose this key, there is no way to retrieve the key from the application. If you reinstall the application without the application key, you will lose all historical data!

? Please enter APPLICATION KEY? **<ENTER APPLICATION KEY; for example: 2613C-5DGAI-UXYX9-D3488-EFAGI-C02C6>**

If MFA is enabled, the user will be prompted to enter a valid email address to receive a one-time password. Once the one-time password is validated, the system will send out an application key to that email.

```
? Please enter Verification Code sent to your email? <ENTER CODE; for example: 032329>
```

Important:

- Email notifications are sent only to Admin Users when MFA is enabled.
- The application key is unique to your installation. Keep the application key in a safe place, as it will be provided to you only once.

Once you enter the application key, the system will validate the application key and create a primary administrator account.

k. Setup Primary Admin Account

```
Running Primary account management utility
? Please select an action from the below Create new primary account
? Enter First Name: <ENTER PRIMARY ADMIN FIRST NAME>
? Enter Last Name: <ENTER PRIMARY ADMIN LAST NAME>
? Enter primary account login id (Login ID is expected to be an email ID): <ENTER PRIMARY EMAIL ID>
? Enter password <ENTER PASSWORD>
? Confirm password <RE-ENTER PASSWORD>
Setting up new primary account with the login id <PRIMARY ADMIN EMAIL ID>
New primary account has been setup successfully
Please login to VCX at: <URL>
```

- I. Restart the application by entering the following PowerShell command

```
Restart-Service vcxservice.exe
```

Access the VCX application by navigating to the URL mentioned.

Note:

- Installation of certificates only must occur once. Replace the TLS certificate if they are about to expire. Visa does not recommend using self-signed certificates.
- If you are using external certificates, put the certificates in the / `config` folder, where the cert is saved as `cert.pem` and the key is saved as `key.pem`. Make sure the key is not encrypted, as VCX does not support encrypted keys.

Chapter 7

VCX Uninstall Procedures for Linux and Windows

This chapter describes the uninstallation procedures for Visa Clearing Exchange (VCX) on the Linux and Windows operating systems.

Note: Users should back up their report data before proceeding with the uninstallation. Uninstalling VCX will delete all previous reports, and the data of the files already processed.

VCX Software Uninstall on Linux

VCX Software Uninstall

1. Check the path where VCX is installed, for example: /opt/app/vcx in Linux. **Ensure you have root access.**
2. Stop the VCX Service:
 - a. If running the application as a daemon service (servicename: VCXService), use the command:

```
systemctl stop <VCXService>
```

3. Stop the VCX Proxy Service:

- a. If running the application as a daemon service (servicename: VCXProxyService), use the command:

```
systemctl stop <VCXProxyService>
```

4. Remove/Delete the VCX folders, using the command:

- a. `rm -rf <VCXFolderPath>`

5. Remove VCX daemon service files: **Ensure you have root access.**

- a. Navigate to the system folder under /etc/systemd, use the command:

```
cd /etc/systemd/system
```

- b. Look for the service files with the names:

- VCXService.service
- VCXProxyService.service

- c. Remove the service files, using the commands:

```
rm VCXService.service
```

```
rm VCXProvider.service
```

VCX Software Uninstall on Windows

VCX Software Uninstall

1. Check the path where VCX is installed, for example: C:/opt/app/vcx
2. Stop the VCX Service (servicename: VCXService), using the PowerShell command:

- a. `Stop-Service -name <VCXService>`

3. Stop the VCX Proxy Service (servicename: VCXProxyService), using the PowerShell command:

- a. `Stop-Service -name <VCXProxyService>`

4. Remove/Delete the VCX folders, using the PowerShell command:

- a. `rm -r <VCXFolderPath>`

Chapter 8

VCX Upgrade Procedures for Linux and Windows

This chapter describes the upgrade procedures for Visa Clearing Exchange (VCX) on the Linux and Windows operating systems.

Upgrade VCX on Linux

Visa Clearing Exchange (VCX) comes as a software installation package. Once the software upgrade is installed, it is available for all users on the machine.

VCX requires certain prerequisites to run on a server. See *Before You Begin with VCX* for system requirements for installation.

Upgrade Steps

1. Create a folder and place the VCX-Linux-nn-nn-nn.tgz file

```
mkdir <folder_name>  
cd <folder_name>  
mv VCX-Linux-nn-nn-nn.tgz <folder_name>
```

2. Extract the Software Package

Make sure to extract the entire VCX-Linux-nn-nn-nn directory to a local directory <folder_name>, for example, /opt/app/VCX.

```
cd <folder_name>  
tar -xvf VCX-Linux-nn-nn-nn.tgz
```

3. Execute the Script with Root Privileges

a. Execute the script.

```
cd <folder_name>  
node Installerbundle.js
```

b. Checking for prerequisite software.

```
Welcome to the Visa Clearing Exchange (VCX) Installation  
process.  
Checking for prerequisite software...  
✓ NodeJS: Installed (Version: 21.6)  
✓ JAVA: Installed (Version: 17.0)  
✓ OpenSSL: Installed (Version: 1.1.1)  
✓ PostgreSQL: Installed (Version: 16) All prerequisite  
software is installed.
```

Note: If PostgreSQL is unavailable on the server, the application displays the following warning message.

```
PostgreSQL: Not Installed or Incorrect Version  
WARNING: If setting up the VCX application and database on  
the same server, ensure PostgreSQL is installed.
```

c. Select the installation type: Upgrade or New Installation.

```
What do you want to do? > Upgrade
```

- d. Select the path to the location where the VCX application was previously installed.

```
✓ Do you want to use the default base path as VCX base
path (/opt/app/vcx)? > (Y/n)... <CONFIRM>
You selected: /opt/app/vcx
Total files to copy: 2537
Files copied successfully!
Creating and copying directories and files...
Files copied successfully.
Files copied successfully.
Script executed successfully.
```

- e. Select the service names to use to execute the VCX application

```
✓ Enter the existing VCX service user name: ... vcxservice
✓ Enter the existing VCX proxy service user name: ...
vcxproxyservice
Upgrade complete. starting services....
```

Upgrade VCX on Windows

Visa Clearing Exchange (VCX) comes as a software installation package. Once the software upgrade is installed, it is available for all users on the machine.

VCX requires certain prerequisites to run on a server. See *Before You Begin with VCX* for system requirements for installation.

Upgrade Steps

1. Create a folder and place the VCX-Windows-nn-nn-nn.tgz file

```
mkdir <folder_name>
cd <folder_name>
mv VCX-Windows-nn-nn-nn.tgz <folder_name>
```

2. Extract the Software Package

Make sure to extract the entire VCX-Windows-nn-nn-nn directory to a local directory <folder_name>, for example, C:\Program Files\VCX.

```
cd <folder_name>
tar -xvf VCX-Windows-nn-nn-nn.tgz
```

3. Execute the Script with Admin Privileges

a. Execute the script

```
cd <folder_name>
node Installerbundle.js
```

b. Checking for prerequisite software.

```
Welcome to the Visa Clearing Exchange (VCX) Installation
process.
Checking for prerequisite software...
✓ NodeJS: Installed (Version: 21.6)
✓ JAVA: Installed (Version: 17.0)
✓ OpenSSL: Installed (Version: 1.1.1)
✓ PostgreSQL: Installed (Version: 16) All prerequisite
software is installed.
```

Note: If PostgreSQL is unavailable on the server, the application displays the following warning message.

```
PostgreSQL: Not Installed or Incorrect Version
WARNING: If setting up the VCX application and database on
the same server, ensure PostgreSQL is installed.
```


- c. Select the installation type: Upgrade or New Installation.

```
What do you want to do? > Upgrade
```

- d. Select the path to the location where the VCX application was previously installed.

```
✓ Do you want to use the default base path as VCX base  
path(C:/vcx)? > (Y/n) ...  
<CONFIRM>  
You selected: C:/vcx  
Total files to copy: 6830  
Files copied successfully
```

- e. Select the service names to use to execute the VCX application

```
✓ Enter the existing VCX service user name: ... vcxservice  
✓ Enter the existing VCX proxy service user name: ...  
vcxproxyservice  
Upgrade complete. starting services....
```

Chapter 9

Getting Started with VCX

This chapter describes registration and login procedures as well as the Dashboard for Visa Clearing Exchange (VCX).

Accessing the VCX Application

1. If you receive the "Your connection isn't private" warning, follow these steps to access the VCX application.
2. Click on the **Advanced** button.
3. Click on the continue to <HOSTNAME> link.

Removing Browser Warnings for a Self-Signed Certificate

To eliminate browser warnings caused by the self-signed certificate from the VCX server, follow these steps using the Command Prompt with administrative permissions:

1. Obtain the Server Certificate
 - a. Request the server certificate from your administrator.
 - b. The certificate (cert.pem) is in the `..\vcx\config` folder on the server.
2. Copy the Certificate to Your Desktop
 - a. Once you have the certificate file, save it to your Desktop.
3. Open the Command Prompt as an Administrator
 - a. Click on the **Start** menu.
 - b. Type `cmd`.
 - c. Right-click on **CommandPrompt** and select "**Run as administrator.**"
 - d. If prompted by User Account Control, click **Yes** to allow.
4. Navigate to the Certificate Directory
 - a. In the Command Prompt window, change the directory to where the certificate is stored:

```
cd Desktop
```


(Replace "Desktop" with the correct path if different.)
5. Ensure the Certificate is in `.cert` format.
 - a. Verify the certificate file has a `.cert` extension.
 - b. If not, rename it using this command:

```
renamefilename.pem certname.crt
```


(Replace `filename.ext` with the current name of your certificate, and `certname.crt` with the desired name.)

6. Install the Certificate into the Root Store

- a. Run this command to add the certificate to the Trusted Root Certification Authorities store:

```
certutil-addstore Root certname.crt
```

(Replace **certname.crt** with your certificate's actual name.)

7. Restart Your Browser

- a. Close all browser windows.
b. Reopen your browser to apply the changes.

The browser should no longer display warnings when accessing the VCX Application.

Register for VCX with MFA

You must first have set up a primary administrator account in Visa Clearing Exchange (VCX) to register users, which should have been completed after installing VCX on the server. All other users must register in VCX before logging in for the first time.

1. From the login page, click **Register**.
2. Enter your name, email, and password in the required fields.

Note: Passwords must be at least twelve characters long, contain at least one number, one special character, an uppercase letter, a lowercase letter, and cannot contain the account name.

3. A registration link will be sent out to the email address
4. Registration link should redirect to registration page
5. Click **Register**.

Register for VCX

A green bar will appear, stating that registration was successful. All users when registering this way in VCX begin as operators. The primary administrator, as well as any other administrator, can promote operators to administrators where required.

Register for VCX without MFA

You must first have set up a primary administrator account in Visa Clearing Exchange (VCX) to register users, which should have been completed after installing VCX on the server. All other users must register in VCX before logging in for the first time.

1. From the login page, click **Register**.
2. Enter your name, email, and password in the required fields.

Note: Passwords must be at least twelve characters long, contain at least one number, one special character, an uppercase letter, a lowercase letter, and cannot contain the account name.

3. Enter your password again in the **Confirm Password** field.
4. Click **Register**.

Register for VCX

The screenshot displays the 'STEP 1 OF 2' registration page for Visa Clearing Exchange. The page has a blue header with the text 'Visa Clearing Exchange'. Below the header, the main content area is light blue. On the right side, there is a white registration form. At the top of the form, it says 'STEP 1 OF 2' and features a logo of a person with a checkmark. Below the logo, the text 'Let's finish setting up your account' is displayed. The form contains five input fields: 'First name*', 'Last name*', 'Email*', 'Create password*', and 'Confirm password*'. Each field has a small blue icon to its right. Below the 'Create password*' and 'Confirm password*' fields, there is a blue button labeled 'NEXT' and a link labeled 'LOG IN'.

A green bar will appear, stating that registration was successful. All users when registering this way in VCX begin as operators. The primary administrator, as well as any other administrator, can promote operators to administrators where required.

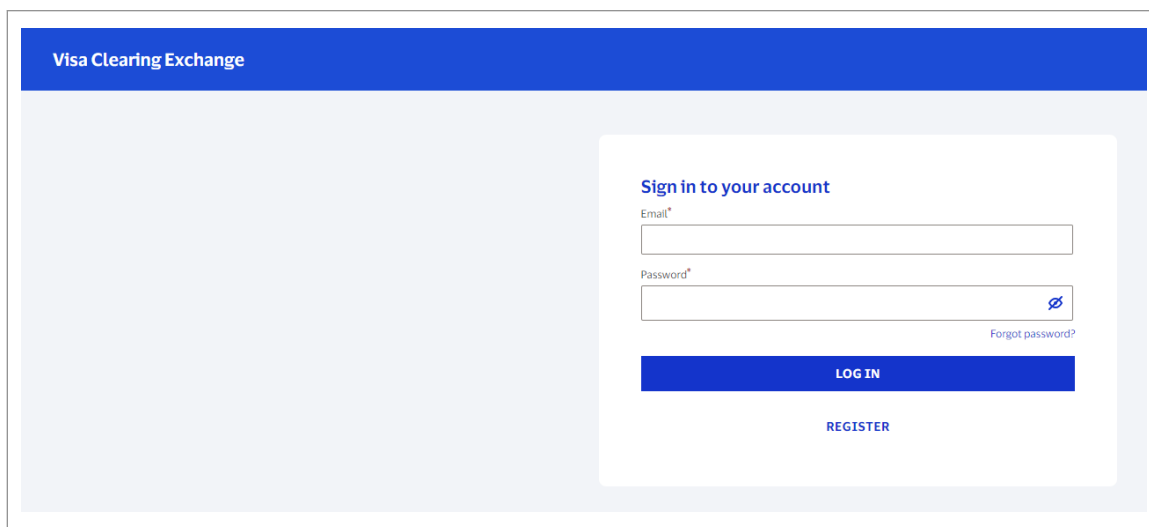
Log In to VCX

This procedure describes how to log in to Visa Clearing Exchange (VCX).

Users must have registered using an approved email and password and have been approved by an administrator before logging in to Visa Clearing Exchange for the first time.

1. Enter your email.
2. Enter your password.
3. Click Log In.

Login

The screenshot shows the login interface for the Visa Clearing Exchange. At the top, there is a blue header bar with the text "Visa Clearing Exchange". Below this, the main content area has a light blue background. On the right side, there is a white box titled "Sign in to your account". Inside this box, there are two input fields: "Email*" and "Password*". The "Password*" field has a small icon of a key and a "Forgot password?" link to its right. Below the input fields is a blue button labeled "LOG IN". At the bottom of the white box, there is a link labeled "REGISTER".

Note: If you are an Operator, you will be taken to the VCX home page. If you are an Administrator, you will be asked for the one-time verification code sent to your email. Enter your verification code and click **Submit**. You will then be taken to the VCX home page.

If you forgot your password:

4. Click **Forgot Password?** under the password field.
5. Enter the email that you used to register for VCX.
6. Click **Next**.
7. Enter the OTP sent to your email.
8. Click **Confirm OTP**.
9. Enter your new password in the box, ensuring that all password criteria are met.
10. Confirm your password by repeating it in the box.

11. Click Update Password.

Note: If your password expires, you will follow the same procedure as the Forgot Password procedure. If you do not log in after 90 days, your account, whether operator or administrator, will be disabled.

If your Operator account is disabled:

12. Contact your administrator to reactivate the operator account.

If your Administrator account is disabled:

13. If there is another administrator, they can reactivate the disabled administrator's account.

14. If there is no other administrator assigned, or if the primary administrator left the application team without passing along the application key, a new administrator will have no way to log in and must run the setup program for VCX.

15. The new administrator will be warned about proceeding with the creation of a new database, but with no application key. You must agree.

16. From there, follow the primary administrator setup procedure.

Choose Release Date

When you log in after you have downloaded and installed the latest version of the VCX software and it is before the effective date, the New Enhancements screen is displayed.

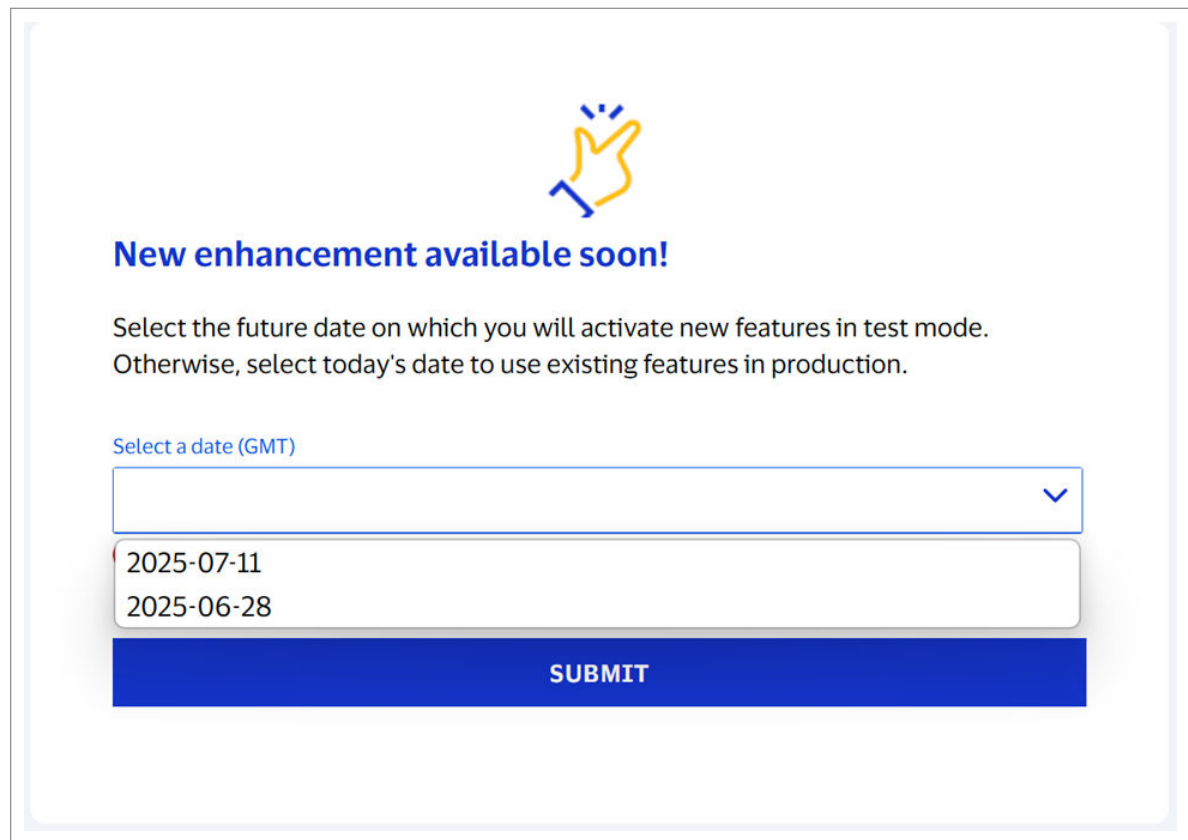
Choose a date from the drop down to either:

a. Select today's date to use the existing features in production.

or

b. Select the future date on which the new features will be activated in test mode.

New Enhancements



The screenshot shows a user interface for selecting a release date. At the top, there is a yellow hand icon pointing to the text "New enhancement available soon!". Below this, a message instructs the user to select a future date for test mode or today's date for production. A dropdown menu is open, showing two date options: "2025-07-11" and "2025-06-28". A blue "SUBMIT" button is at the bottom.

New enhancement available soon!

Select the future date on which you will activate new features in test mode.
Otherwise, select today's date to use existing features in production.

Select a date (GMT)

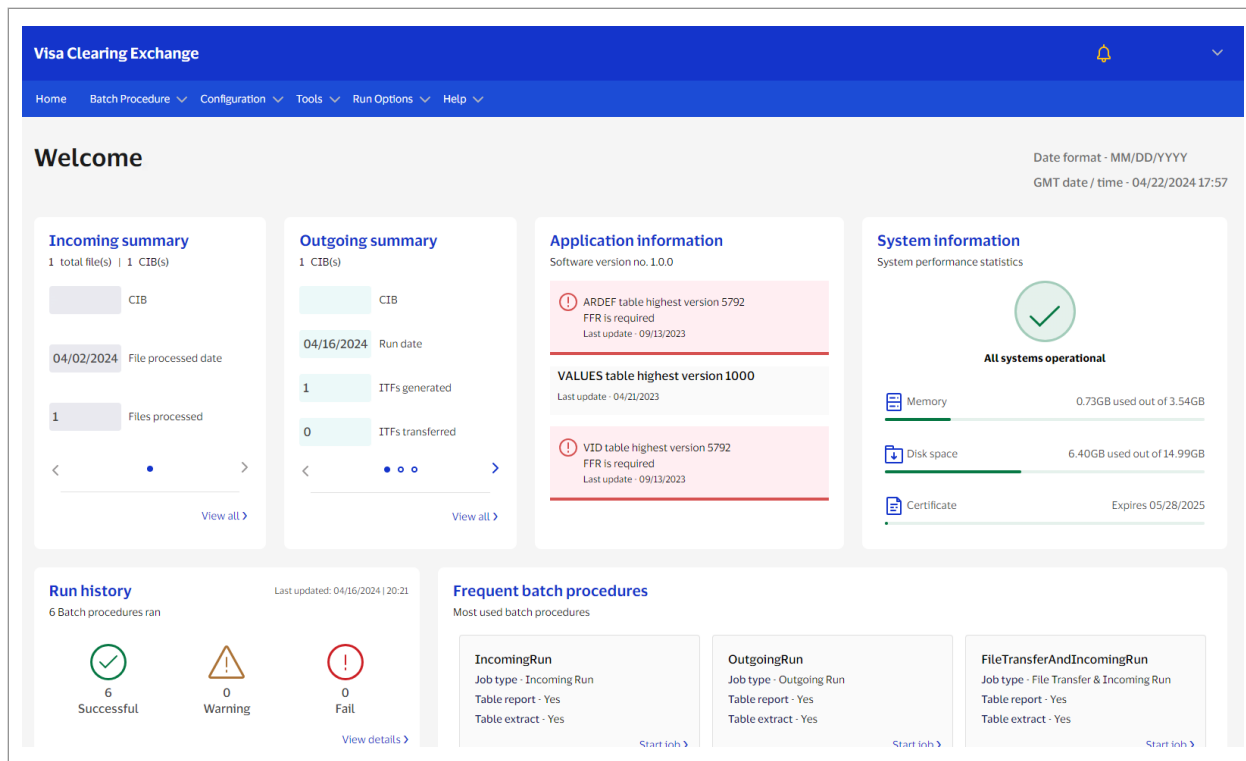
2025-07-11
2025-06-28

SUBMIT

VCX Dashboard

The dashboard is the home page for Visa Clearing Exchange (VCX). The dashboard is divided into different tiles that each display key information.

VCX Dashboard



Incoming Summary

This tile gives a list of the BASE II CIB, run date, and how many files were processed. This information is provided by the ITF, and the tile shows up to five different days.

Outgoing Summary

This tile gives the list of the BASE II CIB, run date, ITFs generated, and ITFs transferred for previously run Outgoing batch jobs.

Application Information

This tile gives the most recent versions of the Values, ARDEF, VID, and software found in the database. If one of the pieces of information is out of date, it will be highlighted in red with an error icon, indicating that information must be updated.

System Information

This tile shows the Memory and Disk Space being used, as well as the expiration date of the TLS certificate.

Run History

This tile shows the last six runs executed in VCX. The tile gives the batch name, end date, return code, and then allows you to view the reports or batch log for that run.

Frequent Batch Procedure

This tile shows the three most frequently used batch procedures in the last 30 days. You can then run one of the frequently used batches from this tile by clicking **Start Job** without needing to go through the process via the **Batch Procedure** page.

Software Version Information

Software version information can be found in the Help drop-down under About.

Notifications

Critical Software Update information can be found by clicking the notifications bell icon.

Note: New notifications are indicated by a number appearing within a circle above and to the right of the notifications bell icon.

Chapter 10

Configuration

The **Configuration** tab of Visa Clearing Exchange (VCX) is available to administrators and allows them to configure various parts of VCX. This includes setting up users, CIBs, secure EAS, secure DEX, secure VFG as well as configuring the scheduler, incoming ITF disposition, outgoing CTF disposition, and view and change application settings.

It is important to configure your CIB, profile, and file interfaces before attempting to create a batch procedure or run a batch job in VCX.

Replacing Expired Certificate and Key Files

When certificate and key files expire they must be replaced. To replace existing certificate and key files, use the following procedure.

1. Find the certificate (cert.pem) and private key (key.pem) files in the config folder under the VCX application, for example, `/opt/app/vcx/config`.
2. Backup the old files (optional, but recommended).

3. Replace them with your new certificate and key: Ensure the new files use the same filenames (cert.pem, key.pem).
4. Restart the VCX Services (Proxy & Application) to use the new certificate and key.
 - For Linux, refer to [Restart the application](#).
 - For Windows, refer to [Restart the application by entering the following PowerShell command](#).

Visa Clearing Exchange User Setup

The Visa Clearing Exchange (VCX) User Setup page is found by navigating to **Configuration > User Setup**. This function is only available to Administrators.

The **User Setup** screen lists the VCX users and their names, email, role, and status. You can edit an existing user by clicking the **Edit** button, or you can delete a user by clicking **Delete** in their specified row. You cannot delete your own user information.

VCX User Setup

Last name	First name	Email	Role	Status	Actions
User	Any	anyuser@mail.com	Administrator	Active	Edit Delete

Visa Clearing Exchange BASE II CIB Setup

The Visa Clearing Exchange (VCX) CIB Setup screen is found by navigating to **Configuration > CIB Setup**.

The **CIB Setup** screen lists the current BASE II CIBs used for the respective user in Visa Clearing Exchange. Additionally, it lists the security code, center name, and mode for each respective BASE II CIB. Users can select BASE II CIBs and edit or delete them. Users may also set up a new BASE II CIB by clicking the **NEW BASE II CIB** button.

BASE II CIB Setup Page

The screenshot shows the 'BASE II CIB setup' page within the 'Visa Clearing Exchange' application. The page has a blue header with the application name and a navigation menu with links for Home, Batch Procedure, Configuration, Tools, Run Options, and Help. The main content area is titled 'BASE II CIB setup' and includes a description: 'Base II CIB is a six-digit system number used by Visa to identify the processing centers and clients. Create a new CIB or use the table below to manage existing ones.' Below the text is a toolbar with five buttons: 'NEW BASE II CIB' (highlighted in blue), 'EDIT', 'DELETE', 'LIST RELATED BATCH PROCEDURES', and 'RECOVER CIB(S)'. At the bottom, there is a table with one row. The table has four columns: 'BASE II CIB' (with a checkbox), 'Security code', 'Center name', and 'Mode'. A 'RECOVER CIB(S)' button is located at the bottom right of the table.

BASE II CIB	Security code	Center name	Mode
☐			

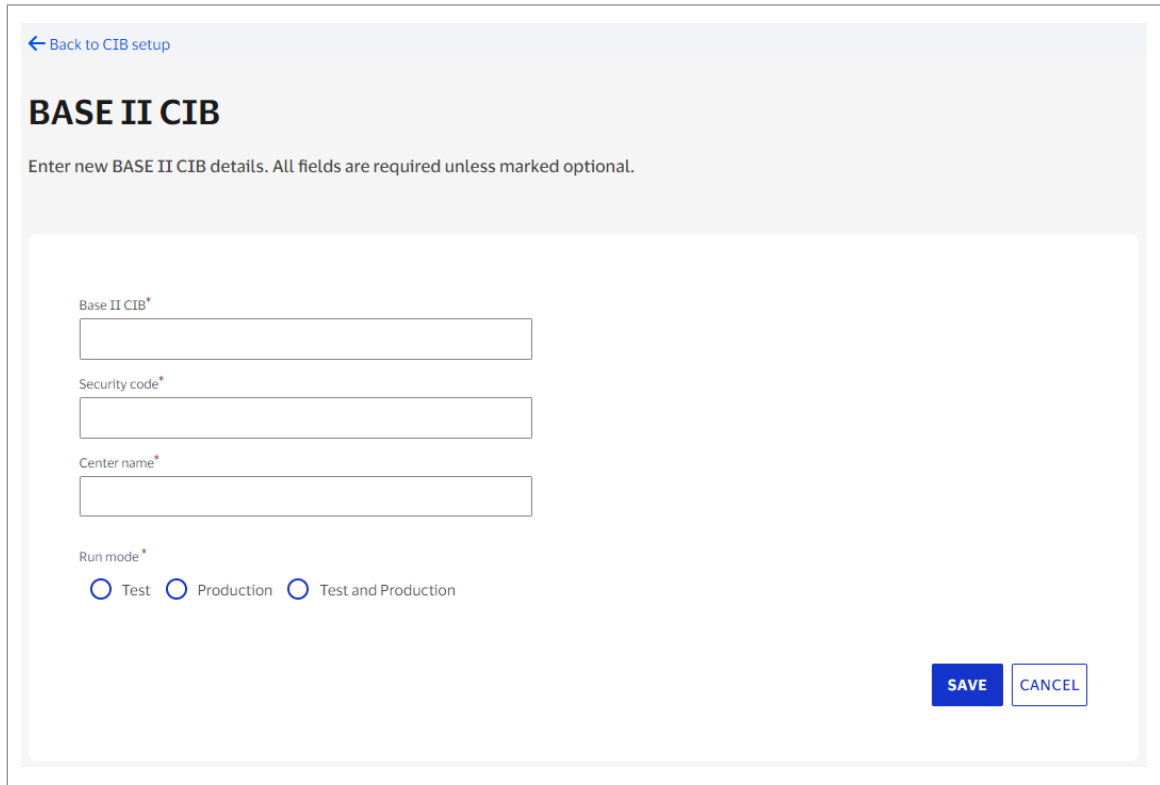
Add a New BASE II CIB

To add a new BASE II CIB, click **NEW BASE II CIB** from the **CIB Setup** page.

1. Enter the new BASE II CIB.
2. Enter the security code.
3. Enter the center name.

4. Select the mode. You can select **Test**, **Production**, or **Test and Production**.
5. Click **SAVE**.

New BASE II CIB



The screenshot shows a web form titled "BASE II CIB" with a back arrow and "Back to CIB setup" link. Below the title is a note: "Enter new BASE II CIB details. All fields are required unless marked optional." The form contains three text input fields: "Base II CIB*", "Security code*", and "Center name*", each with a red asterisk indicating it is required. Below these is a "Run mode*" section with three radio button options: "Test", "Production", and "Test and Production", each with a red asterisk. At the bottom right are two buttons: a blue "SAVE" button and a white "CANCEL" button with a blue border.

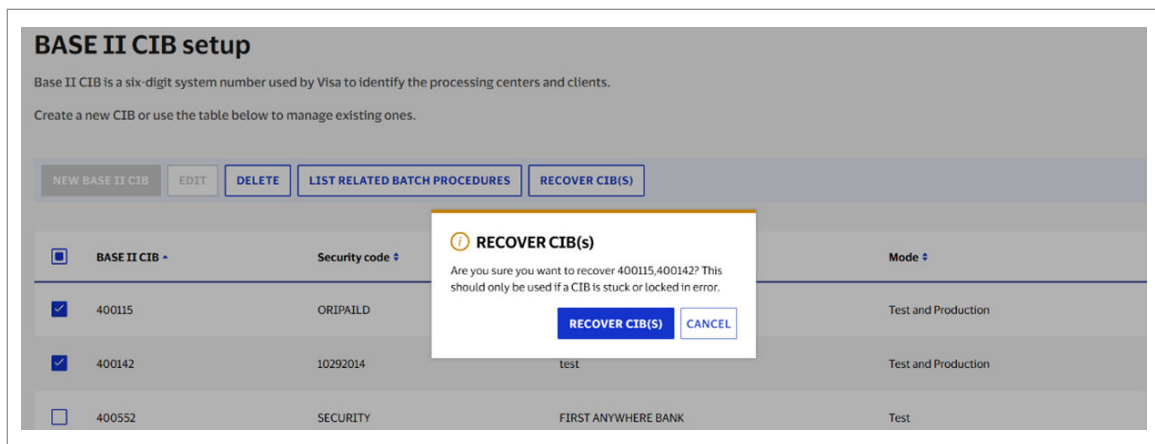
You can also click **CANCEL** to cancel the addition of the new BASE II CIB.

Recover a BASE II CIB

To add recover a BASE II CIB from the **CIB Setup** page.

1. Select the CIB that needs to be recovered.
2. Click **Recover CIB(s)**.
The Recover CIB(s) pop-up window appears.
3. Click **Recover CIB(s)** on the Recover CIB(s) pop-up window.

Recover BASE II CIB



A green bar appears with a message stating that the CIB(s) are recovered successfully.

You can also click **CANCEL** to cancel the recover of the BASE II CIB.

When to Create a New File Transfer Interface

If you are creating a brand-new connectivity path between Visa and a server with Visa Clearing Exchange (VCX) located at your processing center, you must create a new file transfer interface, an Extended Access Server (EAS), Direct Exchange Open File Delivery (DEX OFD), or Visa File Gateway (VFG). If you have an existing connectivity path between Visa and a server with installed VCX, you can use an existing file-transfer interface. See the latest *Visa Direct Exchange Open File Delivery Service Implementation Guide*, *EAS Endpoint Guide*, and *Visa File Gateway Client Implementation Guide* for detailed setup instructions for DEX OFD, EAS, and VFG, respectively. For the process of creating File Transfer Protocol certificates, please refer to the latest *Visa Secure Member Connection Implementation Guide*.

Secure EAS Setup

You can navigate to the Extended Access Server (EAS) File Transfer Interface page by going to **Configuration > Secure EAS Setup** in Visa Clearing Exchange (VCX). From the EAS File Transfer Interface page, you can select a current file transfer interface to edit, delete, or create a new EAS file-transfer interface. This function is available to administrators only.

Create a New EAS File Transfer Interface

To create a new Extended Access Server (EAS) file transfer interface, click **New EAS File Transfer Interface** on the EAS File Transfer Interface page.

1. Enter the file transfer name in the **Interface Name** field.
2. Select whether the interface will be compressed.
3. The FTP Execution path is automatically populated. If you require a different path, click **Browse** to navigate to the proper path.
4. Enter your username in the **Username/Login** field.
5. Enter your password in the **Login Password** field.
6. Enter the IP Address/Hostname and Port in the **IP Address/Hostname** and **Port** fields.

Note: Use Extended Passive Mode (EPSV) by default. If you experience any connection issues, try using Natted Address Translation (NAT).

7. Enter the file path for the server certificate. This field is automatically populated. If your server certificate resides in a different location, click **Browse** to navigate to the proper location.
8. The file path for the client certificate is automatically populated. If your client certificate resides in a different location, click **Browse** to navigate to the proper location.
9. Enter the file path for the client RSA certificate. If your client certificate resides in a different location, click **Browse** to navigate to the proper location.

10. Once all the required fields are complete, you can verify the interface by clicking **Verify Interface**.
11. Once the interface is verified, you can click **Save** to save your new EAS file transfer interface.

New EAS File Transfer Interface

[← Back to EAS File Transfer Interface](#)

Secure EAS setup

To set up a File Transfer Protocol Secure (FTPS) connection through the Extended Access Server (EAS), enter the name of your new interface and fill out the rest of this form. All fields are required unless marked as optional.

Interface information

Fill out the fields below to identify your new interface.

Interface name*

FTPS tool*

☒ VCXFTPSClient Tool(Visa FTPS Client Tool)

FTP execution path*

BROWSE

Username/Login*

Login password*

IP address/hostname*

Port*

☐ Natted IP ☐ Enable EPSV

Server certificate*

BROWSE

Client certificate*

BROWSE

VERIFY INTERFACE

SAVE

The new EAS file transfer interface shows in the list of current EAS file transfer interfaces available.

Secure DEX Setup

You can navigate to the Direct Exchange (DEX) Interface page by going to **Configuration > Secure DEX Setup**. From this page, you can view, edit, or delete the current DEX interfaces, as well as set up a new DEX interface. This function is available to administrators only.

Create a New DEX Interface

To create a new Direct Exchange (DEX) interface, click **New Direct Exchange Interface** on the DEX Interface page.

1. Enter the interface name in the **Direct Exchange Interface Name** field.
2. Select whether the interface will be compressed.
3. Select the BASE II CIB from the drop-down list.

4. Select the Incoming File Directory from your files.

Incoming Only DEX Interface

[← Back to Direct Exchange Interfaces](#)

New direct exchange interface

Enter new DEX details. All fields are required unless marked optional.

Interface details

Direct exchange interface name*

☐ Compression

CIB*

Incoming file directory*

☐ **Outgoing file transfer**

Outgoing file transfers are optional and should only be selected if you're using DEX to transfer files. Choose the FTPS method for your outgoing file transfers. Then, enter or find the server information and certificates needed to process these transfers.

Note: If you use the DEX file transfer interface for only Incoming File Transfer, you do not need to populate the subsequent fields, just save and close the interface.

5. Select Outgoing File Transfer if you're using DEX to transfer files.
6. Enter or find the server information and certificates needed to process the transfers. Additionally, enter the Production and/or Test environment information that the interface should use.

The new DEX interface appears in the list of current DEX interfaces available.

Secure VFG Setup

You can navigate to the Visa File Gateway (VFG) interface page by going to **Configuration > Secure VFG Setup**. From this page, you can view, edit, or delete current VFG interfaces, as well as set up a new VFG interface. This function is available to administrators only.

Note: When VCX sends a compressed outbound file to VFG, the file transfer will fail if VFG is not configured to accept the gzip or zip compression formats.

Create a New VFG Interface

To create a new Visa File Gateway (VFG) interface in Visa Clearing Exchange (VCX), click **New Visa File Gateway Interface** on the VFG Interface page.

1. Enter the interface name in the Visa File Gateway Interface Name field.
2. Select the file compression method.
3. Select the BASE II CIB from the drop-down list.

4. Select the Incoming File Directory from your files. If you use the VFG file transfer interface for only Incoming File Transfer, you do not need to populate the subsequent fields, just save and close the interface.

VFG Interface

The screenshot shows a web application interface for configuring a Visa file gateway. At the top is a blue navigation bar with links: Home, Batch Procedure, Configuration, Tools, Run Options, and Help. Below the navigation bar is a breadcrumb link: < Back to Visa File Gateway Interfaces. The main heading is "Visa file gateway interface". Below the heading is a note: "Enter new VFG details. All fields are required unless marked optional." The form is divided into two sections. The first section, "Interface details", contains four fields: "Visa file gateway interface name" (text input), "Compression Type" (dropdown menu with "Uncompressed" selected), "CIB" (dropdown menu), and "Incoming file directory" (text input with a "BROWSE" button next to it). The second section, "Outgoing file transfer", is preceded by an unchecked checkbox. It contains three fields: "FTPS tool" (radio button selection with "VCXFTPSClient Tool(Visa FTPS Client Tool)" selected), "FTP file directory" (text input with "/opt/app/vcx/ftps" and a "BROWSE" button), and "IP address/hostname" (text input). Below the "IP address/hostname" field is a "Port" label and a text input field. At the bottom right of the form are two buttons: "SAVE" and "CANCEL".

5. Select Outgoing File Transfer if you're using VFG to transfer files.
6. Enter or find the server information and certificates needed to process the transfers.

7. Select Enable PGP Encryption if you are using VFG for encryption during file transfer.

Note: Ensure that you have generated your PGP keypair, passphrase, and shared your public key with your VFG endpoint, before you move on to the next step.

You can generate your PGP information using commonly available resources from the Internet.

8. Provide your Client Public Key.
9. Provide your Client Private Key.
10. Provide your passphrase.
11. You may optionally click Verify Credentials.
12. Enter the login details for the production environment the interface should use.
13. Enter the login details for the test environment the interface should use.
14. Click Save.

The new VFG interface in the list of current VFG interfaces available.

Using the Scheduler

To access the scheduler, navigate to **Configuration > Scheduler**. The scheduler allows you to run or schedule updates to the settings database. The scheduler is available to administrators only.

VCX Scheduler

The screenshot shows the 'Scheduler' page within the 'Visa Clearing Exchange' application. The page has a blue header with the application name and a navigation bar with links: Home, Batch Procedure, Configuration, Tools, Run Options, and Help. The main content area is titled 'Scheduler' and contains a section 'Schedule updates'. This section has a 'Select frequency*' dropdown menu currently set to 'Daily Once'. Below this is a 'Select time of run' section with three input boxes for hours (11), minutes (00), and seconds (00), each with up and down arrow controls. A blue 'SAVE' button is located at the bottom right of the form.

1. You can either run database updates now by selecting **Now** from the drop-down list and clicking **Run** or you can schedule updates by selecting frequency from the drop-down list.
2. Select the time that you want the update to run.
3. Click **Save** to save the Database Schedule.

Saving a schedule allows database updates to run at the time specified.

File Disposition

You can choose what happens to your incoming and outgoing files after a successful run. This function is available to administrators only. You can choose to either archive or delete files upon completion of an incoming or outgoing run.

File Disposition

File disposition

Choose what should happen to your incoming and outgoing center transaction files after a successful run.

Incoming ITF disposition

Select an action

☒ Archive

☐ Delete

SUBMIT

Outgoing CTF disposition

Select an action

☒ Archive

☐ Delete

SUBMIT

Application Settings

The Application Settings can be found by navigating to **Configuration > Application Settings**. From there, you can change the configuration for Visa Clearing Exchange (VCX). After you make

any changes, you must click **Save**, and then restart the server for changes to take effect. This function is available to administrators only.

VCX Application Settings and Configurations

The screenshot displays the 'Visa Clearing Exchange' web application. The top navigation bar includes links for Home, Batch Procedure, Configuration, Tools, Run Options, and Help. The main heading is 'VCX configurations' with a subtext 'Customize application configuration parameters.' Below this, there are four tabs: Service, User & security, Email & notifications, and Archiving & data retention. The 'Service' tab is active, showing 'Service Settings'. A note states: 'The options below are read-only. If you wish to make changes, please refer to the user guide.' The settings are as follows:

Setting Name	Value
VCX base directory	/vcx
Https cert file	/config/cert.pem
Https key file	/config/key.pem
VCX service user	VCXService
VCX service port	8000
VCX proxy service user	VCXProxyService
VCX proxy service port	3000

Update Application Configuration Settings

1. Click the tab corresponding to the type of configuration settings you want to change.
Refer to the Configurable Parameters tables for details on the settings available in each tab.
2. Make changes to the settings as appropriate.
3. Click **SAVE**, or **SET DEFAULT** to save the changes.

Service Tab Configurable Parameters

Parameter	Description	Default Value
VCX Base Directory	The base directory for the VCX application	/vcx
Https Cert File	The https server certificate for the server.	/config/cert.pem
Https Key File	The location of the private key for the server certificate.	/config/key.pem
VCX Service User	TBD	VCXUser
VCX Service Port	TBD	8000
VCX Proxy Service User	TBD	VCXProxyService
VCX Proxy Service Port	TBD	3000

User & Security Tab Configurable Parameters

Parameter	Description	Default Value
User Password Expiry	The duration (in days) that the user's password is valid for.	90
User Account Lock Duration	The duration (in minutes) that the user account is to remain locked after multiple failed password attempts	30
User Attempt Threshold	Number of attempts a user is allowed with an incorrect password	6
User Inactive Days	The number of days that an account can be inactive before the account is deactivated	90
User Idle Duration	The maximum amount of idle time (in seconds) before a user session is automatically logged out	900
User Warning Duration	The duration (in seconds) of the warning dialogue before the session times out.	30
Admin Attempt Timeout	Maximum attempts to verify One-Time Password (OTP) and application key	2
OTP Duration	The number of minutes an OTP is valid	5

Email & Notifications Tab Configurable Parameters

Parameter	Description	Default Value
From Email Address	The email address used by the VCX application to send email notifications.	Blank
Email Notifications	Indicates whether email notifications is enabled.	False
MFA	Indicates whether multi-factor authentication is enabled.	False
Email Reminder for Password Expiry	The number of days before password expiration that reminder emails for changing password will be sent	10
Notification Delete Days	The number of days that a notification is retained in VCX.	7
SMTP Server Name	Domain name or IP address of the server through which emails will be sent, required in applications for sending outgoing emails on a Windows Server. Required for Windows Multifactor Authentication.	Blank
SMTP Server Port	Network access point used by the SMTP server to send or receive mail. Most commonly used is port 25.	Blank

Archiving & Data Retention Tab Configurable Parameters

Parameter	Description	Default Value
Batch Job Archive Days	The number of days before batch job reports and logs that are processed are archived	7
Run Summary Delete Days	The number of days the data on the run summary section of the Dashboard is configured for.	30
FTPS Archive After Days	The number of days before files in the FTP directory will be archived	15
FTPS Archive Days to Keep	The number of days the archived file will be retained. After that, the files will be purged	30
App Log Archive After Days	The number of days before files in the App Log directory will be archived	15

Archiving & Data Retention Tab Configurable Parameters

Parameter	Description	Default Value
App Log Archive Days to Keep	The number of days that the archived logs will be kept.	30
Database Backup Path	The path to the VCX application database backup	/opt/app/vcx/database/ backup
Database Backup Days to Keep	Days to keep the backup file	7

Chapter 11

Run Control Options and Profiles

The **Run Options** menu of Visa Clearing Exchange (VCX) gives operators and administrators the ability to set run control profiles, run control options, environmental options, and table overrides.

Run Control Profiles

Run Control Profiles is a feature that allows saving and naming sets of run control options.

Examples of uses for run control profiles include:

- Saving a test profile and a production profile
- An internal testing profile and a VCMS testing profile
- Profiles for specific transaction types

You can navigate to the **Run Control Profiles** page by going from **Run Options > Run Control Profiles**. From there, you can view, edit, delete, or duplicate current run control profiles or create a new run control profile. This function is available to administrators only.

Run Control Profiles

Run control profiles

Run control profiles allow you to save and name sets of run control options. Use this page to create, delete, or duplicate a run control profile.

NEW RUN CONTROL PROFILE ▾EDITDELETEDUPLICATELIST RELATED BATCH PROCEDURES

☐	Profile name ▾	Profile type ▾	Description
☐	dhsifn	incoming	dcnksdncksd
☐	gmrbc	outgoing	Outgoing Test

Creating a New Incoming Run Control Profile

The functions of incoming run control options are:

- Apply only to incoming edit runs.
- Specify options for returned items, file headers, transaction based reports and fee transaction reports.
- Designate which transaction-based reports the system generates during an incoming edit run.
- Designate which transactions are passed from incoming files Interchange Transaction Files (ITFs) to Center Transaction Files (CTFs).
- Split specific incoming transactions into various CTFs.
- Save multiple run control options settings for future use in the run control profile feature.

To create a new run control profile, navigate to **Home > Run Options > Run Control Profiles**, then clicking **New Run Control Profile**, then select **Incoming** from the drop-down list.

Create New Incoming Run Control Profile – Split by None

To create a new incoming run control profile, click **New Run Control Profile > Incoming**.

1. Type the name of the new incoming run control profile.
Note: There is a maximum character limit of 12 characters.
2. Provide a description of the incoming run control profile.

3. Select to split the incoming interchange by **None** from the drop-down list.
4. Select the incoming returned item file from the drop-down list, either Separate returned File with TCR 9, Separate File in Center Transaction File (CTF) Format, or No Separate Return File.
5. Select whether you want to include the header (TC 90) on incoming CTFs.
6. Click **Next** to go to the pass options and reports. Toggle the option to indicate if you'd like pass option and printer-friendly reports for specific transaction types.
You can select a category of pass options, like all, generic, clearing only, etc. Additionally, you can filter the pass options by showing on/off pass options or on/off reports and clicking **Apply**.
7. Click the switch next to the particular Transaction Code you would like to write (**On**) or omit (**Off**) from the incoming CTF.
Note: Set Default resets any changed pass options to the default values.
8. Click **Finish and Save** to save your new incoming run control profile that will not split the incoming interchange.

Create New Incoming Run Control Profile – Split by Account Number Range

To create a new incoming run control profile, click **New Run Control Profile > Incoming**.

1. Type the name of the new incoming run control profile.
Note: There is a maximum character limit of 12 characters.
2. Provide a description of the incoming run control profile.
3. Select to split the incoming interchange by **Account Number Range** from the drop-down list.
4. Select the incoming returned item file from the drop-down list, either Separate Returned File with TCR 9, Separate File in Center Transaction File (CTF) Format, or No Separate Return File.
5. Select whether you want to include the header (TC 90) on incoming CTFs.
6. Click **Next** to go to the pass options and reports. Toggle the option to indicate if you'd like a pass option and printer-friendly reports for specific transaction types.
You can select a category of pass options, like All, Generic, Clearing Only, etc. Additionally, you can filter the pass options by showing on/off pass options or on/off reports and clicking **Apply**.
7. Click the switch next to the particular Transaction Code you would like to write (**On**) or omit (**Off**) from the incoming CTF.
Note: Set Default resets any changed pass options to the default values.
8. Click **Next** to go to the Split by Account Number Range page.
9. You can select the entries to split the incoming interchange. If there are no entries, click **Add Row**.
10. Enter the Low Range for the account numbers.

11. Enter the High Range for the account numbers.

12. Select the output file from the drop-down list.

Note: Additionally, you can delete an entry by selecting the entry you want to delete and clicking the trash icon.

13. Click **Finish and Save** to save your new incoming run control profile that will split incoming interchange by the account number range defined in your entry.

Create New Incoming Run Control Profile – Split by ACQ/ISS ID

To create a new incoming run control profile, click **New Run Control Profile > Incoming**.

1. Type the name of the new incoming run control profile.

Note: There is a maximum character limit of 12 characters.

2. Provide a description of the incoming run control profile.

3. Select to split the incoming interchange by **ACQ/ISS ID** from the drop-down list.

4. Select the incoming returned item file from the drop-down list, either Separate Returned File with TCR 9, Separate File in Center Transaction File (CTF) Format, or No Separate Return File.

5. Select whether you want to include the header (TC 90) on incoming CTFs.

6. Click **Next** to go to the pass options and reports. Toggle the option to indicate if you'd like a pass option and printer-friendly reports for specific transaction types.

You can select a category of pass options, like all, generic, clearing only, etc. Additionally, you can filter the pass options by showing on/off pass options or on/off reports and clicking **Apply**.

7. Click the switch next to the particular Transaction Code you would like to write (**On**) or omit (**Off**) from the incoming CTF.

Note: **Set Default** resets any changed pass options to the default values.

8. Click **Next** to go to the Split by Acquirer or Issuer Identifier page.

9. You can select the entries to split the incoming interchange. If there are no entries, click **Add Row**.

10. Enter the ACQ/ISS ID that you would like to split the incoming interchange.

11. Select the output file from the drop-down list.

Note: Additionally, you can delete an entry by selecting the entry you want to delete and clicking the trash icon.

12. Click **Finish and Save** to save your new incoming run control profile that will split incoming interchange by the ACQ/ISS IDs defined in your entries.

Create New Incoming Run Control Profile – Split by Transaction Code

To create a new incoming run control profile, click **New Run Control Profile > Incoming**.

1. Type the name of the new incoming run control profile.
Note: There is a maximum character limit of 12 characters.
2. Provide a description of the incoming run control profile.
3. Select to split the incoming interchange by **Transaction Code** from the drop-down list.
4. Select the incoming returned item file from the drop-down list, either Separate Returned File with TCR 9, Separate File in Center Transaction File (CTF) Format, or No Separate Return File.
5. Select whether you want to include the header (TC 90) on incoming CTFs.
6. Click **Next** to go to the pass options and reports. Toggle the option to indicate if you'd like a pass option and printer-friendly reports for specific transaction types.

You can select a category of pass options like all, generic, clearing only, etc. Additionally, you can filter the pass options by showing on/off pass options or on/off reports and clicking **Apply**.

7. Click the switch next to the particular Transaction Code you would like to write (**On**) or omit (**Off**) from the incoming CTF.

Note: **Set Default** resets any changed pass options to the default values.

8. Click **Next** to go to the Split by Transaction Code page.
9. You can select the entries to split the incoming interchange. If there are no entries, click **Add Row**.
10. Select the Transaction Code from the drop-down list that you would like to split the incoming interchange.
11. Select the output file from the drop-down list.

Note: Additionally, you can delete an entry by selecting the entry you want to delete and clicking the trash icon.

12. Click **Finish and Save** to save your new incoming run control profile that will split incoming interchange by the transaction code defined in your entry.

Creating a New Outgoing Run Control Profile

The functions of outgoing run control options are:

- Apply only to outgoing edit runs.
- Specify parameters for duplicate batch detection, error thresholds, batch file capacities, file header information, and rejected item file options.
- Designate which transaction-based reports are generated during an outgoing edit run.

- Specify item tolerance amount settings for various currencies.
- Save multiple run control options settings for future use in the run control profile feature.

To create a new outgoing run control profile, navigate to **Run Options > Run Control Profiles** and then clicking **New Run Control Profile** and selecting **Outgoing** from the drop-down list.

Create New Outgoing Run Control Profile

1. Type the name of the new outgoing run control profile.
Note: There is a maximum character limit of 12 characters.
2. Enter a description of the outgoing run control profile.
3. Select whether you would like to detect duplicate files and batches, then customize the settings for finding and handling duplicates.

Note: Visa highly recommends to select both file and batch duplicate detection.

4. Establish the maximum number of records in a batch.
5. Establish the file type and File ID modifier for file header information.
Note: Use a File ID modifier when you are running files on multiple machines for the same BASE II CIB.
6. Select the option as the preferred method for saving rejected items; either no separate reject file, separate rejected file in Center Transaction File (CTF) format, or separate rejected file with TCR 9.
7. Select the error threshold rejected limit and Interchange Transaction File (ITF) creation limit.

Note: Set Default resets any changed options to the default values.

8. Establish your types of tolerance checking that you'd like to use. This step is optional.

You can establish run level tolerance, item level tolerance, force post tolerance level, and Blank Trans ID tolerance.

Item Level Tolerance: Enables processing centers to specify which currencies they will process. When enabled, add more details, including Security Level, Currency, and amount.

9. Click **Next** to go to the transaction-based reporting page.

This page specifies transaction codes for print, which allows users to override of system default settings in order to control the print of particular transaction codes during outgoing runs. It offers users the privilege of customizing which transaction codes will be reported. You can search by description, or filter by show off/on or hide disabled items. If you turn on the toggle next to off/on, you will turn on reports for all of the listed TCs.

10. Click **Save** to save the new outgoing run control profile.

Run Control Options

Run control options allow processing centers to customize Visa Clearing Exchange (VCX), Release 1.0 software to their specific processing needs. The processing center can update the default options at any time.

All run control options are validated at the beginning of each outgoing and incoming VCX run. If any invalid run control options are detected, warning or error messages are written to the Log File and the run may be discontinued.

All run control option values used for your VCX runs are distributed to Visa via VCX-generated TC 50 environmental records. Visa uses this information for troubleshooting, since it reflects how your VCX environment is set up.

Incoming Processes Controlled by Run Control Options

This table shows the types of incoming run control options that are available in VCX.

Types of Incoming Run Control Options

Incoming Function	Purpose
Identify Processing Center	Defines the processing center's BASE II CIB, Name, and Security Code.
Preform Rerun Processing	Indicates that a rerun is to occur.
Verify Run Date for re-processing incoming files	Verifies the central processing date in the ITF header when files are being reprocessed.
Set Effective Date for VCX Tables.	Specifies effective date used by VCX to establish which table entries to use during processing.
Validation Exception Report Options	Specifies sequencing and formatting of reports.
Identify Run Mode	Specifies whether the interchange to be processed contains production or test data.
Set History Maintenance Parameters	Specifies the number of calendar days that run, file, and batch data are retained on the VCX History File.
Select Report Date, Time, and Numeric Values Format	Specifies format for date, time, and numeric values for reports.
Create Incoming CTF Header Record	Creates a TC 90 Header record on all incoming CTFs.
Split Incoming ITF into Multiple CTFs	Splits an incoming ITF into separate CTFs based on account ranges, destination IDs, and transaction codes.
Create Returned Items File	Writes transactions returned by the Visa Interchange Center (VIC) on incoming interchange to a separate file in a specified format.
Pass Transactions to CTF	Controls which transactions are written to the CTF.

Types of Incoming Run Control Options

Incoming Function	Purpose
Generate Transaction-Based Reports	Indicates whether incoming transaction-based reports are to be generated.
Override Table File Entries	Used to modify table file entries during a run.

Outgoing Processes Controlled by Run Control Options

This table shows the types of outgoing run control options that are available in VCX.

Types of Outgoing Run Control Options

Outgoing Function	Purpose
Identify Processing Center	Defines the processing center's BASE II CIB, security code, and name.
Reprocess Outgoing CTF Files	Identifies the run to be reprocessed.
Set Run Date	Defines the VCX processing date, overriding the system date.
Set Effective Date for VCX Repository Tables	Specifies the effective date used by VCX to establish which table entries and rules to use during processing.
Validation Exception Report Options	Specifies the sequencing and formatting of the reports.
Indicate Run Mode	Specifies whether the interchange to be processed contains production or test data.
Set Size Limits for Batch	Specifies the maximum number of TCR records to be written to a single batch.
Set History Maintenance Parameters	Specifies the number of calendar days that run, file, and batch data are retained on the VCX History File.
Detect Submission of Duplicate Batches	Identifies a duplicate batch and issues a warning or error message to alert the user.
Set File Identification Modifier Character used in the unique file identifier	Can be used to specify a single character included in the BASE II Unique File ID field on the File Header Record. Files can be readily recognized by the Unique File ID modifier.
Define File Type	Specifies File Type populated in the header record file type field.
Set Item Tolerance Levels	Specifies the amount of tolerance levels for Individual transactions by currency, force post, or transaction ID.
Set Reject Limits	Sets the ITF reject limit as a percentage of transactions that are in error.
Create Rejected Items File	Writes transactions rejected by VCX on an outgoing run to a separate file in the specified format.

Types of Outgoing Run Control Options

Outgoing Function	Purpose
Override Table File Entries	Used to modify table file entries during a run.
Generate Transaction-Based Reports	Indicates whether outgoing transaction-based reports are to be generated.

Center BASE II CIB

The Center BASE II CIB run control option defines the processing center's BASE II CIB. It is used in incoming and outgoing runs.

Note: The Center BASE II CIB, Center Security Code, and Center Name run control options are used to identify the processing center for the Visa Clearing Exchange (VCX) software and BASE II System. The Center BASE II CIB and Center Security Code are mandatory values and must be the same as those defined with the VisaNet Interchange Center (VIC) or outgoing interchange files will be rejected by the VIC. Center BASE II CIB and Center Security Code values are passed to the BASE II system in the outgoing Interchange Transaction File (ITF).

The outgoing VCX run validates the Center BASE II CIB against the contents of the VCX Visa Identifier (VID) table. If the BASE II CIB is not found, that run will be terminated with an error.

For incoming ITFs, VCX compares the Center BASE II CIB and Center Security Code values from the ITF TC90 Header record provided by BASE II to the values of these fields defined by the processing centers. If the values do not match, the run will be terminated with errors.

- The CIB must exist in the VCX VID table.

Center Name

The Center Name run control option defines the processing center's name. It can be used in incoming and outgoing runs.

- The Center Name is used on Visa Clearing Exchange (VCX) reports.
- The default for this run control option is all spaces.

Center Security Code

The Center Security Code run control option specifies the processing center's security code. It is used in incoming and outgoing runs.

- This value is a processing center's eight-character alphanumeric security code and must be the same as defined with the VisaNet Interchange Center (VIC).

Date Format

The Date Format run control option specifies how the Gregorian date is to be displayed on Visa Clearing Exchange (VCX) reports. It can be used in incoming and outgoing runs.

- This run control option specifies preference for date format used on VCX reports.
- The DATE run control option can have the following format options:

- YYMMDD
- CCYYMMDD
- DDMMCCYY
- DDMMYY
- MMDDCCYY
- MMDDYY

Note:

CC = Century. The first two digits of the year (20 for 2000). (CCYY is the same as YYYY).

DD = Day of month. For example, 03 for August 3.

MM = Month. For example, 08 for August.

YY = Year. The last two digits of the year (00 for 2000).

- Date Format and Date Separator run control options can be used to select report formatting for date variables on all VCX reports. A combination of the Date Format and Date Separator options can be modified to conform to any national environment.

Date Separator

Date Separator specifies the character to be used to separate the 10-character Gregorian date fields displayed on reports. It can be used in incoming and outgoing runs.

The Date Separator run control option is a single special character used to separate date fields on Visa Clearing Exchange (VCX) reports.

Note: A blank is acceptable but alphabetic or numeric characters such as A–Z or 0–9 are not acceptable.

Duplicate Batch Detection

The Duplicate Batch Detection run control option is used for duplicate batch detection. It can be used in outgoing runs only.

- Duplicate Batch Detection uses one of two different methods to detect duplicate outgoing batches:
 - Batch Hashing detects duplicate batches using a batch hashing technique to check against the contents of the Visa Clearing Exchange (VCX) History File for a specified number of calendar days.
 - Center Batch ID Checking detects duplicate batches by checking the Center Batch ID on the TC 91 batch trailer record.

Note: The Batch Hashing method is the recommended method because it is capable of detecting a greater variety of potential processing errors. In addition, this method does not require the processing center's system to provide a unique TC 91 Center Batch ID.

VCX compares the calculated hash total to the calculated values of all batches that exist in the History File for a specified number of days. The number of days is supplied by the processing center to determine how far back in history to check for duplicate hash totals.

Processing centers can optimize the performance of duplicate batch checking when invoking the Batch Hashing method by using the days parameter to limit the number of batches searched.

Center Batch ID Checking enhances processing efficiency by not calculating the hash value. To check for more than one day of history using this method, the processing center's pre-edit software would have to create unique Center Batch IDs across multiple processing days. Including a date in the Center Batch ID is a preferable technique to use in order to make each ID unique.

The Center Batch ID is supplied by the processing center during pre-edit processing. It can be any value, but it must be unique within the number of days specified by the day parameter. A batch is considered a duplicate if the Center Batch ID exists in the History File for a previous batch within the time period specified by the days parameter.

- If duplicate batch checking is invoked and VCX detects a duplicate batch, the option exists to either have a warning message issued and commit the batch to the outgoing file (ITF), or have the outgoing file rejected.
 - Warning Message – If an error is found, processing continues after a warning message is displayed.
 - Error Message – If an error is found, processing immediately ends and an error message is displayed.

- If duplicate batch checking is invoked, endpoints can specify a number of calendar days to check history.
 - Clients can select from 1 to 100 days.
 - If this value is blank, a default value of 5 days is added.

Error Reporting Threshold

The Error Reporting Threshold run control option specifies the percentage of transaction errors (expressed as a percentage of total transactions processed so far) that may occur before generation of error messages and further processing is terminated. It can be used in outgoing runs.

Percent (1 to 100) (default = 100) – No further errors will be reported and processing will terminate if the percentage of rejected transactions exceeds the percentage specified for Error Reporting Threshold.

- Any valid numeric value from 1 to 100.
- One (1)% ensures that at least some error messages will be reported when errors occur.
- 100% indicates that there is no limit and all transactions in error will be reported.

This option prevents the flooding of the outgoing Validation Exception Detail Report (VX-100A) with redundant messages and avoids lengthy processing when a large number of errors occur.

To prevent exceeding the limit prematurely, the Error Reporting Threshold is not tested for the first 200 transactions. After that, the percentage of rejected transactions is recalculated after each transaction from the Center Transaction File (CTF) is processed. If the percentage of rejected transactions exceeds the percentage of rejects specified by the Error Reporting Threshold, a final error message is displayed to indicate that the limit has been reached, and:

- Processing of remaining transactions is bypassed.
- No outgoing Interchange Transaction File (ITF) is created.
- Reports generated to reflect information about transactions processed up until the threshold has been met.
- A message is logged indicating that the Error Reporting Threshold was exceeded.

Note: This run control option is usually set to a higher value than ITF Error Threshold since it results in bypassing remaining transactions, which means that the outgoing ITF limit would never be reached.

Examples:

- ITF Error Threshold = 0, Error Reporting Threshold = 1
 - An ITF is created only if no errors are detected. Processing stops if 1% of the records on the input has errors.
 - If less than 1% of the records are in error, all reports are produced.
 - If 1% of the records are in error, only the VX-100A Validation Exception Detail Report is produced.
- ITF Error Threshold = 10, Error Reporting Threshold = 100
 - An ITF is created only if less than 10% of the records on the input have errors. Processing stops if 100% of the records on the input have errors.
 - If less than 100% of the records are in error, all reports are produced.
 - If 100% of the records are in error, only the VX-100A Validation Exception Detail Report is produced.
- ITF Error Threshold = 100, Error Reporting Threshold = 10
 - An ITF is created only if less than 100% of the records on the input have errors. However, a value in the ITF Error Threshold higher than the Error Reporting Threshold is not recommended since the Error Reporting Threshold value takes precedence over the ITF Error Threshold.
 - Processing stops if 10% of the records on the input have errors. No output is created. If less than 10% of the records are in error, all reports are produced.
 - If 10% of the records are in error, only the VX-100A Validation Exception Detail Report is produced.

Exception Report Format

The Exception Report Format run control option allows selection of either formatted or unformatted exception report layouts. It can be used in incoming and outgoing runs.

- Validation Exception Detail Reports (VX-100A and VX-200A) list items rejected by Visa Clearing Exchange (VCX). The Exception Report Format allows you to select the report layout of the Validation Exception Detail Reports (VX-100A and VX-200A) that list items rejected by VCX.
- An unformatted report prints the same image as the transactions.
- A formatted report prints the name of each field in the transaction and its contents. This format is similar to transaction-based reports. Formatted versions of the report allow users to easily identify transactions that are in error, without reference to transaction layout documentation. Reports sequenced by ACQ/ISS ID will allow centers processing for multiple ACQ/ISS IDs to route rejected item reports to those responsible for correction and resubmission of the items.

Exception Report Sequence

The Exception Report Sequence run control option allows selection of the exception report sequence. It can be used in incoming and outgoing runs.

Exception Report Sequence allows two selections of the report sequence.

- The FILE Report Sequence option causes rejected items to be reported in the sequence that they occur within the file. Acquiring and issuing identifier (ACQ/ISS ID) causes transactions to be reported in source identifier sequence during the outgoing run and in destination identifier sequence during the incoming run. The source or destination identifier appears in the heading of the report and a page break will occur each time the source or destination identifier changes.
- The ACQ/ISS ID Report Sequence option causes the rejected items to be sequenced by source or destination identifiers.

Exclude Transactions from Output

The Exclude Transactions from Output run control option prevents specific transaction codes from being written to the Center Transaction File (CTF) during incoming runs. It can be used in incoming runs.

Exclude Transactions From Output available options:

Incoming transaction code – specifies a two-digit single transaction code not to be passed to the CTF during an incoming run.

The default is the value set in the Visa Clearing Exchange (VCX) Transaction Table supplied by Visa.

See *VCX Print/Pass Defaults* for a complete list of valid BASE II transaction codes and print defaults.

File Identification Modifier

The File Identification Modifier run control option is used to specify a single character included in the BASE II Unique Field ID on the TC 90 File Header Record during an outgoing run. The BASE II Unique File ID is generated by Visa Clearing Exchange (VCX) for use in history processing by BASE II. The Unique File ID is in positions 79 through 108 of the TC 90 File header Record. The File ID modifier character is in position 94.

The File Identification Modifier is defined as one character, either alphabetic (A to Y) or numeric (0 to 9) character, to be included in the BASE II Unique File ID.

- P – File identification modifier character is specified for a production run.
- T – File identification modifier character is specified for a test run.
- Blank (default) – File identification modifier will be the same as the value specified by the Run Mode run control option.

Note: If File ID Modifier is left blank, position 94 will be based on the value specified in the Run Mode run control option.

File ID Modifier gives the processing center the ability to modify the Unique File ID so that files created in their data center can be easily identified. By specifying different File ID Modifier values for separate runs scheduled during the processing day, files can be readily recognized by the Unique File ID modifier. The Unique File ID generated during the outgoing run appears on VCX reports VX-120F, VX-220, VX-221B, VX-221D, and VX-221E.

File Type

The File Type run control option is used to specify the file type in the TC 90 File Header Record of the outgoing ITF for BASE II Customized Delivery processing. For outgoing runs the File Type run control option should only be used at processing centers where Visa has requested that it be used. Visa will provide the appropriate values.

Note: Processing centers must exercise caution when specifying values for File Type. Visa Clearing Exchange (VCX) does not check for valid values.

Incoming Header Record

The Incoming Header Record run control option allows to create the header records (TC 90) on incoming Center Transaction Files (CTFs). It can be used in incoming runs.

Incoming Header Record available options:

- No (default) – The header record (TC 90) is not written to the incoming CTF file.
- Yes (include) – The header record (TC 90) is written to the incoming CTF file.

Incoming Header Record allows processing centers to have Visa Clearing Exchange (VCX) create a header record on all incoming CTF files. Processing centers can then access this information without the need to read the ITF in their post-edit processing. The default is N.

The following data is available on the incoming CTF header record:

- BASE II CIB and security code
- VIC processing date
- VIC settlement date

- VCX release number
- File type
- Subfile ID number

For the layout of the TC 90 header record, refer to Header Records in *BASE II Clearing Interchange Formats, TC 50 to TC 92*.

Include Transactions on Output

The Include Transactions on Output allows writing of specific transaction codes to the incoming Center Transaction File (CTF). It can be used in incoming runs.

Include Transaction on Output available options:

Incoming transaction code – specifies a two-digit single transaction code to be passed to the CTF during an incoming run.

Include Transactions on Output option causes specific transaction codes to be written to the incoming CTF.

See *VCX Print/Pass Defaults* for a complete list of valid BASE II transaction codes and print defaults.

Item-Level Tolerance Checking

Item-level Tolerance Checking run control option activates item-level tolerance checking. It can be used in outgoing runs.

Item Tolerance Checking run control option specifies all valid currencies processed by your center. It can be used with outgoing runs.

Item Tolerance by Currency available options:

- Currency = Blank (default) – Item-level tolerance checking will not occur.
- Currency = aaa – Where aaa is a valid, three-character alphabetic BASE II currency code.
Note: A currency code table exists in *BASE II Clearing Data Codes* to aid in converting the ISO currency code to the alphabetic currency code.
- Amount = nnn – Where nnn is a 10-digit numeric amount specifying the maximum monetary amount valid for an item, expressed as an integer. (Any amount of 0 through 999999999 is acceptable.)

Once item-level tolerance checking is activated, Item Currency Amount is used to specify all valid BASE II currencies processed by the processing center and a threshold amount for each to be used to determine an out of tolerance condition., Use as many Item Currency Amount entries as necessary in order to specify the currencies for tolerance checking.

If the source amount exceeds the defined tolerance limit or if the transaction currency has not been specified, the transaction is rejected, and the error appears on the VX-100A Validation Exception Detail Report.

ITF Creation Limit

The Interchange Transaction File (ITF) Creation Limit run control option specifies the percentage of transaction errors (expressed as a percentage of total transactions for the file) that may not be exceeded if an ITF is to be produced.

Percent (0 to 100) (default = 100) – No ITF will be created if the percentage of rejected transactions exceeds the percentage specified for the ITF Creation Limit.

- Any valid numeric value from 0 to 100.
- Zero (0)% indicates that a single error will suppress generation of the output.
- 100% indicates that there is no threshold, and the outgoing file will be generated, regardless of the number of transaction errors found.

This option allows the processing center to prevent submission of an outgoing file into VisaNet when a large percentage of the transactions on the outgoing ITF have been rejected.

The ITF Creation Limit is checked after processing all transactions on the Center Transaction File (CTF). If the number of transactions in error exceeds the specified threshold:

- Only the Validation Exception (VX-100A, VX-100B, and VX-100C) and Processing Log (VX-199) Reports are produced.
- A message is logged indicating that the ITF Creation Limit was Exceeded.
- No outgoing ITF is created.

Maximum Number of Records in a Batch

The Maximum Number of Records in a Batch run control option specifies the maximum number of transaction component records (TCRs) that will be included in a batch on the center transaction file (CTF) generated by the processing center's pre-edit software. If a batch contains more TCRs than specified by Maximum Number of Records in a Batch run control option, an error message is generated, and the run is aborted. It can be used in outgoing runs.

The Maximum Number of Records is 999 by default. Values from 100 through 3300 records may be specified.

Note: The TC 91 Batch Trailer Record is included in the TCR count for the batch.

Processing efficiency is enhanced by using larger batches. Visa Clearing Exchange (VCX) does not re-batch data when creating the ITF. To improve processing efficiency, the system which creates the CTF must generate the larger batches and the Maximum Number of Records in a Batch run control option must be set to match the largest possible batch.

Print Incoming Reports

The Print Incoming Reports run control option provides detailed reporting of selected incoming transactions. It can be used in incoming runs.

To manage transaction-based reporting for different transaction types during an incoming edit run, follow these instructions:

Next to each transaction type, there will be a switch. The current state of the switch indicates whether transaction-based reporting is activated (on) or deactivated (off) for that transaction type.

- To Activate: Toggle the switch to the "on" position. This will ensure that transaction-based reports will be generated for this transaction type during the incoming edit run.
- To Deactivate: Toggle the switch to the "off" position. This will prevent transaction-based reports from being generated for this transaction type.

The Selection list includes only those transaction types for which the print option can be modified. Transaction types with fixed print options are displayed in the separate tab for references.

Print Incoming Reports run control options are not used to control printing of TC 45 or TC 47 transactions. TC 45 General Delivery Reports and TC 47 VIC Interchange Reports automatically print. If the processing center wants to be removed from the distribution list for the TC 45 General Delivery Reports, contact Visa.

Note: Some transaction-based reports can take up considerable disk space. This should be accounted for when choosing the report archiving options.

See *VCX Print/Pass Defaults* for a complete list of valid BASE II transaction codes and print defaults.

Print Outgoing Reports

The Print Outgoing Reports run control option provides detailed reporting of selected outgoing transactions. It can be used in outgoing runs.

To manage transaction-based reporting for different transaction types during an incoming edit run, follow these instructions:

Next to each transaction type, there will be a switch. The current state of the switch indicates whether transaction-based reporting is activated (on) or deactivated (off) for that transaction type.

- To Activate: Toggle the switch to the "on" position. This will ensure that transaction-based reports will be generated for this transaction type during the incoming edit run.
- To Deactivate: Toggle the switch to the "off" position. This will prevent transaction-based reports from being generated for this transaction type.

The Selection list includes only those transaction types for which the print option can be modified. Transaction types with fixed print options are displayed in the separate tab for references.

Note: Some transaction-based reports can take up considerable disk space. This should be accounted for when choosing the report archiving options.

See *VCX Print/Pass Defaults* for a complete list of valid BASE II transaction codes and print defaults.

Run Type

The Run Type run control option indicates the type of transaction data (production or test) the interchange contains. It can be used in incoming and outgoing runs.

Run Type options:

- Production (default) – Processed Interchange Transaction File (ITF) contains production data.
 - ITF contains spaces in the Test Option Field of the File Header Record (TC 90)
 - This allows it to be processed only by the production system at the VisaNet Interchange Center (VIC). For incoming VCX runs, only ITF with spaces in the Test Option field are accepted by VCX when Run Type, Run Control Option of Production is specified.
- Test – Processed ITF contains test data.
 - The ITF contains the string "TEST" in the Test Option Field of the File Header Record (TC 90)
 - This allows it to be processed only by the Visa Certification Management System (VCMS) at the VIC. For incoming VCX runs, only ITF with the string "TEST" in the Test Option field are accepted by VCX when Run Type, Run Control Option of Test is specified.

Save File Rejects

The Save File Rejects run control option allows Visa Clearing Exchange (VCX) to write rejected transactions to a separate file. This allows correction of erroneous fields in the transaction without re-generating the entire transaction. It can be used in outgoing runs.

Save File Rejects available options:

Method

- Transaction format (default) – Rejected transactions found during an outgoing run are written to a separate file in transaction format.
- Reject format – Rejected transactions found during an outgoing run are written to a separate file in reject format. Reject format contains TCR 9 data.
- No separate reject file – Rejected transactions found during an outgoing run are not written to a separate file.

If the “Transaction format” option is used, the outgoing interchange transactions rejected by VCX are saved in Center Transaction File (CTF) format and contain TC 91 batch trailer and TC 92 file trailer records. A TC 90 header record is also supplied, if the processing center has provided a TC 90 header record on the original outgoing CTF.

If the “Reject format” option is used, then the rejected transactions are saved with an added TCR 9 record. The TCR 9 record contains the reject reason codes shown on the Validation Exception Detail Report (VX-100A). A maximum of 10 reject codes can be reported for each transaction. See VCX File Layouts for VCX TCR 9 record format.

If the “No” option is used, the outgoing interchange transactions rejected by the VCX are not saved and the rejected transactions are not in the outgoing Interchange Transaction File (ITF). However, rejected transactions will appear on the VX-100A Validation Exception Detail Report, whether the Save File Rejects run control option was set to T, R, or N.

Save Returned Transactions

The Save Returned Transactions Run control options allows Visa Clearing Exchange (VCX) to write returned transactions to a separate file. It can be used in incoming runs.

Save Returned Transactions writes transactions returned by the VisaNet Interchange Center (VIC) on incoming interchange to a separate file as returned item transactions (TC 01, TC 02, or TC 03) or as original transactions.

The separate file is in CTF format and contains batch trailer (TC 91) and file trailer records (T92). A header record (TC 90) is also supplied, if the processing center has switched on the Incoming Record Header run control option.

Save Returned Transactions available options:

- TCR 9 format (default) – Returned transactions from the VIC are written to a separate file in the standard CTF format containing TC 01, 02, and 03 transactions. Each transaction contains TCR 9 data.
- CTF format – Returned transactions from the VIC are written to a separate file in the standard CTF format containing the original transactions. The TCR 9 data is not written to the file.
- No separate return file – Returned transactions are not written to a separate file.

If the “TCR 9 format” option is used, the only change to the records written to the file is that the hash bytes in positions three and four of the Interchange Transaction File (ITF) are removed. Each transaction contains a TCR 9 data with information about the original transaction and the returned item reason codes. See *BASE II Clearing Interchange Formats, TC 01 to TC 49* and *BASE II Clearing Interchange Formats, TC 50 to TC 92*, for TCR 9 returned item format.

If the “CTF format” option is used, the original transaction code from the TCR 9 data or the TC 01, TC 02, and TC 03 return item transaction is restored to positions one and two in the file created. The TCR 9 data from the returned item transaction is not written to the file. The processing center can process the file in an internal correction application. To assist in the correction of returned items, returned item reason codes for each transaction are reported on the VX-204A reports.

If the “No separate return file” option is used, returned items are not saved in a separate file. Returned items are written to the CTF based on the Include Transactions on Output and Exclude Transactions from Output run control options. To ensure that returned items are written to the CTF when Save Returned Transactions=N is used, specify Include Transactions on Output – TC 01, 02, and 03. If the returned items are not saved in a separate file and are not passed on to the CTF, specify Print Incoming Reports for TC 01, 02, and 03 to ensure that the returned items are printed.

Note: Include Transactions on Output is not valid for returned items when Save Returned Transactions is set to R or T.

Split ITF Into Multiple Files

The Split ITF Into Multiple Files run control option allows an incoming Interchange Transaction File (ITF) to be split into separate and distinct output files (Center Transaction Files (CTFs)). Up to 25 separate incoming output files may be created. It can be used in incoming runs.

Split ITF Into Multiple Files available options:

- N (default) – Files are not split. All transactions are written to CTF output file 01.
- Account Range – Splits ITF by account range
- ACQ/ISS ID – Splits ITF by ACQ/ISS ID
- Transaction Code – Splits ITF by transaction code

Note: Before any of Split ITF By options are activated for a run, Split ITF Into Multiple Files must be turned on to a value other than N. Since only one parameter can be specified with Split ITF Into Multiple Files option, only one Split ITF By criterion can be activated for an incoming Visa Clearing Exchange (VCX) run.

The Split ITF Into Multiple Files run control option allows the split to be changed at any time without the need to delete selection criteria entries that have already been defined by various Split ITF By options.

Requirements for this option are:

- If N, all incoming transactions will be written to one CTF.
- If Account Number Range, data will be written to multiple CTFs based on account ranges. Split ITF by Account Range must be used to specify account ranges.
- If ACQ/ISSID, data will be written to multiple CTFs based on acquirer or issuer identifiers. Split ITF by Destination ID must be used to specify ACQ/ISS IDs.
- If Transaction Code, data will be written to multiple CTFs based on transaction codes. Split ITF by Transaction Code must be used to specify transaction codes.

Split ITF by Account Range

The Split ITF by Account Range run control option allows an incoming Interchange Transaction File (ITF) to be split into separate and distinct output files (Center Transaction Files (CTFs)) by specifying the account ranges to be included in the output files. Up to 25 separate incoming files may be created. It can be used in incoming runs.

Split ITF by Account Range options:

- Output file (2 to 25) – A two-digit numeric with values of 02 through 25, specifying the subfile number.
- Low (Start account number) – Starting account number from 9 to 16 digits.
- High (End account number) – Ending account number from 9 to 16 digits.

Requirements for this option are:

- Only one account range can be specified on each option record.
- To specify an additional account range for the same subfile, add another Split ITF by Account Range option record defining the same output file and listing the additional account range.
- Use as many Split ITF by Account Range records as necessary to identify all accounts to be written to output file 02 through 25.
- Account ranges may not overlap.
- Any account number not specified in output file 02 through 25 has its transactions written to output file 01.
- Account numbers can be from 9 to 16 digits in length.
- Any transaction that does not contain an account number or contains an account number that is outside of a specified range is written to output file 01.

Split ITF by Destination Identifier

The Split ITF by Destination Identifier Run control option allows an incoming Interchange Transaction File (ITF) to be split into separate and distinct output files (Center Transaction Files

(CTFs)) by specifying the destination identifier to be included in the output file. Up to 25 separate incoming output files may be created. It can be used in incoming runs.

Split ITF by Destination Identifier available options:

- Output file (2 to 25) – A two-digit numeric with values of 02 through 25 specifying the output file number.
- ID (Any valid ACQ/ISS ID) – A valid six-digit ACQ/ISS Identifier.

Requirements for this option are:

- Only one ACQ/ISS ID can be specified on each option record.
- To specify an additional ACQ/ISS ID for the same subfile, add another Split ITF by option record defining the same output file and listing the additional ACQ/ISS ID.
- The destination identifier is the issuing identifier for original, dispute response financial, and reversal transactions and the acquiring identifier for dispute financial and dispute financial reversal transactions.
- Use as many Split ITF by Destination ID records as necessary to identify all IDs to be written to output file 02 through 25.
- Destination identifiers may not be specified more than once.
- Any ACQ/ISS identifier not specified in output file 02 through 25 has its transactions written to output file 01.

Split ITF by Transaction Code

The Split ITF by Transaction Code run control option allows an incoming Interchange Transaction File (ITF) to be split into separate and distinct output files (Center Transaction Files (CTFs)) by specifying the transaction code to be included in the output file. Up to 25 separate incoming output files may be created. It can be used in incoming runs.

Split ITF by Transaction Code available options:

- Output file (2 to 25) – A two-digit numeric with values of 02 through 25 specifying the output file number.
- Transaction Code (Any valid incoming transaction code) – A valid two-digit transaction codes.

Requirements for this option are:

- To specify additional transaction codes for the same subfile, use another Split ITF by Transaction Code option record defining the same subfile and listing the additional transaction code.
- Use as many Split ITF by Transaction Code records as necessary to identify all transaction codes to be written to output files 02 through 25.

- No transaction code may be specified more than once.
- Any transaction code not specified in output file 02 through 25 has its transactions written to subfile 01.

When selecting draft data transactions (TC 05, 06, 07, 15, 16, 17, 25, 26, 27, 35, 36, 37), you can split them by usage (original or dispute). For instance, specify that the output file should include only original sales drafts by choosing TC05 Sales Draft, Original, or only dispute drafts by choosing TC05 Sales Draft, Dispute Response Financial.

Also, the splitting of TC 33 Multipurpose Message transactions is now possible into BASE II Dispute Financial Status Advice and other TC 33 Advices. The Dispute Financial Status Advice transactions can be split with a value of V and the non-Dispute Advices can be split with a value of N.

Note: A transaction must be passed to the CTF before it will appear on the output file. Refer to the *VCX Print/Pass Defaults* for the specified TC to determine if the Include Transactions on Output override is required.

Stop If Return Code High

The Stop If Return Code High run control option is used to control the maximum return code that will cause Visa Clearing Exchange (VCX) to stop processing. It can be used in incoming and outgoing runs.

Stop If Return Code High available options:

(0–16) – (default is 08) – A value that is used to identify a maximum return code that is accepted by VCX.

Table Date

The Table Date run control option defines the effective date for table rules and values used by Visa Clearing Exchange (VCX). It can be used in incoming and outgoing runs.

Table Date available options:

Date:

MM/DD/YYYY – The effective date to be used for loading VCX tables rules and values.

This option forces the VCX to use edits that it would otherwise not select. During outgoing runs, the default is the system date or Run Date, if specified. During incoming runs, the default is one day less than the Center Processing Date (CPD) of the Interchange Transaction File (ITF) header.

This run control option should only be used when testing BASE II modifications against future date edits. The use of this option for production data may result in an increase in returned items from the VisaNet Interchange Center (VIC).

Note: Table Date will always be used as the table effective date, but if Table Date is not entered then the Run Date will be used. If neither Table Date nor Run Date are available, the computer system date will be used during outgoing runs and one day before the CPD of the ITF header during incoming runs.

Table Override

The Table Override run control options can be used to temporarily override the one record in the Visa Clearing Exchange (VCX) tables. There are three VCX table files for which table overrides can be done. The following table describes each type.

Any VCX table can be overridden with these run control options. However, Visa recommends that you only use these run control options to override the VID and ARDEF tables. All other table overrides should be done only with information supplied by Visa support staff.

ACQ/ISS ID and ARDEF Table Overrides

Table overrides are used to temporarily override one record in the VCX VID and ARDEF tables. The overrides do not change the file copy of any table. They simply override the file copy of the record that would be effective for the current run. Table overrides can be used to add, change, or delete a table entry.

Manual Entry Overrides

Both key and data table overrides must be specified together. They cannot be used individually.

Table Key Override

The Table Key specifies the key value of a table entry in the repository value table. It can be used in incoming and outgoing runs.

Table Key available options:

- tablename – Visa-specified values. One-to eight-character names identifying the table that will be overwritten.
- Key – Visa-specified values. Key value of the table entry in the target table.

Table Data Override

The Table Data specifies VCX table information to be used to override entries in a repository value table. A maximum of 25 Table override statements may be present. It can be used in incoming and outgoing runs.

Table Data available options:

Data – Visa-specified values. Table information to override entries in a repository table. The option can be used to override, delete, or add entries.

Note: Create an ACQ / ISS ID table override for a New BASE II CIB.

If a client did not participate in Edit Package 4.0 and has a new CIB the client needs to create an acq/iss table override for their CIB to be added to the table. Then, they can execute their FFR. If their CIB is present in the FFR, they can disable this override, otherwise they will need to keep it active until their CIB is added.

Transaction Report Sequence

The Transaction Report Sequence run control options allows clients to print VX-700 series reports in ACQ/ISS ID or File sequence. By default, transaction-based reports (in the VX-700 series) show transaction detail information in the same sequence as transactions occur in the incoming ITF or outgoing CTF. By using the Transaction Report Sequence option (ACQ/ISS ID) clients can cause transactions to be reported in sequence by ACQ/ISS ID. During the outgoing run, transactions will be in source identifier sequence. During the incoming run, transactions will be in destination identifier sequence. The ACQ/ISS ID is used for sequencing, will appear in the header of the report and page breaks will occur between ACQ/ISS ID. It can be used in Incoming and Outgoing runs.

Transaction Report Sequence available options:

- File (default) – Prints reports in file sequence.
- ACQ/ISS ID – Prints reports in acquirer or issuer identifier sequence.

Note: Transactions must be selected for print by default or by use of the Print Incoming Reports or Print Outgoing Reports before this option can work.

During incoming runs, reports received from Visa services as TC 45 General Delivery transactions will also be sequenced according to the value of the Transaction Report Sequence option. The following rules apply:

- If the FILE option is selected, reports will be grouped in the VX-745 file by service. Within service, most reports are sequenced by acquirer or issuer identifier.
- If the ACQ/ISS ID option is selected, reports from multiple services will be grouped together by acquirer or issuer identifier.

Validation Errors Reported

The Validation Errors Reported run control option is used to identify the maximum number of errors per transaction reported. During validation it is possible to identify many exception conditions. Only the first 10 are reported by default. It can be used in incoming and outgoing runs.

Validation Errors Reported available options:

Number (1 to 1,000) (default = 10) – Maximum number of errors reported for any single transaction.

Mainframe to Visa Clearing Exchange: Run Control Options Transition

Mainframe to Visa Clearing Exchange: Run Control Options Transition

Edit Package Run Control Option	VCX setting	Where to locate in VCX
centrbin	Base II CIB	Configuration -> CIB Setup
centrcode	Security Code	Configuration -> CIB Setup
centrname	Center Name	Configuration -> CIB Setup
runmode	Run Mode	Configuration -> CIB Setup
splitinfil = N	Incoming Interchange 'None'	Run Options -> New Run Control Profile -> Profile Type Incoming
splitinfil = A	Incoming Interchange 'Account Number Range'	Run Options -> New Run Control Profile -> Profile Type Incoming
splitinfil = B	Incoming Interchange 'ACQ/ISS ID'	Run Options -> New Run Control Profile -> Profile Type Incoming
splitinfil = T	Incoming Interchange 'Transaction Code'	Run Options -> New Run Control Profile -> Profile Type Incoming
savereturn=N	returned item file 'Separate Returned File in CTF Format'	Run Options -> New Run Control Profile -> Profile Type Incoming
savereturn=T	returned item file 'Separate Returned File in CTF Format'	Run Options -> New Run Control Profile -> Profile Type Incoming
savereturn=R	returned item file 'Separate Returned File with TCR9'	Run Options -> New Run Control Profile -> Profile Type Incoming
writeheadr = Y	check Include Header (TC90) in Incoming CTF file	Run Options -> New Run Control Profile -> Profile Type Incoming
splitbyard	splitbyard 02,405266001,405266006	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split By Account Number Range
Low Account Range	Low Range	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split By Account Number Range
High Account Range	High Range	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split By Account Number Range
File Number	Output to	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split By Account Number Range

Mainframe to Visa Clearing Exchange: Run Control Options Transition

Edit Package Run Control Option	VCX setting	Where to locate in VCX
splitbybin		Run Options -> New Run
ACQ/ISS Value 1	ACQ/ISS ID	Run Options -> New Run
File Number	Output to	Control Profile -> Profile Type Incoming -> Split by Acquirer or Issuer Identifier
splitbytc	Select	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split by Transaction Code
Transaction Code TC Value 1	Transaction Code entry	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split by Transaction Code
File Number	Output to	Run Options -> New Run Control Profile -> Profile Type Incoming -> Split by Transaction Code
	Pass Options	Run Options -> New Run Control Profile -> Profile Type Incoming -> Pass Options
Trnpass=01	01 - Return Credit – pass options - On	Run Options -> New Run Control Profile -> Profile Type Incoming -> Pass Options
Trnopass=46	46 - Member Settlement Data – pass options - Off	Run Options -> New Run Control Profile -> Profile Type Incoming -> Pass Options
	Print Options	Run Options -> New Run Control Profile -> Profile Type Incoming -> Transaction Based Reporting
Trninprt=46	46 - Member Settlement Data – print option - On	Run Options -> New Run Control Profile -> Profile Type Incoming -> Transaction Based Reporting
Trnnoinprt=40	40 – Fraud Advice – print option - Off	Run Options -> New Run Control Profile -> Profile Type Incoming -> Transaction Based Reporting
dupbatch = N	Duplicate Batch Detection not checked	Run Options -> New Run Control Profile -> Profile Type Outgoing
dupbatch = E,HASH,5	Duplicate Batch Detection checked	Run Options -> New Run Control Profile -> Profile Type Outgoing
	Security Level - Error	Run Options -> New Run Control Profile -> Profile Type Outgoing

Mainframe to Visa Clearing Exchange: Run Control Options Transition

Edit Package Run Control Option	VCX setting	Where to locate in VCX
	Detection Method – Hashing Technology	Run Options -> New Run Control Profile -> Profile Type Outgoing
	History Search Days - 5	Run Options -> New Run Control Profile -> Profile Type Outgoing
dupbatch = W,CENTERID,10	Duplicate Batch Detection checked	Run Options -> New Run Control Profile -> Profile Type Outgoing
	Security Level - Warning	
	Detection Method – Center Batch ID	Run Options -> New Run Control Profile -> Profile Type Outgoing
	History Search Days - 10	Run Options -> New Run Control Profile -> Profile Type Outgoing
Maxbatrecs	'Max Number of Records in Batch'	Run Options -> New Run Control Profile -> Profile Type Outgoing
Filetype	'File Type'	Run Options -> New Run Control Profile -> Profile Type Outgoing
Fileidmod	'File ID Modifier'	Run Options -> New Run Control Profile -> Profile Type Outgoing
	'Rejected Item File'	Run Options -> New Run Control Profile -> Profile Type Outgoing
savereject=N	'No Separate Reject File'	Run Options -> New Run Control Profile -> Profile Type Outgoing
savereject=T	'Separate Rejected File in CTF Format'	Run Options -> New Run Control Profile -> Profile Type Outgoing
savereject=R	'Separate Rejected File with TCR9'	Run Options -> New Run Control Profile -> Profile Type Outgoing
rejitflim	'ITF Creation Limit'	Run Options -> New Run Control Profile -> Profile Type Outgoing
rejmsglim	'Rejected Message Limit'	Run Options -> New Run Control Profile -> Profile Type Outgoing
	Print Options	Run Options -> New Run Control Profile -> Profile Type Outgoing -> Transaction Based Reporting
Trnoutprt=40	40 – Fraud Advice – print option - On	Run Options -> New Run Control Profile -> Profile Type Outgoing -> Transaction Based Reporting
internal	Internal Application ID	Run Options -> Environmental Options

Mainframe to Visa Clearing Exchange: Run Control Options Transition

Edit Package Run Control Option	VCX setting	Where to locate in VCX
dateformat	'Date Format' from the list	Run Options -> Environmental Options
datesep	Date Separator	Run Options -> Environmental Options
timesep	Time Separator	Run Options -> Environmental Options
numdecsep	Decimal Separator	Run Options -> Environmental Options
numhousep	Thousand Separator	Run Options -> Environmental Options
tranrptseq	Transaction Report Sequence	Run Options -> Environmental Options
excp rptseq	Exception Report Sequence	Run Options -> Environmental Options
excp rptfmt	Exception Report Format	Run Options -> Environmental Options
histrepd	History File Retention Period	Run Options -> Environmental Options
valerror	Validation Errors Reported	Run Options -> Environmental Options

Environmental Options

The Environmental Options in Visa Clearing Exchange (VCX) are used to specify details about the BASE II CIB, designate the formats for dates, times, and numbering on reports, set various options for how reports are sorted and how exception items are reported, and set retention period for the VCX History File. You can find this page by navigating to **Run Options > Environmental Options**.

If you select **Set Default** on the Environmental Options page, any changes made to the environmental run control options will be reset to their default values.

VCX Environmental Options Page

Environmental run control options

Environmental options are used to designate the formats for dates, times, and numbering on reports, set various options for how reports are sorted and how exception items are reported, and set retention period for the VCX History File. Enter the details and all fields are required unless marked optional.

General

Select the BASE II CIB for your processing center. Then, choose the internal application identifier that should use the PC VCX for Windows.

Base II CIB*

Internal application identifier

Report date and time formats

Select your preferred formatting for dates and time in your reports.

Date format*

SET DEFAULT

SAVE

History Retention Period

The History Retention Period run control option specifies the number of calendar days that run, file, and batch data (history information) is retained on the Visa Clearing Exchange (VCX) History File before it is deleted during purge processing. Up to 100 days of history can be retained on the History File. It can be used in incoming and outgoing runs.

History Retention Period available options:

Days – Retain History File data for a minimum of 10 days (default) to a maximum of 100 days.

Note: Storage of history information is required by VCX in order to perform duplicate file and batch checking. It is also required for reruns of files previously processed by VCX.

Increasing the retention period increases the amount of disk space required for the History Files and may also increase processing time.

Number Decimal Separator

The Number Decimal Separator run control option specifies the decimal character to separate number fields on Visa Clearing Exchange (VCX) reports. It can be used in incoming and outgoing runs.

Separator (". " or ",")– The single character used to separate number fields on VCX reports.

Number Thousands Separator

The Number Thousands Separator run control option specifies the thousands separator character. It can be used in incoming and outgoing runs.

Separator (". " or ",")– The single character used to separate number fields on VCX reports.

Time Separator

The Time Separator specifies the character to separate time fields on Visa Clearing Exchange (VCX) reports. It can be used in incoming and outgoing runs.

Time Separator available options:

Separator (Any special character) – The single special character used to separate hours, minutes, and seconds on VCX reports. (A blank is acceptable but alphabetic or numeric characters such as A–Z or 0–9 are not acceptable.)

Time appears on VCX reports in HHMMSS format:

- HH = hours
- MM = minutes
- SS = seconds

Table Overrides

Visa Clearing Exchange (VCX) table files are used to validate transaction data during edit runs and data entry. They contain current data regarding ACQ/ISS ID, card account ranges, and validation edits. Updates to the table files are made daily and the changes placed on the Extended Access Server (EAS), Direct Exchange (DEX), or Visa File Gateway (VFG). During an incoming edit run, the updates are applied automatically to the original table files in each user's system. Table overrides can only be used by administrators. Administrators can access the table overrides page by going to **Run Options > Table Overrides**.

Table overrides are specific to a particular ACQ/ISS ID, account range (ARDEF), or manual entry.

Table overrides are optional and can be used to supersede entries in the original table files or add new ones. They can be used to add, change, or delete an original table entry. A table override does not change the entry in the original table file; instead, when a table override is invoked, an entry is made into a separate file. Table overrides simply override the entry in the original copy of the table file while keeping it intact.

To create a new table override, click **New** on the Table Override page.

Table Overrides

Add a New ACQ/ISS ID Table Override

Select **ACQ/ISS ID** when selecting a new table override type.

1. Add the ACQ/ISS ID you want to override.
2. Enter the BASE II CIB for that ACQ/ISS ID.
3. Select the country that the ACQ/ISS ID is located in from the drop-down list.
4. Select the region from the drop-down list.
5. Select the Acquirer Indicators from the drop-down list.
6. Select the ACQ/ISS ID Type from the drop-down list.
7. Select any additional attributes, if desired.
8. Click **Save** to save your ACQ/ISS ID table override.

Add a New ARDEF Table Override

Select **ARDEF** when selecting the new table override type.

1. Enter the High ARDEF range.
2. Enter the Low ARDEF range.
3. Enter the Issuer Identifier for the range that you would like to override.
4. Select whether you would like to use the **Check Digit Algorithm**.

5. Select the account length from the drop-down list.
6. Enter the BASE II CIB that you would like to apply the table override to.
7. Select the Domain from the drop-down list.

Important: Do not change Domain unless instructed to do so by Visa.

8. Select the Issuer ID region from the drop-down list.
9. Select the Issuer ID country from the drop-down list.
10. Select the ARDEF region from the drop-down list.
11. Select the ARDEF country from the drop-down list.
12. Select the Product ID from the drop-down list.
13. Select the Account Funding Source from the drop-down list.
14. Select any of the additional attributes, if desired.
15. Click **Save** to save your new ARDEF table override.

Add a New Manual Entry Table Override

Select **Manual Entry** when selecting the new table override type.

1. Enter the Table Key in the text box.
2. Enter the Table Data in the text box.
3. Click **Save** to save the new manual entry table override.

Chapter 12

Batch Procedure

The **Batch Procedure** tab of Visa Clearing Exchange (VCX) contains the **Batch Procedures** and **View Batch Logs** functions and is available to administrators and operators. The **Batch Procedures** function allows you to run or create batch procedures. The **View Batch Logs** function allows you to view the log after the batch procedure is run.

Batch Procedure

You can navigate to the Batch Procedure page by going to **Batch Procedure > Batch Procedures**. From there, you can view, edit, duplicate, or delete the current batch procedures that can be run or create a new batch procedure.

Important: You must set up your BASE II CIB, profile, and file transfer interface prior to creating a batch procedure.

Creating a New Batch Procedure

To create a new batch procedure, navigate to **Home > Batch Procedure > Batch Procedures** and click the **New Batch Procedure** button. The procedures differ based on whether the new

batch procedure is incoming or outgoing, as well as if it is a run, file transfer, or run and file transfer.

Batch Procedure Setup

[← Back to Batch procedures](#)

Batch procedure setup

Enter new batch procedure details. All fields are required unless marked optional.

General

Enter a name for your new batch procedure and select its job type. Batches can only be either incoming or outgoing.

Name (maximum: 50 characters)*

Job type*

Scheduler

☐ Generate script for scheduler

Reporting options

Select your preferred reporting format for this batch procedure. Then, enter an amount for many days to retain these logs.

☐ Table extract: create a data file a computer can automatically process

NEXT →

CANCEL

Create a New Incoming Run Batch Procedure

To create a new incoming run batch procedure, click **New Batch Procedure** from the Batch Procedure page.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a 50-character maximum for this field. Names with underscores are not allowed.
2. Select the **Incoming Run** job type from the drop-down list.
3. Select whether you would like to overwrite the incoming CTF.

4. Select the run mode.
5. Optionally, you can select the **Generate Script for Scheduler** option that generates a run script on Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time. If you select this option, you can then select the table effective date for the scheduler.
6. Select the reporting options.
7. Enter the logs retention period. The default is 30 days.
8. Select the continue return code from the drop-down list.
9. Check the 'Output Files in EBCDIC format' box to save the output files in both EBCDIC and ASCII formats.

Note:

- By default, output files created in ASCII format.
- This option is not available for Table Extracts or Reports.

10. Click **Next** to go to the BASE II CIB Selection page.
11. Select the BASE II CIBs to be used for the incoming run from the list of available CIBs.
12. Select the profile from the profile drop-down list for each BASE II CIB selected.

Note: When you select the BASE II CIB, you can see the default location from which input ITF files for an edit run will be taken. You may change the location of the input ITF file.

13. Click **Next** to go to the reporting options you selected; either Table Extract, Table Report, or both.

For Table Extract Option:

14. Select the Table Extracts you would like with the incoming run.
15. Click **Next**.

For Table Report Option:

16. Select the Table Reports you would like with the incoming run.
17. Click **Next**.
18. Review the information you have entered.

Note: The input location listed is the directory path from which input ITF files for an Edit Run will be taken. The output location listed is the output CTF location. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

19. Click **Save Batch Procedure** to save the incoming run batch procedure.

The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Create a New Incoming File Transfer Batch Procedure

To create a new incoming file transfer batch procedure, click **New Batch Procedure** from the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface, secure Direct Exchange (DEX) file transfer interface, or secure Visa File Gateway (VFG) file transfer interface configured to use an EAS, DEX, or VFG file transfer type.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a 50-character maximum for this field. Names with underscores are not allowed.
2. Select **Incoming File Transfer** from the job type drop-down list.
3. Select the file transfer type.
4. Select whether you're going to use custom file delivery.
5. Select the run mode.
6. Optionally, you can select the **Generate Script for Scheduler** option that generates a run script on Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time. If you select this option, you can then select the table effective date for the scheduler.
7. Select the reporting options.
8. Select the logs retention period. The default is 30 days.
9. Select the continue return code from the drop-down list.
10. Click **Next** to continue to the BASE II CIB selection page.
11. Select the BASE II CIBs to be used for the incoming file transfer from the list of available CIBs.
12. If you selected that you would use custom file delivery, click **Next** to continue to the Custom File Delivery page. If you selected **No**, continue to the File Transfer page.
13. Select the file type that will be created from the incoming file transfer.
14. Click **Add File Type** if more customization is required.
15. Click **Next** to continue to the File Transfer page.
16. Select the File Transfer Interface to be used for the incoming file transfer from the existing interfaces in the drop-down list. If you selected a secure EAS file transfer type, the list of existing EAS file interfaces is listed in the drop-down list. If you selected a secure DEX file transfer type or VFG file transfer type, you must select a DEX or VFG interface per CIB you selected.
Note: There must be a secure DEX interface or a secure VFG interface set up for the CIBs you selected. If not, set up a new secure DEX or VFG interface for those CIBs.
17. Click **Next** to continue to the reporting options pages you selected; either Table Extract, Table Report, or both.

For Table Extract Option:

18. Select the Table Extracts you would like with the incoming file transfer.

19. Click **Next**.

For Table Report Option:

20. Select the Table Reports you would like with the incoming file transfer.

21. Select whether you would like your report selections to be formatted or unformatted from the drop-down list.

22. Click **Next**.

23. Review the information you have entered.

Note: The location listed on the review page next to the BASE II CIB is the directory path where the output Interchange Transaction Files (ITFs) from a file transfer will be placed. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

24. Click **Save Batch Procedure** to save the incoming file transfer batch procedure.

The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Create a New File Transfer and Incoming Run Batch Procedure

To create a new file transfer and incoming run batch procedure, click **New Batch Procedure** on the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface, secure Direct Exchange (DEX) file transfer interface, or secure Visa File Gateway (VFG) file transfer interface configured to use an EAS, DEX, or VFG file transfer type.

1. Type the name for the new batch procedure in the **Name** field.

Note: There is a 50-character maximum for this field. Names with underscores are not allowed.

2. Select **File Transfer & Incoming Run** from the job type drop-down list.

3. Select your file transfer type.

4. Select whether you will overwrite your incoming Center Transaction File (CTF).

5. Select whether you will use custom file delivery.

6. Select your run mode.

7. Optionally, you can select the **Generate Script for Scheduler** option that generates a run script on Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time. If you select this option, you can then select the table effective date for the scheduler.

8. Select your reporting options.

9. Enter your logs retention period. The default is 30 days.

10. Select your continue return code from the drop-down list.

11. Click **Next** to continue to the BASE II CIB selection page.

12. Select the BASE II CIBs to be used for the incoming run and file transfer from the list of available CIBs.
13. Select the profile from the profile drop-down list for each BASE II CIB selected.
Note: When you select the BASE II CIB, you can see the default location from which input Interchange Transaction Files (ITF) files for an edit run will be taken. You may change the location of the input ITF file.
14. If you selected that you would use custom file delivery, click **Next** to continue to the Custom File Delivery page. If you selected **No**, continue to the File Transfer page.
15. Select the file type that will be created from the incoming file transfer.
16. Click **Add File Type** if more customization is required.
17. Click **Next** to continue to the File Transfer page.
18. Select the File Transfer Interface to be used for the incoming file transfer from the existing interfaces in the drop-down list. If you selected a secure EAS file transfer type, the list of existing EAS file interfaces is listed in the drop-down list. If you selected a secure DEX file transfer type, you must select a DEX interface per CIB you selected. If you selected a secure VFG file transfer interface type, you must select a VFG interface per CIB you selected.
Note: There must be a secure DEX interface or VFG interface set up for the CIBs you selected. If not, set up a new secure DEX interface or VFG interface for those CIBs.
19. Click **Next** to continue to the reporting options pages you selected, either Table Extract, Table Report, or both.

For Table Extract Option:

20. Select the Table Extracts you would like with the incoming run and file transfer.
21. Click **Next**.

For Table Report Option:

22. Select the Table Reports you would like with the incoming run and file transfer.
23. Click **Next**.
24. Review the information you have entered.

Note: The input file location listed on the review page is the directory path where ITFs from a file transfer will be placed and from which input ITF files for an edit run will be taken. The output file location is where the output Center Transaction Files (CTFs) can be found. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

25. Click **Save Batch Procedure** to save the incoming run and file transfer batch procedure.

The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Create a New Outgoing Run Batch Procedure

To create a new outgoing run batch procedure, click **New Batch Procedure** from the Batch Procedure page.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a maximum character limit of 50 for this field.
2. Select the **Outgoing Run** job type from the drop-down list.
3. Select the run mode.
4. Optionally, you can select the generate script for scheduler option which generates a run script on a Linux or Windows server so that a scheduler can call it to run a batch with specific date and time.
5. Select your preferred reporting format for the batch procedure, either Table Extract or Table Report, or both.
6. Enter the logs retention period. The default is 30 days.
7. Select the continue return code from the drop-down list.
8. Click **Next** to go to the BASE II CIB Selection page.
9. Select the BASE II CIBs to be used for the outgoing run from the list of available CIBs and indicate which profile and folder you'd like to store your data.
Note: When you select the BASE II CIB, you can see the default location from which input CTF files for an Edit Run will be taken. You may change the location of the input CTF file.
10. Click **Next** to go to the reporting options you selected, either Table Extract, Table Report, or both.

For Table Extract Option:

11. Select the Table Extracts you would like with the outgoing run.
12. Click **Next**.

For Table Report Option:

13. Select the Table Reports you would like with the outgoing run.
14. Click **Next**.
15. Review the information you have entered.

Note: The output ITF location is listed on the review page. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

16. Click **Save** to save the outgoing run batch procedure.

The batch procedure is saved to the list of batch procedures.

Create a New Outgoing File Transfer Batch Procedure

To create a new outgoing file transfer batch procedure, click **New Batch Procedure** from the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface, secure Direct Exchange (DEX) file transfer interface, or secure Visa File Gateway (VFG) file interface configured to use a file transfer batch procedure.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a maximum character limit of 50 for this field. Names should not contain any special characters except spaces.
2. Select **Outgoing File Transfer** from the job type drop-down list.
3. Select the file transfer type.
4. Select the run mode.
5. Optionally, you can select the generate script for scheduler option which generates a run script on a Linux or Windows server so that a scheduler can call it to run a batch with specific date and time.
6. Select the reporting options, either Table Extract, Table Report, or both.
7. Select the logs retention period. The default is 30 days.
8. Select the continue return code from the drop-down list.
9. Click **Next** to continue to the BASE II CIB selection page.
10. Select the BASE II CIBs to be used for the outgoing file transfer from the list of available CIBs.
11. Click **Next** to continue to the File Transfer page.
12. Select the File Transfer Interface to be used for the outgoing file transfer from the existing interfaces in the drop-down list. You should have an interface already set up for the selected BASE II CIB, otherwise you will receive an error. If you selected a secure EAS file transfer type, the list of existing EAS file interfaces is listed in the drop-down list. If you selected a secure DEX file transfer type, you must select a DEX interface per CIB you selected. If you selected a secure VFG file transfer type, you must select a VFG interface per CIB you selected.
13. Click **Next** to continue to the reporting options pages you selected, either Table Extract, Table Report, or both.

For Table Extract Option:

14. Select the Table Extracts you would like with the outgoing file transfer.
15. Click **Next**.

For Table Report Option:

16. Select the Table Reports you would like with the outgoing file transfer.
17. Click **Next**.

18. Review the information you have entered.

Note: The input file location listed on the review page is the directory path from which input ITF files will be taken for that CIB. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

19. Click **Save** to save the incoming file transfer batch procedure.

The new batch procedure is saved to the list of batch procedures.

Create a New Outgoing Run and File Transfer Batch Procedure

To create a new outgoing run and file transfer batch procedure, click **New Batch Procedure** from the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface, secure Direct Exchange (DEX) file transfer interface, or secure Visa File Gateway (VFG) file transfer interface configured to use an EAS, DEX, or VFG file transfer type.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a maximum character limit of 50 for this field. Names with special characters other than spaces are not allowed.
2. Select **Outgoing Run & File Transfer** from the job type drop-down list.
3. Select your file transfer type.
4. Select your run mode.
5. Optionally, you can select the generate script for scheduler option which generates a run script on a Linux or Windows server so that a scheduler can call it to run a batch with specific date and time.
6. Select your reporting options.
7. Enter your logs retention period. The default is 30 days.
8. Select your continue return code from the drop-down list.
9. Click **Next** to continue to the BASE II CIB selection page.
10. Select the BASE II CIBs to be used for the outgoing run and file transfer from the list of available CIBs.
11. Select the profile from the profile drop-down list for each BASE II CIB selected.
12. Click **Next** to continue to the File Transfer page.
13. Select the File Transfer Interface to be used for the outgoing file transfer from the existing interfaces in the drop-down list. You should have a file transfer interface already set up for the selected CIB, otherwise you will receive an error. If you selected a secure EAS file transfer type, the list of existing EAS file interfaces is listed in the drop-down list. If you selected a secure DEX file transfer type, you must select a DEX interface per CIB you selected. If you selected a secure VFG file transfer type, you must select a VFG interface per CIG you selected.
14. Click **Next** to continue to the reporting options pages you selected, either Table Extract, Table Report, or both.

For Table Extract Option:

15. Select the Table Extracts you would like with the outgoing run and file transfer.
16. Click **Next**.

For Table Report Option:

17. Select the Table Reports you would like with the outgoing run and file transfer.
18. Click **Next**.
19. Review the information you have entered.

Note: The location listed on the review page is the directory path from which input CTF files will be taken for a run. The output run and file transfer ITF files are listed in the output locations for the CIB. Additionally, the output table extracts and reports are found in the output locations listed on the review page.

20. Click **Save** to save the outgoing run and file transfer batch procedure.

The new batch procedure is saved to the list of batch procedures.

Create a New Table Reports/Extracts Only Batch Procedure

To create a new table reports/extracts only batch procedure, click **New Batch Procedure** from the Batch Procedure page.

Note: In VCX, CIB selection is not supported and files are saved in /vcx/tableReportsExtracts Folder.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a 50-character maximum for this field. Names with underscores are not allowed.
2. Select the **Table Reports/Extracts Only** job type from the drop-down list.
3. Optionally, you can select the **Generate Script for Scheduler** option that generates a run script on a Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time. If you select this option, you can then select the table effective date for the scheduler.
4. Select the reporting options.
5. Enter the logs retention period. The default is 30 days.
6. Select the continue return code from the drop-down list.
7. Click **Next** to go to the Table Extract or Table Report page, depending on which reporting options you selected.

For Table Extract Option:

8. Select the Table Extracts you would like to run.
9. Click **Next**.

For Table Report Option:

10. Select the Table Reports you would like to run.
11. Click **Next**.
12. Review the information you have entered.

Note: The output table extracts and reports are found in the output locations listed on the review page.

13. Click **Save Batch Procedure** to save the table reports/extracts only batch procedure.

The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Create a New Incoming Generic File Transfer Batch Procedure

To create a new incoming generic file transfer batch procedure, click **New Batch Procedure** on the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface, secure Direct Exchange (DEX) file transfer interface, or secure Visa File Gateway (VFG) file transfer interface configured to use an EAS, DEX, or VFG file transfer type.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a 50-character maximum for this field. Names with underscores are not allowed.
2. Select **Incoming Generic File Transfer** from the job type drop-down list.
3. Select the File Transfer Type as **Extended Access Server (EAS)** from the drop-down list.
4. Select whether you would like to overwrite the incoming CTF.
5. Select the run mode.
6. Optionally, you can select the Generate Script for Scheduler option that generates a run script on Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time.
7. Enter the logs retention period. The default is 30 days.
8. Select the continue return code from the drop-down list.
9. Click **Next** to the File Transfer page.
10. Select the File Transfer Interface to be used for the incoming generic file transfer from the existing EAS interfaces in the drop-down list.
11. Click **Next**.
12. Review the information you have entered.
Note: The Incoming Generic Files will be saved in the location /opt/app/vcx/datafiles/IGF
13. Click **Save Batch Procedure** to save the incoming generic file transfer batch procedure. The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Create a New Generate Inventory File Batch Procedure

To create a new generate inventory file batch procedure, click **New Batch Procedure** on the Batch Procedure page.

You must have a secure Extended Access Server (EAS) file transfer interface.

1. Type the name for the new batch procedure in the **Name** field.
Note: There is a 50-character maximum for this field. Names with underscores are not allowed.
2. Select **Generate Inventory File** from the job type drop-down list.
3. Select the File Transfer Type as **Extended Access Server (EAS)** from the drop-down list.
4. Optionally, you can select the Generate Script for Scheduler option that generates a run script on Linux or Windows server so that a scheduler can call it to run a batch with a specific date and time.
5. Enter the logs retention period. The default is 30 days.
6. Select the continue return code from the drop-down list.
7. Click **Next** to the File Transfer page.
8. Select the File Transfer Interface to be used for the generate inventory file batch procedure from the existing EAS interfaces in the drop-down list.
9. Click **Next**.
10. Review the information you have entered.
Note: The Generate Inventory File will be saved in the location <folder>/vcx/ (for example - on Linux its /opt/app/vcx/ & Windows its C:\\vcx)
11. Click **Save Batch Procedure** to save the generate inventory file batch procedure. The new batch procedure is saved to the list of batch procedures. A green bar appears with a message stating that the batch procedure has been saved successfully.

Running a Batch Procedure

To run a batch procedure, navigate to **Batch Procedure > Batch Procedures**.

1. Click **Run** for your preferred batch procedure.
Note: Only one batch procedure can be selected.
2. Click **Next**.
3. Select the optional parameters for this batch run. If the input file is more than 90 days old, select today's date.

4. If you want to rerun the batch run, select **Run previously processed ITF**, and then select the Run Date that the ITF was processed on.
5. Click **Run Batch Procedure** to start the run.

Note:

- The database must be promptly updated after each incoming RUN to ensure the latest changes are applied. For instructions on updating the database, please see the 'Using the Scheduler' section.
- If you cancel a batch job while it is in progress, it may cause errors in future batch runs because the current job might still be holding the resources it was using.
- Ensure there is sufficient hard disk space available before starting a batch run. Insufficient disk space will prevent the application from processing the batch run successfully.
- Processed ITFs will remain in the same folder.

Run Batch Procedure

[← Back to all Batch procedures](#)

Run batch procedure

Run the selected batch procedure or add additional parameters to customize this run.

Batch job
[rtfidsrls](#)

Job type
Incoming Run

Mode
TEST

Table report
Yes

Table extract
Yes

Created by
[User]

Last modified
02/13/2024 | 17:37

Optional parameters

Enter the information below to start an incoming batch run with alternative parameters than above.

Table effective date (MM/DD/YYYY)

☐ Run previously processed ITF

Run Date (MM/DD/YYYY)

☒ If input file is more than 90 days old please select today's date

☐ Stay on this page while the batch is running.

Run a Previously Processed File

The Run a Previously Processed File (Rerun) option indicates that a rerun is to occur. It can be used in incoming and outgoing runs.

A processing center can select these options:

- Not Selected (default) – A rerun will not occur.
- Selected (Outgoing Only) – A Run Number and Run Date need to be specified. Processing for the processing day will start over from run number.
- Selected (Incoming Only) – All the files in the run that have been previously processed.

Rerun – Outgoing Processing

The combination of the Run Number and the Visa Clearing Exchange (VCX) processing date (Run Date) identifies the run to be reprocessed. The run number is a three-digit numeric field previously assigned by VCX during the original run. VCX verifies that the run number is present in the History File for the processing date.

Within a given day, outgoing VCX allows previous runs to be rerun. VCX history data is cleared out from the point where the rerun begins based on the Run Type value. Subsequent runs are treated as original runs and do not require the use of the Rerun run control option record.

When the ALL parameter is used, all outgoing history records for the VCX processing date in effect are deleted from the History File without regard to Run Type. The beginning batch file and run number are set to 1.

Rerun – Incoming Processing

Reruns are allowed for any file that was previously processed by Visa Clearing Exchange (VCX) and still exists in the History File. If the rerun flag is not set and the Interchange Transaction File (ITF) header record matches the History File, then the run will abort unless the file is a nonfinancial file.

If the Central Processing Date (CPD) in the ITF header record is not the current system date, then the Run Date run control option must be used in order to specify the correct CPD.

Run Date

The Run Date processing option defines the Visa Clearing Exchange (VCX) processing date. It can be used in incoming and outgoing runs.

Run Date options:

- date nnn
 - Outgoing Only – The VCX processing date to be used to create an outgoing Interchange Transaction File (ITF),
 - Incoming Only – Specifies the TC 90 File Header record CPD in order to verify that the correct file is being processed.

The date is in MM/DD/YYYY format.

For outgoing runs, the Run Date run control option defines the VCX processing date, overriding the system date default.

For incoming runs, the Run Date run control option can only be used for rerun processing. It must match the Central Processing Date (CPD) on the TC 90 header record of the output being processed.

The run control option should only be used for rerun processing or testing.

Note: The specified run date cannot be greater than the current date for production runs or more than 44 days prior to the current date. When Run Type Run Control Option is Test, the run date can be greater than the current date to allow future dated testing.

If the Table Date run control option is specified, it will always be used as the table effective date. If Table Date is not specified, then the Run Date will be used as the table effective date.

View Batch Jobs

You can navigate to the **View Batch Jobs** page by going from **Batch Procedure > View Batch Jobs** from the navigation bar. From there, you can view archived jobs, filter by name, filter by start and end date, as well as view details for a specific job.

By toggling **View Archived Jobs**, you are able to view the jobs previously archived in Visa Clearing Exchange (VCX). Batch jobs are automatically archived by a scheduler. The default timeframe is 7 days. See *Configurable Fields in the VCX Config File* to modify the parameter.

Clicking **View Details** for a specific job will trigger a pop-up window showing the reports and batch log for that batch run. To print the report or log, click **Print**.

Schedule a Batch Job in Linux

This procedure describes how you can schedule a Visa Clearing Exchange (VCX) batch job under a Cron Table in Linux.

1. To see what's in the cron table existing on the system, run **crontab -l**.
You will see your cron table.
2. Edit your cron table by typing **crontab -e**.
This opens the cron jobs in edit mode.
3. You can add jobs after the comment out lines end.

Added Batch Job using CronTab

```
#####  
#Added by UNIX Engineering-avoid ldap lookups for local users  
#Do not remove  
0 0 * * * /usr/local/bin/update_SSSD_user_group_v2.0.sh 2>&1  
#####  
*/2 * * * * /opt/app/vcx/batch/IncomingRun/scheduler/scheduledBatch.sh  
~
```

The syntax of this is very important. This shows what each asterisk represents.

Syntax

```
* * * * * [path to the script/command to execute]  
- - - - -  
| | | | |  
| | | | | ----- day of week (0 - 7)  
| | | | | ----- month (1 - 12)  
| | | | | ----- day of month (1 - 31)  
| | | | | ----- hour (0 - 23)  
| | | | | ----- minute (0 - 59)
```

4. In the VCX user interface, when you select **Generate Script for Scheduler** while creating a Batch Procedure, VCX will create a scheduler folder under the batch procedure name that contains the `schedulerBatch.sh` file. Use this file in the crontab and schedule the job using the proper syntax.

Note: This script file was created by the web service account as owner. It has executable permission for this owner only. If you want to run the script manually, you either need an Administrator account, or to change permissions to allow another account to run it.

For example, a batch procedure with the name "FTP and Incoming" will be located in the path `/opt/app/vcx/batch/FTP_and_Incoming/scheduler`.

Schedule a Batch Job in Windows

This procedure describes how to schedule a batch job in Visa Clearing Exchange (VCX) on Windows.

1. Open Command Prompt as an Administrator.
2. Use the **schtasks** command to create a new task.

```
schtasks /create /sc MINUTE /mo 1 /tn "scheduledBatch" /tr  
"C:\path\to\your\script.bat"
```

The breakdown of the command is:

- **/create** creates a new scheduled task
- **/sc MINUTE /mo 1** sets the schedule for the task to run every minute. Adjust this interval to fit your needs.
- **/tn "scheduledBatch"** sets the name of the task to "MyTask". You can choose a different name if preferred.
- **/tr "C:\path\to\your\script.bat"** sets the task to run the script located at "C:\path\to\your\script.bat". Replace this with the path to your .bat file.

A new task is scheduled and will run at the set interval. If you want to check the status of the task, use the **/query** option with **schtasks**, like

```
schtasks /query /tn "MyTask"
```

This will provide information about the task named "MyTask," such as its next run time and status.

EXAMPLE

Run a task every day at 3 PM:

```
schtasks /create /sc DAILY /st 15:00 /tn "MyDailyTask" /tr  
"C:\path\to\your\script.bat"
```

Run a task every Monday at 3 PM:

```
schtasks /create /sc WEEKLY /d MON /st 15:00 /tn  
"MyWeeklyTask" /tr "C:\path\to\your\script.bat"
```

Run a task every 5 minutes:

```
schtasks /create /sc MINUTE /mo 5 /tn "MyFrequentTask" /tr  
"C:\path\to\your\script.bat"
```

Run the Scripted Batch Job

This describes how you can execute a batch procedure from the console. You must have access to the server and have sufficient privileges to execute scheduler files generated by Visa Clearing Exchange (VCX). This procedure describes how to run a scripted batch job without scheduling it in the user interface for VCX. The script is generated in the same location under the batch procedure name. For example, the script for running a batch procedure called "FTP and Incoming" will be found in the file path `opt/app/vcx/batch/FTP_and_Incoming`.

1. Navigate to the folder where the application is installed, for example, `cd /opt/app/vcx`.

Application Location

```
vcx]$ ls  
batch  Collected_logs  copyTestFiles.sh  database  efs  ftps  history  logs  startWebserver.sh  tableReportsExtracts  
certs  confio             cron              datafiles  ep   ftpsLogs  jers     package  stopWebServer.sh  testFiles
```

2. Navigate to the scheduler file folder (`$applicationFolder/batch/$BatchProcedureName/scheduler/`), for example `cd /opt/app/vcx/batch/<BatchProcedureName>/scheduler/`

Scheduler Folder

```
vcx]$ cd batch/commandLineExecution/  
commandLineExecution]$ ls  
scheduler  
commandLineExecution]$
```

3. Execute `./scheduler.sh` on Linux or `./scheduler.bat` on Windows.

Execute Script

```
> .\scheduledbatch.bat  
% Total    % Received % Xferd  Average Speed   Time    Time       Time  Current  
           100    105    0     0      22      0  0:00:04  0:00:04  --:--:--   23  
Status: 0  
> |
```


If the run is successful, the output files and reports will be generated in a subfolder.

Note: Scheduler output will be the return code from the batch procedure execution. If there are multiple jobs or CIBs in the scheduled batch procedure, the output return code will be the highest value of all the return codes.

Stop Running a Cron Job in Linux

There are two options for stopping a Visa Clearing Exchange (VCX) job from running in Linux.

1. Run the command **crontab -e** and remove the job from the Cron Table.
2. Put **#** before the job to comment it out temporarily.

Stop a Task Running in Windows

To stop or delete a task you have created, you need to run commands in Command Prompt or PowerShell with Administrator privileges. You can use the **/end** or **/delete** options with **schtasks** command. Replace "MyTask" with the task you want to stop or delete.

1. End a running task using the **/end** option:

```
schtasks /end /tn "MyTask"
```

This stops the task named "MyTask" if it is currently running.

2. Delete a task using the **/delete** option:

```
schtasks /delete /tn "MyTask" /f
```

This deletes the task named "MyTask". The **/f** option is used to force the command to run without asking for user confirmation.

Chapter 13

Tools

The **Tools** tab of Visa Clearing Exchange (VCX) gives operators and administrators the ability to initialize history files, view and purge files via the EFS tool, and view job and application logs via the job collector.

EFS Tool

The Extended Access Server (EAS) File Services (EFS) tool for BASE II files is used to view Ready Files (incoming files that are available to download from EAS to the VCX Application Server) or Purge Files under a specific EAS profile and BASE II CIB. The EFS tool can also be used to cancel Ready Files or restore Purge Files. If a cancel Ready Files or restore Purge Files request fails, the errors can be viewed in the FTP log. The Incoming File ID will be the same as EAS File ID.

EFS tool

EFS tool for secure EAS

Date format - MM/DD/YYYY
GMT date / time - 06/28/2025 21:55

Use the Extended Access Server (EAS) File Services (EFS) tool to view ready or purge files under a specific EAS profile and BASE II CIB.

Use the table below to view, cancel or restore files.

BASE II files

Generic files

EAS profile*

Base II CIB*

File status*

☐ Ready Files

☐ Purge Files

LIST FILES

CANCEL READY FILES

RESTORE PURGE FILES

VIEW FTP LOG

☐ File num ▾

Status ▾

Batch ▾

Date ▾

BASE II CIB ▾

Run mode ▾

Pullkey ▾

File name ▾

No records found.

Items per page

10 ▾

1 - 10 of 0

How to List Files:

1. Select the EAS profile.
2. Select the BASE II CIB from the drop-down list.
3. Click whether you want to list **Ready Files** or **Purge Files**.
4. Click **List Files**.

Note: Data can be sorted in ascending or descending order for each column.

How to Cancel Ready Files:

5. After listing Ready files, select the files you would like to cancel.
6. Click **Cancel Ready Files**.
This changes the status of the files from Ready to Purge.

How to Restore Purge Files:

7. After listing purge files, select the files you would like to restore.
8. Click **Restore Purge Files**.

EFS Tool for Generic Files

The Extended Access Server (EAS) File Services (EFS) tool for Generic Files is used to view Ready Files (incoming files that are available to download from EAS to the VCX Application Server) or Purge Files under a specific EAS profile. The EFS tool for Generic Files can also be used to cancel Ready Files. If a cancel Ready Files request fails, the errors can be viewed in the FTP log.

EFS tool for generic files

The screenshot shows the 'EFS tool for secure EAS' web interface. At the top, it says 'Use the Extended Access Server (EAS) File Services (EFS) tool to view ready or purge files under a specific EAS profile and BASE II CDB.' and 'Use the table below to view, cancel or restore files.' There are two tabs: 'BASE II files' and 'Generic files', with 'Generic files' selected. Below the tabs, there is a dropdown menu for 'EAS profile' and a radio button group for 'File status' with 'Ready Files' selected. A toolbar contains four buttons: 'LIST FILES', 'CANCEL READY FILES', 'RESTORE PURGE FILES', and 'VIEW FTP LOG'. To the right of these buttons is a search bar with a magnifying glass icon and the word 'Search'. Below the toolbar is a table header with columns: 'File name', 'Status', 'Processor ID', 'Run mode', 'Priority', and 'File name'. The table body is empty, and a message 'No records found.' is displayed at the bottom.

How to List Files:

1. Select the EAS profile.
2. Click **List Files**.

View Application Log

The **View Application Log** tool lists all of the actions completed by the user in Visa Clearing Exchange (VCX) as well as their timestamps.

VCX History Files

The Visa Clearing Exchange (VCX) History Files page allows you to run the History Report or initialize the history files.

To run a History Report:

1. From the Report tab on the History Files page, select the BASE II CIB you want to report for.
2. Select the reporting direction you 'd like to run the report for (incoming, outgoing, or incoming and outgoing).
3. Select the reporting level.
4. Select the reporting period's start (either start of file or a specific date) and end (either end of file or a specific date).
5. Click **View Report**.

To Initialize history

6. Select the BASE II CIB from the drop-down list that you want to initialize the history files for.
7. Click **Submit**.

Log Collector

The log collector collects the logs for a batch procedure ran on a certain date. You can then download said collection of logs.

Collecting Logs

1. In the navigation bar at the top of the page, click on **Tools** and choose **Log Collector** from the menu.
2. In the **Batch Procedure** drop-down list, select the name of the procedure you want to run.
3. In the **Date** field, choose the date when the batch procedure was run.
4. Click the **Submit** button to start collecting logs.

Downloading the Log File

5. Once the log collection is complete.
6. When the file status shows **Completed**, click the **Download** button to save the log file to your computer.

Appendix A

VCX File Layouts

Record layout and pertinent data for each of the following Visa Clearing Exchange (VCX) files are detailed. Users of VCX for Windows will find that the records are standard ASCII format, have no trailing spaces, and each record is terminated with an ASCII carriage return/line feed character. Users of VCX for Linux will find that the Linux records are standard ASCII format and have no trailing spaces. Each Linux record is terminated with an ASCII line feed character.

- Currency Conversion Rates File
- History File
- History Backup File
- TC 33 Extract File
- Rejected Item File
- Returned Item File
- VCX Table Files
 - Account Range Definition (ARDEF) File
 - VID Table File
- ARDEF Extract File

- ACQ/ISS ID Extract File
- Value Table Extract Files

Note: The formats for the file layouts can be numeric (N), alphanumeric (AN), or unpacked numeric (UN).

Currency Conversion Rates File Layout

The Currency Conversion Rates File contains all valid BASE II Counter and Base currency codes, their effective date, Buy and Sell scale factors, and Buy and Sell Conversion Rates. This file is updated by TC 56 currency rate update transactions during Incoming Process.

Important: The contents of the Visa Currency Conversion Rate file (i.e., Rate Update Records or TC 56 file) are Confidential and may be used only by Visa and its authorized client financial institutions. Any other use is strictly prohibited unless expressly authorized by Visa.

Record Length:

80

File Structure:

Contains only one record type, namely, the currency conversion rates record. There are no header or trailer records.

Currency Conversion Rates Record Layout

Position	Length	Format	Field Name	Description
01-03	03	UN	Counter Currency Code	This field will contain the ISO numeric currency code of the Counter Currency.
04-06	03	UN	Base Currency Code	This field will contain the ISO numeric currency code of the Base Currency.
07-10	04	UN	Effective Date	The format of this field is MMDD where: MM = 01-12 DD = 01-31

Currency Conversion Rates Record Layout

Position	Length	Format	Field Name	Description
11–12	02	UN	Buy Scale Factor	Number of decimal places implied with the Buy Conversion rate. For example, a scale factor of 00(zero) indicates the conversion rate is an integer (NNNNNN) with no decimal points, while a scale factor of 02 indicates the conversion rate has 2 decimal points (NNNN.NN). The field must be 00 through 99.
13–18	06	UN	Buy Conversion Rate	Base Currency equivalent to buy one unit of Counter Currency.
19–20	02	UN	Sell Scale Factor	Number of decimal places implied with the Sell Conversion rate. For example, a scale factor of 00(zero) indicates the conversion rate is an integer (NNNNNN) with no decimal points, while a scale factor of 02 indicates the conversion rate has 2 decimal points (NNNN.NN). The field must be 00 through 99.
21–26	06	UN	Sell Conversion Rate	Base Currency equivalent to Sell one unit of Counter Currency.
27–31	05	AN	Rate Table ID	Rate Table ID The format will be ANNNN, (a leading 1-byte alphabetic character A-Z, followed by a 4-byte numeric value.)
32–80	49	AN	Reserved	Reserved for future use; set to spaces.

History File Layout

This file contains a history of batches, files, and runs for outgoing and incoming Visa Clearing Exchange (VCX) processing.

The file has five record types:

- Header Record (Record Type HH)
This record controls reorganization of the history file. The History Header record appears once, and is the first record of the file.
- Incoming Data Run Summary Record (Record Type ID)
Run Summary Record—This incoming data record summarizes incoming run information. There is one Run Summary Record for each incoming run.

- Incoming Data File Summary Record (Record Type IF)
File Summary Record—This incoming data record summarizes incoming file information. There is one File Summary Record for each incoming file.
- Outgoing Data Run Summary Record (Record Type OD)
Run Summary Record—This outgoing data record summarizes outgoing run information. There is one Run Summary Record for each outgoing run.
- Outgoing Data File Summary Record (Record Type OF)
File Summary Record—This outgoing data record summarizes outgoing file information. There is one File Summary Record for each outgoing file.
- Outgoing Unique Batch Record (Record Type OU)
These records summarize batch information. There is one Outgoing Unique Batch record for each outgoing batch.

Record Length and Format:

252

File Structure:

Includes a header record and detail records. A trailer record is not associated with this file.
The header record in this table determines when the file is reorganized.

History Header (HH) Record Layout

Position	Length	Format	Field Name	Description
01-02	02	AN	Record Type	HH = Header record
03-10	08	UN	Processing Date	Contains zeros.
11	01	AN	Mode	Set to spaces.
12-14	03	UN	Run Number	Contains zeros.
15-17	03	UN	File Number	Contains zeros.
18-25	08	UN	Reorg Date	Date that VCX was last reorganized. The history file is in CCYYMMDD format.
26-29	04	AN	Status	GOOD = file is not corrupted BAD = file is corrupted
30-252	223	AN	Reserved	Set to spaces.

The history incoming and outgoing run summary records in this table summarize incoming and outgoing run information.

History File Incoming and Outgoing Data Run Summary Record (ID, OD)

Position	Length	Format	Field Name	Description
01-02	02	AN	Record Type ²	ID = Incoming Data OD = Outgoing Data
03-10	08	UN	Processing Date ²	Processing date of the run in CCYYMMDD format. Incoming = BASE II Central Processing Date Outgoing = VCX Processing Date
11	01	AN	Mode ²	P = Production T = Test
12-14	03	UN	Run Number ²	Sequential number of the run within processing date.
15-17	03	UN	File Number ²	This field contains zeros for run summary records.
18-25	08	UN	Run Start Date	System date at the start of the run expressed in CCYYMMDD format.
26-31	06	UN	Run Start Time	System time at the start of the run expressed in HHMMSS format.
32-37	06	UN	Last ITF Batch Number	The last ITF batch number used for the run.
38-43	06	UN	Last Center Batch Number	Incoming = Zeros Outgoing = The last Center Batch Number used for the run
44-46	03	UN	Last File Number	The last file number used for the run.
47	01	AN	Rerun Flag	Space = No rerun R = Rerun was performed
48-49	02	UN	Number of Reruns	00-99 = Number of reruns performed
50-252	203	AN	Reserved	Set to spaces.

The history incoming and outgoing file summary records in the following table summarize incoming and outgoing file information.

History File Incoming Data and Outgoing Data File Summary Record (Record Type IF, OF)

Position	Length	Format	Field Name	Description
01-02	02	AN	Record Type ²	IF = Incoming Data OF = Outgoing Data
03-10	08	UN	Processing Date ²	Processing date of the run in CCYYMMDD format. Incoming = BASE II Central Processing Date Outgoing = VCX Processing Date
11	01	AN	Mode ²	P = Production T = Test
12-14	03	UN	Run Number ²	Sequential number of the run within processing date.
15-17	03	UN	File Number ²	File number assigned by VCX. Incoming = ITF file numbers are assigned sequentially starting from one for each VCX run Outgoing = CTF file numbers are assigned sequentially starting from one for each VCX processing day
18-19	02	AN	Collection-Only Indicator	CO = Collection-Only File Space = Financial and Non-Financial File
20-49	30	AN	Unique File ID	Assigned by Visa.
50-54	05	UN	CTF Processing Date	Incoming = Processing date from the ITF header in YYDDD format Outgoing = Processing date from the CTF header in YYDDD format, or zeros
55-59	05	AN	BASE II File Type	Incoming = Assigned by Visa. Valid values: ICRPT Interchange Report File NONIC Non-Interchange Report File Spaces Interchange Data Outgoing = Spaces

History File Incoming Data and Outgoing Data File Summary Record (Record Type IF, OF)

Position	Length	Format	Field Name	Description
60–62	03	UN	CTF ID	Incoming = Zeros Outgoing = Client-defined in outgoing CTF TC 90 record (Outgoing File ID), or zeros
63–77	15	UN	File Summary Destination Amount	Incoming = Total destination amount for the file Two-digit decimal place is implied Outgoing = Zeros
78–86	09	UN	File Summary Monetary Transaction Count	Number of monetary transactions in the file.
87–96	10	UN	File Summary TCR Count	Number of TCRs in the file.
97–102	06	UN	File Summary Block Count	Number of blocks in the file.
103–108	06	UN	File Summary Batch Count	Number of batches in the file.
109–117	09	UN	File Summary Transaction Count	Number of transactions in the file.
118–132	15	UN	File Summary Source Amount	Total source amount for the file. Two-digit decimal place is implied.
133–138	06	UN	File Starting Batch Number	Starting batch number for the file.
139	01	AN	File Continuation Indicator	Incoming C = continuation of transmission space = no continuation Outgoing = Space
140–144	05	AN	BASE II Customized Delivery File Type	Customized Delivery file type from the ITF TC 90 header record. (Refer to the BASE II Clearing Interchange Formats manuals for valid values.)
145	01	AN	File Financial Flag	Y = Indicates file contains financial transactions N = Indicates file does not contain financial transactions

History File Incoming Data and Outgoing Data File Summary Record (Record Type IF, OF)

Position	Length	Format	Field Name	Description
146–153	8	AN	First Central Processing Date	Incoming = Central Processing Date of the first file processed during the run Outgoing = Spaces
154–252	99	AN	Reserved	Set to spaces.

The history outgoing unique batch record in this table contains control data for outgoing interchange at the batch level.

History File Outgoing Unique Batch Record (Record Type OU)

Position	Length	Format	Field Name	Description
01–02	02	AN	Record Type ²	OU = Outgoing Unique Batch
03–32	30	AN	Unique File ID ²	Assigned by Visa.
33–38	06	UN	ITF Batch Number ²	Batch number assigned by Visa to the outgoing ITF batch.
39	01	AN	Mode ²	Run Control Option UNMODE P = Production T = Test
40–45	06	UN	CTF Batch Number	CTF Batch Number from Batch Trailer and zero if it is a Visa Batch at file level.
46	01	AN	Batch Status	A = Accepted by the VIC R = Rejected by the VIC Space = No TC 44 acknowledgment has been received from the VIC
47–61	15	UN	Batch Destination Amount	Total destination amount for the batch. Two decimal places are implied.
62–70	09	UN	Batch Summary Monetary Transaction Count	Number of monetary transactions in the batch.
71–80	10	UN	Batch Summary TCR Count	Number of TCRs in the batch.
81–86	06	UN	Batch Summary Block Count	Number of blocks in the batch.
87–94	08	UN	Center Batch ID	Center Batch ID from Batch Trailer.

History File Outgoing Unique Batch Record (Record Type OU)

Position	Length	Format	Field Name	Description
95–103	09	UN	Batch Transaction Count	Accepted Number of transactions (TC/TT) in the batch that excludes control records.
104–118	15	UN	Batch Source Amount	Computed Total source amount for the batch for accepted transactions. Two decimal places are implied.
119	01	AN	Batch Hash Flag	Space or N = Batch no hashed Y = Batch hashed
120–151	32	AN	Batch Hash Total	Hash value for the batch assigned by Visa.
152	01	AN	Visa-Generated Batch Flag	N = Batch generated by user
153–154	02	AN	Record Type	OF = Outgoing Data
155–162	08	UN	Processing Date	Processing date from in CCYYMMDD format.
163	01	AN	Mode	P = Production T = Test
164–166	03	UN	Run Number	Sequential number of the run within processing date.
167–169	03	UN	File Number	Sequential number of the file within processing date.
170–171	02	AN	Collection-Only Indicator	CO = Collect Only File Space = Financial and Non-Financial File
172–252	81	AN	Reserved	Set to spaces.

History Backup File Layout

Serves as a backup file for the History File. There are two History Back up files:

First backup (backup-a.dat): It is overlaid by new History file at the end of every successful outgoing and incoming run. This file is used to restore the History File in case of an error on the keyed file.

Second backup (backup-b.dat): It is a copy of the History file taken before the start of an outgoing and incoming run.

Record Length:

252

File Structure:

It has the same file structure as the History file.

TC 33 Extract File Layout

This file is only used at processing centers that subscribe to receive information from Visa that is delivered on designated TC 33 records. Information currently delivered using this technique includes:

Information currently delivered using this technique includes:

- Plus Identifier File
- Combined Visa/Plus Routing Table
- Visa Transaction Routing tables
- Interlink Identifier tables
- Visa Electron Routing Table
- Mastercard/CIRRUS Routing Table
- CIRRUS-Only Routing Table
- AFFN Routing Table
- VisaVue Solution Series Data
- ABA Table File
- POS Acceptance Types File
- ATM No Surcharge Routing File

Processing centers that receive more than one type of data in this file will have to use a client-written program to separate the information.

Record Length:

168

File Structure:

Visa Clearing Exchange (VCX) writes the entire TC 33 record in CTF format to this file. See *BASE II Clearing Interchange Formats* manual and to documentation for the services subscribed to by your center for more information.

Rejected Item File Layout

This file is created during outgoing Visa Clearing Exchange (VCX) runs. It will contain each transaction that is rejected (not written to the ITF) due to errors found during editing. A processing center must use the save rejected items run control option to retain rejected transactions.

With this file, processing centers can correct and resubmit invalid transactions to the edit process without re-keying them. Correction is done either in the processing center's pre-edit system or via VCX.

Record Length:

168, Fixed Blocked for TC format

File Structure:

The file is in a CTF format including:

- A file header (if the processing center has supplied a CTF file header).
- An individual rejected transactions.
- A batch trailer following each batch submitted by the processing center in which there were errors.
- A file trailer at the end of the file.

If the save rejected items run control option is used with a value of R, the transaction in the Rejected Item File will include a TCR 9 containing the validation message numbers indicating what caused the transaction to reject.

See *BASE II Clearing: Interchange Formats, TC 01 to TC 49* and *BASE II Clearing: Interchange Formats, TC 50 to TC 92* for the layout of other CTF file records.

Rejected Item File TCR 9 Record

Position	Length	Format	Field Name	Description
01-02	02	AN	Transaction Code	BASE II transaction code.
03	01	UN	Transaction Code Qualifier	0, 1, or 2
04	01	UN	Transaction Component Sequence Number	9
05-10	06	UN	Destination Identifier	BASE II Destination Identifier
11-16	06	UN	Source Identifier	BASE II Source Identifier

Rejected Item File TCR 9 Record

Position	Length	Format	Field Name	Description
17–20	04	AN	Reserved	Set to spaces.
21–25	05	AN	Run Date	Date the transaction was processed, in YYDDD format.
26–31	06	UN	Batch Number	Batch number in which the transaction was located.
32–35	04	UN	Batch Sequence	Sequence number of the rejected record within the batch.
36–38	03	AN	Reserved	Set to spaces.
39–50	12	UN	Source Amount	Source amount of the rejected transaction.
51–53	03	UN	Source Currency	Source currency code of the rejected transaction.
54	01	AN	Settlement Flag	Settlement flag of the rejected transaction.
55–67	13	AN	Reserved	Set to spaces.
68–71	04	AN	Validation Message Code 1	1st error Validation message number causing the transaction to be rejected
72–75	04	AN	Validation Message Code 2	2nd error Validation message number causing the transaction to be rejected.
76–79	04	AN	Validation Message Code 3	3rd error Validation message number causing the transaction to be rejected.
80–83	04	AN	Validation Message Code 4	4th error Validation message number causing the transaction to be rejected.
84–87	04	AN	Validation Message Code 5	5th error Validation message number causing the transaction to be rejected.
88–91	04	AN	Validation Message Code 6	6th error Validation message number causing the transaction to be rejected.
92–95	04	AN	Validation Message Code 7	7th error Validation message number causing the transaction to be rejected.
96–99	04	AN	Validation message Code 8	8th error Validation message number causing the transaction to be rejected.

Rejected Item File TCR 9 Record

Position	Length	Format	Field Name	Description
100–103	04	AN	Validation Message Code 9	9th error Validation message number causing the transaction to be rejected.
104–107	04	AN	Validation Message Code 10	10th error Validation message number causing the transaction to be rejected.
108–168	61	AN	Reserved	Set to spaces.

Returned Item File Layout

This file, which is created during incoming Visa Clearing Exchange (VCX) runs, contains each transaction that is returned by Visa as a result of editing at the VIC. With this file, processing centers can correct and resubmit returned transactions to the edit process without rekeying them.

The contents of the file depend on the save return items run control option used. The file is always written in CTF format.

For TC formats, if the save return items run control option is used with a value of R, transactions retain the transaction codes returned by the VIC (TC 01, TC 02, and TC 03). The TCR 9 added by the VIC is included in the file. If the save return values run control option is used with a value of T, the transaction code from the TCR 9 is restored and the TCR 9 is not included in the file.

Record Length:

168

File Structure:

The file is in a CTF format including:

- a file header (if the user has chosen to create file headers on incoming CTFs)
- individual returned items
- batch trailers
- a file trailer at the end of the file

See *BASE II Clearing: Interchange Formats, TC 01 to TC 49* and *BASE II Clearing: Interchange Formats TC50 to TC 92* for layouts of CTF file records. The layout of the TCR 9 generated by the VIC is different from the layout of the TCR 9 generated by VCX for rejected items. See the section for TC 01, TC 02, and TC 03 for returned item TCR 9 layout.

Repository Value Table File Layout

Visa Clearing Exchange (VCX) Repository Value table files control transaction editing and report generation.

Usage:

The files contain the following types of information:

- Transaction mapping tables
- Value tables
 - BAI Data
 - Country Code
 - Currency Code
 - Merchant Category Code
 - Fee Reason Code
 - Holiday
 - Product ID
 - Product Sub Type
 - Return Item Reason Code
 - Regulated ARDEF

File Structure:

Each table contains a table header record, comment records, and one or more table detail records.

The first 29 positions of any record are referred to as the common prefix. This prefix, except for the last position (the delete indicator), is the key for the record. The last field of the key contains a table effective date. Not only does this technique assign a unique key to each record, it also permits multiple future and past logical images of each file to be maintained in each physical file. Remaining portions of a record may vary from table to table and from entry to entry.

Table Records:

Table records contain values (for system data tables) unique to a file table. Each record can have multiple occurrences depending on the number of effective dates.

Repository Table File – Account Range Definition (ARDEF) Table Layout

The repository file contains the ARDEF Table where ARDEF values are defined.

Record Format and Length:

Record Format = Fixed length, maximum record length = 200

File Structure:

Includes a file header record, comment record, and multiple detail records. Each record may have multiple occurrences with different effective dates including comment records. Character “C” in the 11th position of the file signifies Comment records. Comment records are for comment purposes only and are used to display the name for a field or column.

See the *ARDEF Table* section for layouts for the various records comprising the ARDEF table.

ARDEF File Header Record Layout

ARDEF File Header record layout details include position, length, format, field name, and description.

ARDEF File Header Record Layout

Position	Length	Format	Field Name	Description
01 – 02	02	AN	Table Type	AA = File Header Record
03 – 10	08	AN	Table Mnemonic	ACTRNG (left-justified, space-filled).
11	01	AN	Record Type	A = Table Header Record
12 – 23	12	AN	Table Key	EPACRN (left-justified, space-filled).
24 – 31	08	UN	Effective Date	Date in CCYYMMDD format.
32	01	AN	Delete Indicator	Blank = Not a delete record D = Delete record (to be deleted after the effective date specified in the preceding field.)
33 – 36	04	AN	Filler	Set to spaces.
37 – 39	03	UN	Release Number	100 for Release 1

ARDEF File Header Record Layout

Position	Length	Format	Field Name	Description
40	01	AN	Separator	Comma
41 – 47	07	AN	Filler	Set to spaces.
48	01	AN	Separator	Comma
49	01	AN	Filler	Set to spaces.
50	01	AN	Separator	Comma
51	01	AN	Filler	Set to spaces.
52	01	AN	Separator	Comma
53	01	AN	Filler	Set to spaces.
54	01	AN	Separator	Comma
55 – 62	08	AN	Filler	Set to spaces.
63	01	AN	Separator	Comma
64 – 67	04	UN	Version Number	Version number of the table update.
68 – 200	133	AN	Reserved	Set to spaces.

ARDEF Table

Valid values for the ARDEF Table are specified in a value table.

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
1–2	02	AN	Table Type	VL = Value Table Record
3–10	08	AN	Table Mnemonic	ACTRNG
11	01	AN	Record Type	R = Detail Record
12–23	12	AN	Table Key	The key for the record is the high range ARDEF value. For example, 401299999.

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
24–31	08	UN	Effective Date	Date in CCYYMMDD format. Comment records are included in the file by setting this field to OCMMNTnn, where nn is a number from 00 to 99. Format records are included in the file by setting this field to OFRMATnn, where nn is a number from 00 to 99. These records control the formatting of the heading and report information.
32	01	AN	Delete Indicator	Blank = Not a delete record D = Delete record (to be deleted after the effective date specified in the preceding field)
33–44	12	AN	Low Key for Range	The low range ARDEF value. For example, 401200000.
45–50	06	UN	Issuer Identifier	Reporting Issuer Identifier. For example, 401200.
51	01	UN	Check Digit Algorithm	0 = No check digit validation performed 1 = Visa modulus 10 performed
52–53	02	UN	Account Number Length	The valid values are: - Any specific value between 10 and 19 00 which means 13/16
54	01	AN	Token Indicator	Valid values: Space = Not applicable Y = Token Range
55	01	AN	Clearing Only Indicator	This field indicates participation in Clearing Only Service: Valid Values: C = Participate Space = Not Participate
56–61	06	UN	BASE II CIB	Member processing CIB.

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
62	01	AN	Domain	W = Worldwide R = Regional N = National D = Domestic
63	01	UN	Region	Visa region of issuer identifier in position 45.
64–65	02	AN	Country	Country of issuer identifier in position 45.
66	01	AN	Large-Ticket	Valid values: L = Large-Ticket Space = Not applicable This field is formerly Issuer Commercial Card Service Indicator (ICCSI)
67	01	AN	Technology Indicator	A = Chip Card
68	01	UN	ARDEF Region	Account range (ARDEF) region of issuer
69–70	02	AN	ARDEF Country	Account range (ARDEF) country of issuer
71	1	AN	Commercial Card Level 2 Data Indicator	Y = Level 2 supported space = Level 2 not supported
72	1	AN	Commercial Card Level 3 Enhanced Data Indicator	Y = Level 3 supported space = Level 3 not supported
73	1	AN	Commercial Card POS Prompting Indicator	Y = POS prompting required Note: Y is only valid if Level 2 data is supported. space = POS prompting is not required
74	1	AN	Commercial Card Electronic VAT Evidence Indicator	Y = Electronic VAT evidence supported space = Electronic VAT evidence not supported

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
75	01	AN	Original Credit	For US issuers: OC flag Space = Reject OC transactions A = Accept OC transactions For Non-US issuers: OC flag Space = Accept OC transactions N = Reject OC transactions
76	01	AN	Account Level Processing Indicator	This field was formerly the Product Type Extension field. Valid values: Y = Card-Level Identification Space = Not participating in ALP
77	01	AN	Original Credit Money Transfer	For US issuers: MT flag Space = Reject MT transactions A = Accept MT transactions For Non-US issuers: MT flag Space = Accept MT transactions N = Reject MT transactions

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
78	01	AN	Original Credit Online Gambling	For US issuers: OG flag Space = Reject OG transactions A = Accept OG transactions For Non-US issuers: OG flag Space = Accept OG transactions N = Reject OG transactions
79–80	02	AN	Product ID	For a complete list of valid values, refer to <i>BASE II Clearing Data Codes</i> .
81	01	AN	Combo Card	Valid values: Y = Combo card, credit and debit C = Combo card, credit and prepaid Space = Not a combo card
82	01	AN	Fast Funds	Valid values: Y = Domestic and Cross-Border Fast Funds C = Cross-Border only D = Domestic Fast Funds only Space = Does not participate in Fast Funds
83	01	AN	Travel Indicator	Valid values: Y = Travel card-capable Space = Not applicable

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
84–85	02	AN	B2B Program ID	This field indicates Business-to-Business (B2B) Program Identification (ID): Valid values: • A1 (B2B Program 1) • A2 (B2B Program 2) • A3 (B2B Program 3) • A4 (B2B Program 4) • A5 (B2B Program 5) • A6 (B2B Program 6) • Spaces (Not Specified)
86	01	AN	Prepaid Program Indicator	This field provides additional information for certain prepaid products. Valid values: 1 = Consumer program in selected markets Spaces = Not specified
87–88	02	AN	RR 12	Spaces (Reserved for future use.)
89	01	AN	Multi Account Access Indicator	This field indicates participation in the Multi Account Access. Valid Values: M = Multi-Account Managed S = Multi-Account Self-Serve Space = Not Participate
90	01	AN	Account Funding Source	C = Credit D = Debit P = Prepaid H = Charge R = Deferred Debit
91	01	AN	Settlement Match	B = B2B E = B2E Space = Not applicable

ARDEF Table Detail Record Layout

Position	Length	Format	Field Name	Description
92	01	AN	Travel Account Data	A = CTA Air H = CTA Hotel S = Central travel account (CTA) Switzerland Space = Not applicable
93	01	AN	Account Restricted Use	A = ATM Space = Not applicable
94	01	AN	NNSS Indicator	Y = Issuing Identifier participates in an NNSS Space = Issuing Identifier does not participate in an NNSS Note: The NNSS indicator will not provide information as to which NNSS the Issuing Identifier participates in nor will it indicate if the Issuer Identifier participates in more than one NNSS or if a transaction will qualify for NNSS settlement.
95–96	02	AN	Product Subtype	For a complete list of valid values, refer to <i>Clearing User's Guide – for the Configuration Repository</i> .
97	01	AN	Alternate ATM	This field indicates participation in Alternate ATM program: Valid values: Y = Participate Space = Not Participate
98	01	AN	Reserved	Reserved for Future Use
99	01	AN	Test Indicator	ARDEF Test Indicator.
100–200	101	AN	Reserved	Set to spaces.

Repository Table File – VID (Visa Identifier) Table Layout

The repository file contains the VID Table which is a list of valid ACQ/ISS IDs and their BASE II processing status codes that are used to control the accuracy of Visa Clearing Exchange (VCX) Processing.

Record Format and Length:

Record Format = Fixed length, maximum record length = 200

File Structure:

Includes a file header record, comment record and multiple detail records. Each record may have multiple occurrences with different effective dates including comment records. Character "C" in the 11th position of the file signifies Comment records. Comment records are for comment purposes only and are used to display the name for a field or column.

VID Table File Header Layout

VID Table file header record layout details include position, length, format, field name, and description.

VID Table File Header Record Layout

Position	Length	Format	Field Name	Description
01 – 02	02	AN	Table Type	AA = File Header Record
03 – 10	08	AN	Table Mnemonic	BINTBL (left-justified, space-filled.)
11	01	AN	Record Type	A = Table Header Record
12 – 23	12	AN	Table Key	EPBINT (left-justified, space-filled.)
24 – 31	08	UN	Effective Date	Date in CCYYMMDD format.
32	01	AN	Delete Indicator	Blank = This is not a delete record D = This is a delete record (to be deleted after the effective date specified in the preceding field)
33 – 36	04	AN	Filler	Set to spaces.
37 – 39	03	UN	Release Number	100 for Release 1
40	01	AN	Separator	Comma

VID Table File Header Record Layout

Position	Length	Format	Field Name	Description
41 – 47	07	AN	Filler	Set to spaces.
48	01	AN	Separator	Comma
49	01	AN	Filler	Set to spaces.
50	01	AN	Separator	Comma
51	01	AN	Filler	Set to spaces.
52	01	AN	Separator	Comma
53	01	AN	Filler	Set to spaces.
54	01	AN	Separator	Comma
55 – 62	08	AN	Filler	Set to spaces.
63	01	AN	Separator	Comma
64 – 67	04	UN	Version Number	Version number of the table update.
68 – 200	133	AN	Reserved	Set to spaces.

VID Table Detail Record Layout

Valid values for the VID Table are specified in a value table. Fields in positions 31 through 77 of this record contain the attributes that are used to do edits.

VID Table Detail Record Layout

Position	Length	Format ¹	Field Name	Description
1–2	02	AN	Table Type	VL = Value Table Record
3–10	08	AN	Table Mnemonic	BINTBL
11	01	AN	Record Type	R = Detail Record
12–23	12	AN	Table Key	ACQ/ISS ID for the table key.

¹ UN = unpacked numeric; AN = alphanumeric.

VID Table Detail Record Layout

Position	Length	Format ¹	Field Name	Description
24–31	08	UN	Effective Date	Date in CCYYMMDD format. Comment records are included in the file by setting the field to OCMNTnn, where nn is a number from 00 to 09. Format records are included in the file by setting this field to OFRMTnn, where nn is a number from 00 to 99. These records control the formatting of heading and report information.
32	01	AN	Delete Indicator	Blank = Not a delete record D = Delete record (to be deleted after the effective date specified in the preceding field)
33–44	12	AN	Filler	Set to spaces.
45–50	06	UN	BASE II CIB	Processing center number.
51	01	UN	Region	Visa region of ACQ/ISS ID in position 12.
52–53	02	AN	Country	Country of ACQ/ISS ID in position 12.
54	01	AN	VID Type	F = Full Service I = Issuer Only M = Merchant Only N = Nonfinancial O = On Behalf Of (OBO) P = Processor S = SMS Financial entity V = Visa Internal entity
55	01	AN	Participation Flag	Blank = Not participating A = GSA IGOTS B = Airline

¹ UN = unpacked numeric; AN = alphanumeric.

VID Table Detail Record Layout

Position	Length	Format ¹	Field Name	Description
56	01	AN	Visa Direct	Blank = Not participating Y = Participating
57	01	AN	TIF Bypass Flag	Valid values: Y = bypass TIF Space = do not bypass TIF
58	01	AN	Multinational Merchant Acceptance	This field will contain an indicator to identify that the enrolled Multinational Merchant Acceptance Acquiring Identifier may qualify for domestic processing. Valid values: Y = Acquirer is participating Space = Acquirer is not participating
59	01	AN	Domestic OCT Program Information Form (PIF)	This field indicates whether the Acquiring Identifier supports the origination of domestic OCTs other than Money Transfer OCTs: Valid values: Y = Acquirer can originate domestic Non-Money Transfer OCTs Space = Acquirer cannot originate domestic Non-Money Transfer OCTs
60	01	AN	Cross-Border OCT Program Information Form (PIF)	This field indicates whether the Acquiring Identifier supports the origination of Cross-Border OCTs other than Money Transfer OCTs: Valid values: Y = Acquirer can originate Cross-Border Non-Money Transfer OCTs Space = Acquirer cannot originate Cross-Border Non-Money Transfer OCTs

¹ UN = unpacked numeric; AN = alphanumeric.

VID Table Detail Record Layout

Position	Length	Format ¹	Field Name	Description
61	01	AN	OCT GEO Blocking	This field indicates type of OCT GEO Blocking: Valid values: D = Domestic Blocking X = Cross-Border Blocking B = Block all OCT Space* = No Blocking (Default)
62–98	37	AN	Reserved	Set to spaces
99	01	AN	Test Indicator	Blank = ACQ/ISS ID is valid for production or test T = ACQ/ISS ID is valid only for RUNMODE=TEST R = Responder ID Note: Responder IDs are present in VCMS VID Tables only
100	01	AN	OBO Indicator ²	Blank = Not an OBO identifier - = OBO identifier ! = OBO Processor
101–200	100	AN	Reserved	Set to spaces.

VCX Extract File Layouts

This section details the Visa Clearing Exchange (VCX) extract file layouts.

¹ UN = unpacked numeric; AN = alphanumeric.

² On Behalf Of indicator.

ARDEF Extract File Layout

This file is produced when the extract function is used. It contains the Account Range Definition (ARDEF) information used by Visa Clearing Exchange (VCX). This file was designed for use in client processor systems that need ARDEF information.

Record Length:

140

File Structure:

This file contains one record for every ARDEF that is current on the table date in effect during the extract process.

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
1–9	09	UN	High Key for Range	The high range ARDEF value, for example, 401299999.
10–12	03	AN	Reserved	Blanks
13–21	09	UN	Low Key for Range	The low range ARDEF value, for example, 401200000.
22–24	03	AN	Reserved	Blanks
25–30	06	UN	Issuing Identifier	Reporting Issuing Identifier, for example, 401200.
31	01	UN	Check Digit Algorithm	0 = No check digit validation performed 1 = Visa modulus 10 performed
32–33	02	UN	Account Number Length	The valid values are: - Any specific value between 10 and 19 00 which means 13/16
34	01	AN	Token Indicator	Valid values: Space = Not applicable Y = Token Range

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
35	01	AN	Clearing Only Indicator	This field indicates participation in Clearing Only Service. Valid Values: C = Participate Space = Not Participate
36–41	06	UN	BASE II CIB	Client processing center CIB
42	01	AN	Domain	W = Worldwide R = Regional N = National D = Domestic
43	01	UN	Region	Visa region of Issuing Identifier in position 25
44–45	02	AN	Country	Country of Issuing Identifier in position 25
46	01	AN	Large-Ticket	Valid values: L = Large-Ticket Space = Not applicable Note: This field is formerly Issuer Commercial Card Service Indicator (ICCSI)
47	01	AN	Technology Indicator	A = Chip Card
48	01	N	ARDEF Region	Account range region of issuer
49–50	02	AN	ARDEF Country	Account range country of issuer
51	1	AN	Commercial Card Level 2 Data Indicator	Y = Level 2 supported space = Level 2 not supported
52	1	AN	Commercial Card Level 3 Enhanced Data Indicator	Y = Level 3 supported Spaces = Level 3 not supported
53	1	AN	Commercial Card POS Prompting Indicator	Y = POS prompting required Note: Y is only valid if Level 2 data is supported. space = POS prompting not required

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
54	1	AN	VAT Evidence	Commercial Card Electronic VAT Evidence Indicator
55	01	AN	Original Credit	For US issuers: OC flag Space reject OC transactions A = accept OC transactions For Non-US issuers: OC flag Space accept OC transactions N = reject OC transactions
56	01	AN	Account-Level Processing Indicator	This field was formerly the Product Type Extension field. Valid values: Y = Card-Level Identification Space = Not participating in ALP
57	01	AN	Original Credit Money Transfer	For US issuers: MT flag Space reject MT transactions A = accept MT transactions For Non-US issuers: MT flag Space Accept MT transactions N = Reject MT transactions
58	01	AN	Original Credit Online Gambling	For US issuers: OG flag Space reject OG transactions A = accept OG transactions For Non-US issuers: OG flag Space accept OG transactions = N = reject OG transactions
59–60	02	AN	Product ID	For a complete list of valid values, refer to <i>BASE II Clearing Data Codes</i> .

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
61	01	AN	Combo Card	Valid values: Y = Combo card, credit, and debit C = Combo card, credit, and prepaid Space = Not a combo card
62	01	AN	Fast Funds	Valid values: Y = Domestic and Cross-Border Fast Funds C = Cross-Border only D = Domestic Fast Funds only Space = Does not participate in Fast Funds
63	01	AN	Travel Indicator	Valid values: Y = Travel card-capable Space = Not applicable
64-65	02	AN	B2B Program ID	This field indicates Business-to-Business (B2B) Program Identification (ID): Valid values: • A1 (B2B Program 1) • A2 (B2B Program 2) • A3 (B2B Program 3) • A4 (B2B Program 4) • A5 (B2B Program 5) • A6 (B2B Program 6) • Spaces (Not specified)
66	01	AN	Prepaid Program Indicator	This field provides additional information for certain prepaid products. Valid values: 1 = Consumer program in selected markets Spaces = Not specified
67-68	02	AN	RR 12	Spaces (Reserved for future use.)

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
69	01	AN	Multi Account Access Indicator	This field indicates participation in the Multi Account Access. Valid Values: M = Multi-Account Managed S = Multi-Account Self-Serve Space = Not Participate
70	01	AN	Account Funding Source	C = Credit D = Debit P = Prepaid H = Charge R = Deferred Debit
71	01	AN	Settlement Match	B = B2B E = B2E Space = Not applicable
72	01	AN	Travel Account Data	A = CTA Air H = CTA Hotel S = Central travel account (CTA) Switzerland Space = Not applicable
73	01	AN	Account Restricted Use	A = ATM Space = Not applicable
74	01	AN	NNSS Indicator	Valid values: Y = Issuer participates in an NNSS Space = Issuer does not participate in NNSS Note: The NNSS indicator will not provide information as to which NNSS the issuer participates in nor will it indicate if the issuer participates in more than one NNSS or if a transaction will qualify for NNSS settlement.

ARDEF Extract File Table Layout

Position	Length	Format	Field Name	Description
75–76	02	AN	Product Subtype	For a complete list of valid values, refer to <i>Clearing User's Guide – for the Configuration Repository</i> .
77	01	AN	Alternate ATM	This field indicates participation in Alternate ATM program: Valid values: Y = Participate Space = Not Participate
78	01	AN	Reserved	Reserved for Future Use
79	01	AN	Test Indicator	ARDEF Test Indicator.
80–140	61	AN	Reserved	Set to spaces

ACQ/ISS ID Extract File Layout

This file is produced when the extract function is used. It contains the Acquiring Identifier/ Issuing Identifier (ACQ/ISS ID) information used by Visa Clearing Exchange (VCX). This file was designed for use in client processor systems that need Acquiring Identifier/Issuing Identifier information.

Record Length:

140

File Structure:

This file contains one record for every ACQ/ISS ID that is current on the table date in effect during the extract process.

ACQ/ISS ID Extract File Table Layout

Position	Length	Format	Field Name	Description
1–6	06	UN	ACQ/ISS ID	Six-digit ACQ/ISS ID.
7–24	18	AN	Reserved	Blanks
25–30	06	UN	BASE II CIB	BASE II CIB for ACQ/ISS ID in Position 1. This field will be the same as ACQ/ISS ID field for processing centers.
31	01	UN	Region	Visa region of ACQ/ISS ID in Position 1.

ACQ/ISS ID Extract File Table Layout

Position	Length	Format	Field Name	Description
32–33	02	AN	Country	Country of ACQ/ISS ID in Position 1.
34	01	AN	VID Type	F = Full Service I = Issuer Only M = Merchant Only N = Nonfinancial O = On Behalf Of (OBO) P = Processor S = SMS Financial entity V = Visa Internal entity
35	01	AN	Participation Flag	Blank = Not participating A = GSA IGOTS A = Airline
36	01	AN	Visa Direct	Blank = Not participating Y = Participating
37	01	AN	TIF Bypass Flag	Valid values: Y = bypass TIF Space = do not bypass TIF
38	01	AN	Multinational Merchant Acceptance	This field will contain an indicator to identify that the enrolled Multinational Merchant Acceptance acquirer may qualify for domestic processing. Valid values: Y = (Acquirer is participating) Space = (Acquirer is not Participating)

ACQ/ISS ID Extract File Table Layout

Position	Length	Format	Field Name	Description
39	01	AN	Domestic OCT Program Information (PIF)	This field indicates whether the Acquirer supports the origination of domestic OCTs other than Money Transfer OCTs: Valid values: Y = Acquirer can originate domestic Non-Money Transfer OCTs Space = Acquirer cannot originate domestic Non-Money Transfer OCTs
40	01	AN	Cross-Border OCT Program Information Form (PIF)	This field indicates whether the Acquirer supports the origination of Cross-Border OCTs other than Money Transfer OCTs: Valid values: Y = Acquirer can originate Cross-Border Non-Money Transfer OCTs Space = Acquirer cannot originate Cross-Border Non-Money Transfer OCTs
41	01	AN	OCT GEO Blocking	This field indicates type of OCT GEO Blocking: Valid values: D = Domestic Blocking X = Cross-Border Blocking B = Block all OCT Space* = No Blocking (Default)
42-78	37	AN	Reserved	Blanks
79	01	AN	Test Indicator	Blank = BASE II CIB is valid for Test or Production T = BASE II CIB is valid only for RUNMODE=TEST R = Responder ID Note: Responder IDs are present in VCMS VID Tables only

ACQ/ISS ID Extract File Table Layout

Position	Length	Format	Field Name	Description
80	01	AN	OBO Indicator	Blank = Not an OBO Identifier - = OBO Identifier ! = OBO Processor
81–140	50	AN	Reserved	Set to spaces

Value Table Extract Files Layouts

This file is produced when the extract function is used. The file can contain the following value table information based on parameter records used.

Record Length:

140

File Structure:

The first eight positions indicate the type of information contained on the record. There is no header or trailer record.

Value Table Extract Country Code Table

Table Name	Release 1 Table ID
BUSINESS Application ID Table	BAI
Country Code Table	COUNTRY
Currency Code Table	CURRENCY
Fee Collection Reason Code Table	FEERSNCD
Holiday Table	HOLIDAY
List of Tables	LIST
Merchant Category Code Value Table	MCC
Product ID Value Table	PRODID
Product Sub Type Value Table	PRODSUBT
Regulated Account Range Table	REGARDEF
Release Type and Description Table	RELEASE

Value Table Extract Country Code Table

Table Name	Release 1 Table ID
Release Module Status Table	RLSTATUS
Return Item Reason Code Table	RTNITEM

Country Code Table

Contains the values of the Visa Clearing Exchange (VCX) Country Code table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Country Code table record.

Value Table Extract Country Code Table

Position	Length	Format	Field Name	Description
01–08	08	AN	Table ID	COUNTRY
09–10	02	AN	Country Code	Country Code
11–32	22	AN	Reserved	Set to spaces
33	01	UN	Foreign Indicator	1 = Foreign 2 = U.S. Territory 3 = U.S.
34–36	03	UN	Currency Code	Default currency code for the country.
37	01	AN	CPS Participation Indicator	C = CPS participating country
38	01	AN	ePay Participation Indicator	E = ePay participating country
39	01	AN	Reserved	Set to spaces
40	01	AN	Area Net Participation Indicator	A = Area Net participating country
41	01	UN	Region	Country Region (System Internal Use)
42	01	AN	CPS/ATM ARN ACQ/ID Participant	System Internal Use

Value Table Extract Country Code Table

Position	Length	Format	Field Name	Description
43	01	AN	VCR Country Migration Status	P = Eligible to submit VCR transactions in production mode. T = Eligible to submit VCR transactions in test mode. Space = Not eligible to submit VCR transactions in production or test mode.
44	1	AN	EEA Flag Note: EEA is "European Economic Area".	Y = Country falls within European Economic Area Space = Country does not fall within EEA
45–140	96	AN	Reserved	Set to spaces

Currency Code Table

Contains the values of the Visa Clearing Exchange (VCX) Currency Code table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Currency Code table record.

Value Table Extract Currency Code Table

Position	Length	Format	Field Name	Description
01–08	08	AN	Table ID	CURRENCY
09–11	03	UN	Currency Code	ISO numeric currency code.
12–14	03	AN	Currency Code	ISO alpha currency code.
15	01	UN	Decimal Place Indicator	The number of minor units for the currency.
16–140	125	AN	Reserved	Set to spaces.

Holiday Table

Contains the values of the Visa Clearing Exchange (VCX) Holiday table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Holiday table record.

Value Table Extract Holiday Table

Position	Length	Format	Field Name	Description
01-08	08	AN	Table ID	HOLIDAY
09	01	UN	Usage	0 = Extended Service Module 1 = Airline Program 2 = European Central Bank calendar
10-12	03	AN	Table Type	If the usage field is 0, this field will contain the ISO alpha country code. If the usage field is 1, this field will contain AI. If the usage field is 2, this field will contain EC.
13-17	05	UN	Holiday date	Julian date format YYDDD.
18-140	123	AN	Reserved	Set to spaces.

Merchant Category Code Table

Contains the values of the Visa Clearing Exchange (VCX) Merchant Category Code table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Merchant Category Code table record.

Value Table Extract Merchant Category Code Table

Position	Length	Format	Field Name	Description
01–08	08	AN	Table ID	MCC
9–12	04	UN	Merchant Category Code: Low Range	Four-digit numeric category code. Lowest code in the range.
13–16	04	UN	Merchant Category Code: High Range	Four-digit numeric category code. Highest code in the range.
17	01	AN	Category Valid Indicator	Space valid for all drafts. 0 Not valid for any drafts D Not valid for original drafts.
18	01	AN	Service Category	1 = Air/Passenger Transport 2 = Car Rental 3 = Lodging 5 = ATM 7 = Amtrak
19	01	UN	Travel & Entertainment (T&E) Indicator	0 = Non-T&E 1 = T&E
20	01	UN	International Airline Indicator	0 = Non-international airline 1 = International airline
21	01	AN	B2B Settlement Mach	Space = not eligible 1 = Eligible
22–140	119	AN	Reserved	Set to spaces.

Fee Reason Code Table

Contains the values of the Visa Clearing Exchange (VCX) Fee Reason Code Table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Fee Reason Code table record.

Value Table Extract Fee Reason Code Table

Position	Length	Format	Field Name	Description
01–08	08	AN	Table ID	FEERSNCD
09–12	04	UN	Fee Reason Code	Four-digit numeric fee reason code.
13–20	08	AN	Reserved	Set to spaces
21–32	12	UN	Max Dollar Amount	999999999999 = No limit 000000000000–999999999998 = Dollar limit (for example, 0010000 = 100 dollar limit)
33	01	AN	State/Country Code Indicator	0 = State or country code not allowed 1 = State or country code is required
34	01	AN	Incoming/Outgoing Indicator	0 = Valid for incoming or outgoing 1 = Valid for incoming only
35	01	AN	Account Number Indicator	0 = Account number must be valid 1 = Account number must be zeros
36–42	7	AN	Reserved	Set to spaces
43	01	AN	Visa-Generated Indicator	0 = Non-Visa-generated 1 = Visa-generated
44–140	97	AN	Fee Reason Code	Code Description

Return Item Reason Code Table

Contains the values of the Visa Clearing Exchange (VCX) Return Item Reason Code table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Return Reason Code table record.

Value Table Extract Return Reason Code Table

Position	Length	Format	Field Name	Description
01-08	08	AN	Table ID	RTNITEM
09-11	03	AN	Return Reason Code	Two-character return reason code.
12-32	21	AN	Reserved	Set to spaces.
33-72	40	AN	Return Reason Code Description	Code description.
73-140	68	AN	Reserved	Set to spaces.

Regulated Account Range Table Extract

Contains Regulated Account Range Table in sequential format.

Record Length:

140

File Structure:

Includes only one record type, the Regulated Account Range table record.

Regulated Account Ranges Table

Position	Length	Format	Field Name	Description
01-08	08	AN	Table ID	REGARDEF
09-26	18	UN	Regulated ARDEF High Range	Regulated ARDEF High Range
27-44	18	UN	Regulated ARDEF Low Range	Regulated ARDEF Low Range.
45-140	96	AN	Reserved	Set to spaces.

Business Application ID (BAI) Table Extract

Contains Business Application ID (BAI) Table in sequential format

Record Length:

140

File Structure:

Includes only one record type, the Business Application ID table record.

Business Application ID (BAI) Table

Position	Length	Format	Field Name	Description
01–03	03	AN	Table Name	“BAI”
04	01	AN	Space	Space
05–06	02	AN	BAI Value	BAI Value
07	01	AN	Space	Space
08–09	02	AN	BAI Class	BAI Class
10	01	AN	Space	Space
11–50	40	AN	BAI Name	BAI Name/Description
51–140	90	AN	Reserved	Space

Appendix B

VCX Print/Pass Defaults

BASE II Clearing transaction codes have print/pass defaults in Visa Clearing Exchange (VCX). The pass default identifies a transaction that must be passed to the Center Transaction File (CTF) before it will appear on the output file.

Print/Pass Defaults for BASE II Transaction Codes

Transaction Code and Type	Financial Transaction	Incoming or Outgoing	Outgoing Print Default	Incoming Print Default	Incoming Pass Default
00: Short Block	No	Incoming	Not Applicable	No Print	No Pass
01: Returned Credit	Yes	Incoming	Not Applicable	Print	No Pass
02: Returned Debit	Yes	Incoming	Not Applicable	Print	No Pass
03: Returned Nonfinancial	No	Incoming	Not Applicable	Print	No Pass
04: Reclassification Advice	No	Incoming	Not Applicable	Print	No Pass
05: Sales Draft or Representment	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹

Print/Pass Defaults for BASE II Transaction Codes

Transaction Code and Type	Financial Transaction	Incoming or Outgoing	Outgoing Print Default	Incoming Print Default	Incoming Pass Default
06: Credit Voucher	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
07: Cash Disbursement	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
10: Fee Collection	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
15: Sales Draft Chargeback	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
16: Credit Voucher Chargeback	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
17: Cash Disbursement Chargeback	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
20: Funds Disbursement	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
25: Sales Draft Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
26: Credit Voucher Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
27: Cash Disbursement Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
30: ICS Input	No	Outgoing	No Print	n/a	n/a
31: ICS Response Transaction File	No	Incoming	Not Applicable	No Print	Pass
33: RDMS Message or CRS Advices	No	Incoming	Not Applicable	Print	No Pass
35: Sales Draft Chargeback Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
36: Credit Voucher Chargeback Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹

Print/Pass Defaults for BASE II Transaction Codes

Transaction Code and Type	Financial Transaction	Incoming or Outgoing	Outgoing Print Default	Incoming Print Default	Incoming Pass Default
37: Cash Disbursement Chargeback Reversal	Yes	Incoming and Outgoing	No Print	No Print	Pass ¹
40: Fraud Advice	No	Incoming and Outgoing	No Print	Print	No Pass
42: Merchant File Update	No	Outgoing	No Print	Print	No Pass
43: Merchant File Update	No	Outgoing	No Print	Print	No Pass
44: Collection Batch Acknowledgment	No	Incoming	Not Applicable	No Print ¹	No Pass ¹
45: General Delivery Report	No	Incoming	Not Applicable	Not Applicable ²	No Pass
46: Member Settlement Data	No	Incoming	Not Applicable	No Print ²	Pass ¹
47: Report Generation	No	Incoming	Not Applicable	Not Applicable ²	No Pass ¹
48: BASE I Advice	No	Incoming	No Print	Print	No Pass
49: Local Batch Acknowledgment	No	Incoming	Not Applicable	No Print	Not Applicable
50: Free Text Message	No	Incoming and Outgoing	No Print	Print	No Pass
54: Table Update Data	No	Incoming	Not Applicable	No Print	No Pass ¹
55: RCRF Update	No	Incoming	Not Applicable	Print	No Pass
56: Currency Conversion Rate Update	No	Incoming	Not Applicable	Print	No Pass
57: Data Capture Transaction Advice	No	Incoming and Outgoing	No Print	No Print	Pass
58: National Settlement Advice	No	Incoming and Outgoing	No Print	Print	Pass

Print/Pass Defaults for BASE II Transaction Codes

Transaction Code and Type	Financial Transaction	Incoming or Outgoing	Outgoing Print Default	Incoming Print Default	Incoming Pass Default
59: Interface Transaction Advice	No	Incoming and Outgoing	No Print	No Print	Pass
90: File Header	No	Incoming and Outgoing	No Print	No Print	Not Applicable ³
91: Batch Trailer	No	Incoming and Outgoing	No Print	No Print ¹	Pass ¹
92: File Trailer	No	Incoming and Outgoing	No Print	No Print ¹	Pass ¹

¹The default value cannot be changed.

²These reports are produced automatically and are not applicable for the Print Incoming Reports and Stop Incoming Report Printing run control options.

³These transactions are controlled by the Incoming Header Record run control option and are not applicable for the Include Transactions on Output and Exclude Transactions from Output run control options.

Appendix C

VCX Folder Structure

VCX Folder Structure

Scenario	VCX	Comments
Incoming Batch run	Input location - /vcx/datafiles/IITF/<CIB>/ For example: • Linux: /opt/app/vcx/datafiles/IITF/<CIB>/ • Windows: C:\vcx\datafiles\IITF\<CIB>\ Output location - /vcx/datafiles/ICTF/<CIB>/	Only the input location can be custom path
Incoming Reports	/vcx/batch/<batchnumber>/incoming/reports	
Processed ITF files	/vcx/batch/processedITFs	
Incoming Run Options	/vcx/batch/<batchnumber>/	
Incoming logs	/vcx/batch/<batchnumber>/	
Incoming history	/vcx/history/<CIB>/	

VCX Folder Structure

Scenario	VCX	Comments
Outgoing Batch Run	Input location - /vcx/datafiles/ OCTF/<CIB>/ Output location - /vcx/datafiles/ OITF/<CIB>/	Only the input location can be custom path
Outgoing Reports	/vcx/batch/<batchnumber>/ outgoing/reports	
Processed CTF files	/vcx/batch/processedCTFs	
Outgoing Run Options	/vcx/batch/<batchnumber>/	
Outgoing logs	/vcx/batch/<batchnumber>/	
Outgoing history	/vcx/history/<CIB>/	
Scheduler	/vcx/batch/<batchnumber>/ scheduler/	
Table Extracts/Reports	/vcx/tableExtractsReports	Custom path cannot be provided
Incoming File Transfer (EAS)	/vcx/datafiles/IITF/<CIB>/	Custom path can be provided
FTPS logs	/vcx/ftpsLogs/	
FTPS jars	/vcx/ftps/	

Appendix D

Manage PostgreSQL DB Backup

The VCX application includes a backup and encryption feature for PostgreSQL databases. Backup files are encrypted to ensure data security and compliance with privacy regulations. When clients need to restore or access the backup data, they must first manage the file to retrieve the SQL queries.

How to retrieve PostgreSQL DB Backup:

1. Navigate to the VCX Application Installed folder, ex: `/opt/app/vcx` on Linux or `C:\vcx` on Windows.

2. Navigate to the main folder.

3. Execute the following command:

```
node decryptDbBackup.bundle.js
```

4. The user will be prompted to answer:

? Enter the postgresQL encrypted database backup file path:

```
ENTER <VCX Application Folder>/database/backup/  
encrypted_backup_<TIMESTAMP>.txt
```

5. After successful execution, the utility confirms with a similar message:

```
"Database backup file is available at <VCX Application Folder>/  
database/backup"
```

Note: The PostgreSQL dump file is a plain-text SQL file. Once you have finished using this file, you are responsible for securely deleting it from the server to prevent unauthorized access.

Glossary

acquirer	A client financial institution that has agreements with merchants to accept Visa card transactions or offers cash disbursement services to cardholders, or both. The acquirer is responsible for: • Accepting card transaction data from merchants and its own ATMs and bank branches. • Providing authorization decisions to those card-accepting locations. • Conveying transaction information to Visa as interchange transactions.
acquirer center	A BASE II processing center supporting one or more Visa acquirers. The processing center receives transaction information from merchants and cash dispensing locations on behalf of acquirers; processes local transactions and sends interchange transactions to a VIC for distribution to the issuer processing centers; and settles the value of transactions with merchants and agents and, for interchange transactions, with other clients through the BASE II System.
Acquirer Reference Number	A 23-digit identification number associated with every draft and voucher. It consists of a Format Code, Acquiring Identifier, Capture Date, Film Locator, and Check Digit.
ARDEF File	The permanent file for the ARDEF (Account Range Definition) Table, which is used to control the accuracy of Visa Clearing Exchange (VCX) processing. The table contains all valid ARDEF entries, namely: the account prefix range, its associated Issuing Identifier, card-number length indicator, check-digit indicator, product ID, and account funding source.
BASE II CIB	A six-digit system number used by Visa to identify the processing centers and clients. Center Identification Blocks (CIBs) are assigned to processing centers operated by clients, nonclient processing centers designated by clients, and the clients that operate processing centers. Note: The BASE II CIB of a given processing center does not necessarily appear in the cardholder account numbers processed by that center.
BASE II processing center	See processing center.

BASE II System	An electronic batch transmission system primarily used for the exchange of Visa interchange transaction data and for settlement of the value of those transactions between acquirers and issuers. This system is also used by centers to retrieve records from the Advice File and by Visa to settle various fees with clients.
batch	A set of transaction records, terminating with a batch trailer, sent through BASE II.
batch acknowledgment transaction	A receipt confirmation generated at a VIC of each batch of outgoing transactions. These transactions are received in the center's incoming Interchange Transaction File.
batch reject	See rejected batch.
batch trailer record	A record designating the end of a batch of BASE II transactions. It contains count and monetary totals used to control the integrity of the batch's transaction data. See also merchant batch trailer record.
billing currency	The currency in which the issuer bills the cardholder for transactions.
card issuer processing center	See issuer center.
cardholder processing center	See issuer center.
Center Transaction File (CTF)	The outgoing Center Transaction File contains interchange transactions generated by a processing center's pre-edit program. If the format is acceptable to Visa Clearing Exchange (VCX), it is converted to an ITF and is submitted to the VIC. The incoming Center Transaction File contains ITF data transmitted from the VIC through the VAP to VCX for processing. If there are no errors, the ITF is converted to a CTF and used as input to the post-edit program.
central processing date (CPD)	The date (based on GMT) when the ITF or report in question was generated at a VIC.
Clear Only	Clearing Service Only Transactions.
clearing	All of the functions required to collect a transaction from an acquirer in the merchant's currency and deliver it to the issuer in the cardholder's currency.
collection-only transactions	An intraprocessor transaction submitted to BASE II for collection only (not settlement or delivery). Normal BASE II processing charges and interchange reimbursement fees do not apply to collection-only transactions.

CPD	See central processing date (CPD).
CPS	See Custom Payment Service (CPS).
credit voucher	Sometimes referred to as credit return, it is the record of a return or price adjustment of a purchase.
CTF	See Center Transaction File (CTF).
currency conversion rate	This rate is applied by Visa International to certain transactions (original sales drafts, representments, travel vouchers, credit vouchers, and cash disbursements) and the reversal of such transactions.
currency of purchase	See transaction currency.
currency trading cutoff	The time at which currency conversion rates expire.
Custom Merchant Service	A service that tailors interchange reimbursement fees to specific merchant categories.
Custom Payment Service (CPS)	A Visa payment service that minimizes chargebacks and facilitates transaction clearing and settlement by assigning a unique identifier that stays with the transaction throughout its life cycle.
designated currency	One of the currencies that may be chosen by a client for settlement and funds transfer.
destination Identifier	An Identifier to which a BASE II transaction message is sent.
destination currency	The currency type presented to the client on incoming transactions. For most transactions (that is, drafts), it is the billing currency. For other transactions, for example, fee collection and chargeback, it is the settlement currency of the destination.
DEX	See Direct Exchange
Direct Exchange (DEX)	Direct Exchange provides a single, secure, and extensible Internet Protocol (IP)-based connection to the VisaNet network for all Visa USA endpoints.
dispute financial	A sales draft or other item that has been examined by the issuer center, found to be improper, and sent back to the acquirer center with other outgoing interchange.
dispute financial reversal	The cancellation of a dispute financial sent in error to the acquirer center.

dispute response financial	Paper (or a transaction) submitted by an acquirer to an issuer a second time, following receipt of a dispute financial.
draft data transaction	A BASE II financial transaction that contains data for a cardholder transaction and results in a debit or a credit to clearing clients during the settlement process.
EA Server	See Visa Extended Access Server.
Extended Access Server	See Visa Extended Access Server.
fee collection transaction	A BASE II transaction representing a miscellaneous financial charge assessed by one client or by Visa against another client.
file trailer record	A record designating the end of a CTF or ITF. It contains count and monetary totals used to control the integrity of file content. A CTF is terminated with one file trailer record regardless of the number of volumes used to contain the data; an ITF on tape is terminated with a file trailer record at the end of each volume, if multiple volumes are needed.
file transfer	Electronic transfer of an ITF between Visa Clearing Exchange and Visa.
file header record	A record designating the beginning of an CTF or ITF. It contains the processing center ID, security code, and relevant control information.
funds disbursement transaction	A BASE II transaction used to transfer monetary credit from one BASE II entity to another or to reverse a fee collection transaction.
incoming interchange	All BASE II transactions transmitted from a VIC to a processing center, or the entire process of receiving incoming interchange transaction data from a VIC.
interchange processing	The electronic movement of transaction data between acquirers and issuers.
Interchange Reimbursement Fee (IRF)	A fee paid by issuers and acquirers to each other for transactions entered into interchange (and their reversals) to balance the cost of doing business.
interchange transaction	Any transaction where the client that signed the cardholder submits transactions through a different processing center than the client that signed the merchant.

Interchange Transaction File (ITF)	The <i>outgoing</i> Interchange Transaction File (ITF) contains transactions sent to VisaNet by an endpoint. This file may be created by the outgoing Visa Clearing Exchange (VCX) after the endpoint's pre-edit processing, or it may be sent directly to VisaNet. The <i>incoming</i> ITF contains transactions delivered to an endpoint by VisaNet. This file may be read by VCX prior to the transaction being processed by the endpoint's post-edit processing, or it may be directly processed by the endpoint.
issuer	A client financial institution that issues Visa cards. For a given transaction, the institution that issued the card used for that transaction to the cardholder. The issuer is responsible for maintaining the accounts of its cardholders, for providing authorization decisions, for cardholder billing, and for settlement of transactions its cardholders have with merchants and cash dispensing locations of other clients. Each issuer operates or designates an issuer center to perform the functions related to clearing and settlement of interchange transactions.
issuer center	A BASE II processing center acting in support of one or more issuers. The processing center processes completed cardholder transactions (local and interchange) for cardholder account posting and billing. For completed interchange transactions, the center is also responsible for receiving and processing incoming transactions for the cardholders of the issuer or issuers.
ITF	See Interchange Transaction File (ITF).
Julian date	A date expressed as the day's position in a year rather than in a particular month. The format is YDDD or YYDDD.
National Net interchange	Transactions that are exchanged between processing centers whose issuers and acquirers are located in the same country where settlement is accomplished through a central agent bank for each country using the service. BASE II clears the transactions and records transaction totals on National Net interchange reports.
national settlement transaction	A National Net transaction.
net settlement amount	The currency amount representing the difference between a settlement entity's outgoing and incoming interchange for a given day plus or minus fees and charges. May be a debit or a credit.
outgoing interchange	All BASE II transactions transmitted from a client's processing center to a VIC. Both acquirer and issuer centers send outgoing interchange.

post-edit program	Software written and maintained by a processing center to restructure an incoming Center Transaction File into a format acceptable for the local posting and billing process and to apply client-unique edit criteria against the transactions. This program is executed following the incoming Visa Clearing Exchange run.
pre-edit program	Software written and maintained by a processing center to separate on-us items from interchange items and to apply client-unique edit criteria against the transactions. This program also formats outgoing interchange into the outgoing Center Transaction File for processing by Visa Clearing Exchange according to BASE II specifications. This program is executed before the outgoing Visa Clearing Exchange run.
presentment	Paper (or a transaction) submitted for the first time by an acquirer to an issuer and processed through VisaNet interchange.
processing center	The entity, operated or designated by a clearing client of Visa, responsible for processing of interchange transactions. It executes Visa Clearing Exchange (VCX) and the pre- or post-edit programs, or both, and sends and receives interchange transactions to and from a VIC. A single processing center may function as an acquirer center, an issuer center, or both. It performs interchange transaction services for one client or a multiple number of clients. Most BASE II processing centers are operated by Visa clients; nonclient processing centers may be authorized to process Visa transactions.
processing date	See central processing date (CPD) or Visa Clearing Exchange (VCX) processing date.
reimbursement attribute	A one-digit alphanumeric code designating reimbursement fees applicable to a specific transaction.
reimbursement fee	Amount paid by one client to another (usually by the acquirer to the issuer), and can vary according to market requirements.
rejected batch	An interchange batch that is not accepted by the VIC due to an error in the audit integrity of that batch.
rejected transaction	An outgoing BASE II transaction record in which the Visa Clearing Exchange (VCX) detected an error that affects the financial integrity of the batch. The VCX excludes such transactions from outgoing interchange, that is, the transaction is not included in the outgoing Interchange Transaction File (ITF) forwarded to a VIC. (The batch is not rejected; all valid transactions in the batch are included in the outgoing ITF.) Unless the transaction is a batch or file trailer record, the run aborts.

returned transaction	A cardholder transaction record in which the VIC edit function detected an error that does not violate the financial integrity of the batch. When such an error is detected, the transaction is included in the outgoing batch interchange totals (in a separate category), but it is not forwarded to the issuer center. The transaction is placed in a new BASE II transaction, with a new transaction code, and returned to the originating center with incoming interchange. (On the incoming reports for an originating center, the transactions appear both in outgoing and in incoming totals.)
reversal	A BASE II transaction used to negate or cancel a transaction that has been sent through interchange in error.
settlement	The actual transfer of funds from the issuing bank to the acquiring bank through a wire transfer to a settlement account, and the total amount owed by one Visa client to another. See also net settlement amount.
settlement currency	The currency used by the BASE II System to calculate a settlement entity's daily net settlement position.
source currency	The currency type associated with the amount of a transaction entered into interchange.
source Identifier	The Identifier from which a BASE II transaction message is sent.
special airline fee	A fee charged on transactions from International Airlines whenever the issuer, acquirer, and transaction countries are not all the same. This fee is collected instead of the IAF fee and is paid to the transaction region.
TC 33 Extract File	A file containing Routing table update records that is created through incoming Visa Clearing Exchange (VCX) processing for all clients subscribing to the Plus ATM system. The TC 33 Extract Table contains Plus card issuing Identifiers.
TCR	See Transaction Component Record (TCR).
transaction	<i>BASE II transaction.</i> The record or records that make up a single financial, administrative, or text message, as required for transmission between a processing center and a VIC. BASE II transactions are identified by transaction codes. <i>Cardholder transaction.</i> The use of a card by a customer (normally assumed to be the cardholder) to purchase goods or services from a merchant or secure cash from an ATM or financial institution.
transaction code (TC)	A two-digit code that identifies a specific type of BASE II transaction.

Transaction Component Record (TCR)	A fixed-length record used to contain a component portion of a BASE II transaction. A single BASE II transaction may consist of multiple TCRs.
transaction component sequence number	A single digit placed in each TCR so multiple records (TCRs) can be combined into a single BASE II transaction.
transaction currency	The currency of the purchase, as agreed to by the cardholder and the merchant.
VFG	See Visa File Transfer (VFG).
VIC	See VisaNet Interchange Center (VIC).
VIC processing date	See central processing date (CPD).
VID Table File	The permanent file for VID Table, which is used to control the accuracy of Visa Clearing Exchange (VCX) processing. The table contains all valid Acquiring/Issuing Identifiers and their BASE II processing status codes.
V.I.P. System	An electronic data transmission system for the real-time delivery and processing of messages related to authorization of bank, T&E, private label, and proprietary card and check acceptance transactions. It accepts authorization requests from acquirer and merchant authorization centers and either provides authorization decisions or secures authorization decisions from the issuer authorization centers.
Visa Extended Access Server	The Visa Extended Access Server is the next-generation gateway to Visa products and services, replacing legacy VisaNet Access Points. The EA Server offers improved security and a flexible platform for future updates.
Visa File Gateway (VFG)	Visa File Gateway (VFG) is an interface type used to electronically transfer interchange data and Visa Clearing Exchange (VCX) maintenance files between your processing center and Visa.
Visa Smart Debit/Visa Smart Credit (VSDC)	A payment service involving the use of chip cards and chip capable card acceptance devices providing a variety of features to support offline authorization, protect against fraud, enhance cardholder verification, and provide a platform for multifunction programs.
VisaNet Interchange Center (VIC)	The computers and all associated peripheral devices and telecommunications support facilities needed for the V.I.P. System, the BASE II System, related systems, and backup for these systems.

Notices

User interfaces, menu options and steps depicted herein are illustrative only, may vary and are subject to change in Visa's sole discretion.

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